

CHAPTER THREE

AC-DC CONVERSION: CONTROLLED RECTIFICATIONS

3.1 INTRODUCTION

The disadvantages of the uncontrolled rectifiers discussed in the previous chapter are overcome if the diodes are replaced by controllable power semiconductor switch; the resulting converters are called controlled converters. In Controlled Rectifications, the generated d.c. power is controllable and variable. They usually use SCRs as their power switches. For fast switching operation, MOSFETs and IGBTs are used. The following sections deal with the basic operation of some examples of controlled rectifiers starting with the simplest type which is the single-phase half-wave controlled rectifier loaded with resistive load.

3.2 SINGLE-PHASE, HALF-WAVE, CONTROLLED RECTIFIER LOADED WITH PASSIVE LOADS

3.2.1 Case of Resistive Load

Fig.3.1(a) shows the basic circuit for a single-phase, half-wave, controlled rectifier loaded with a resistive load. For this configuration, during the positive half cycle of the supply voltage v_s , the anode of the thyristor is connect to the positive terminal of the supply, the cathode is connect to negative terminal of supply, and the thyristor is forward biased. Hence, the thyristor will conduct only when triggered using gate pulses.

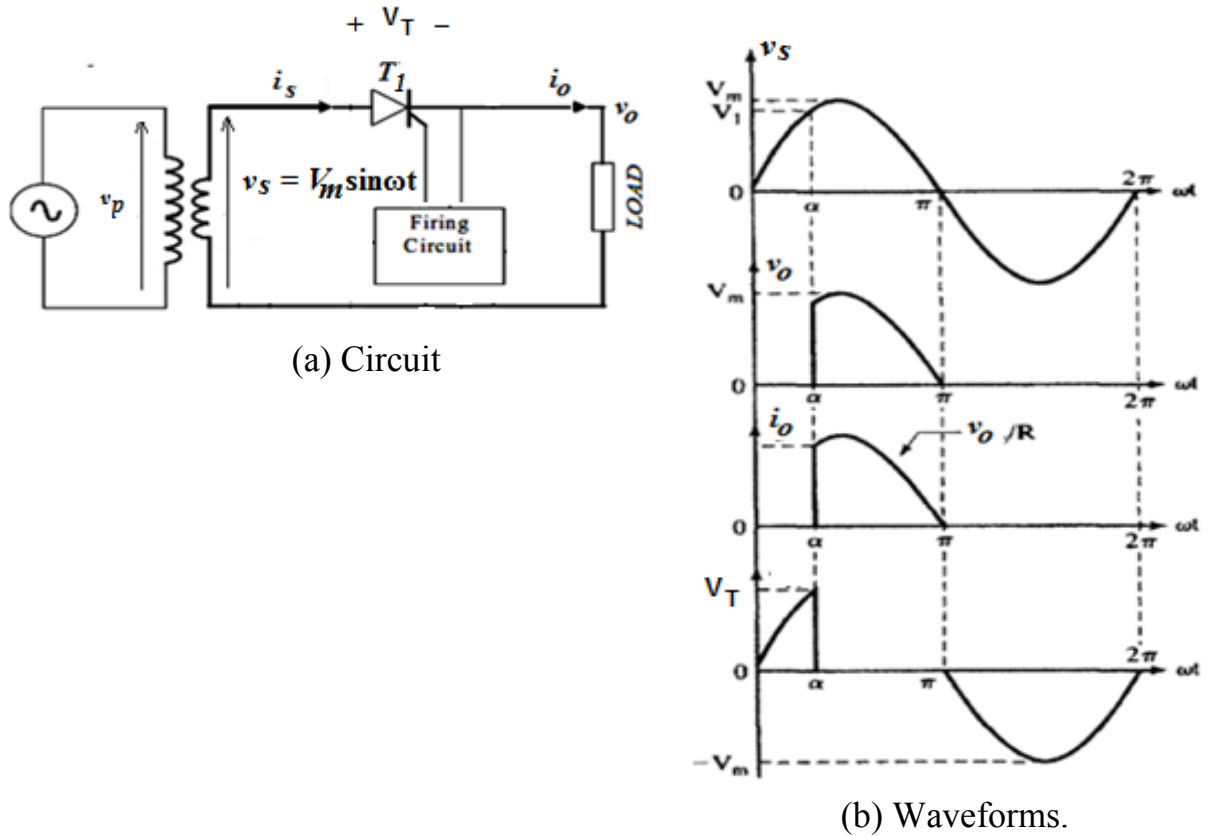


Fig.3.1 Single-phase half-wave controlled rectifier.

The thyristor is fired at $\omega t = \alpha$, and the input voltage appears across the load. The voltage across the thyristor collapses to almost zero and the full supply voltage appears across the load. From this point onwards, the load voltage follows the supply voltage. The load being purely resistive the load current i_o is proportional to the load voltage.

At $\omega t = \pi$, T_1 is reverse-biased by the negative half-cycle of the supply and during the period $\pi < \omega t \leq 2\pi$, the thyristor is turned off and blocks the supply voltage and the load voltage remains zero as shown in Fig.3.1(b). Consequently, no load current flows during this interval. Since the output voltage and current are both positive, the converter is said to operate in the first quadrant. This converter is not normally used in the industrial application because of its high ripple current and low ripple frequency. The waveforms for one total period of operation of this circuit are shown in Fig.3.1(b).

The average value of the load voltage V_{dc} can be calculated as follows:

$$V_{dc} = \frac{1}{2\pi} \int_{\alpha}^{\pi} v_s(\omega t) d\omega t = \frac{1}{2\pi} \int_{\alpha}^{\pi} V_m \sin \omega t d\omega t$$

$$\therefore V_{dc} = \frac{V_m}{2\pi} (-\cos\pi + \cos\alpha) = \frac{V_m}{2\pi} (1 + \cos\alpha) \quad (3.1)$$

Therefore, the average output voltage can vary from 0 to V_m/π and the average load current will vary from 0 to $V_m/\pi R$ when varying α from π to 0, respectively.

Since the load is resistive, therefore the load voltage and current are in phase and they are related by $i = v/R$. Consequently, the average value of the load current I_{dc} is

$$I_{dc} = \frac{V_{dc}}{R} = \frac{V_m}{2\pi R} (1 + \cos\alpha) \quad (3.2)$$

The output d.c. power is given by:

$$P_{dc} = V_{dc} I_{dc} = \frac{V_{dc}^2}{R} \quad (3.3)$$

The *rms* value of the load voltage V_{orms} can be calculated as follows:

$$\begin{aligned} V_{orms} &= \sqrt{\frac{1}{2\pi} \int_{\alpha}^{\pi} v_s^2(\omega t) d\omega t} = \sqrt{\frac{1}{2\pi} \int_{\alpha}^{\pi} (V_m \sin\omega t)^2 d\omega t} \\ &= \sqrt{\frac{(V_m)^2}{2\pi} \int_{\alpha}^{\pi} \frac{1}{2} (1 - \cos 2\omega t) d\omega t} \\ &= \frac{V_m}{2} \sqrt{\frac{1}{\pi} (\pi - \alpha + \frac{1}{2} \sin 2\alpha)} \end{aligned} \quad (3.4)$$

$$\text{since : } (\sin\omega t)^2 = \frac{1}{2} (1 - \cos 2\omega t)$$

Therefore the *rms* value of the load current I_{orms} is :

$$I_{orms} = \frac{V_{orms}}{R} = \frac{V_m}{2R} \sqrt{\frac{1}{\pi} (\pi - \alpha + \frac{1}{2} \sin 2\alpha)} \quad (3.5)$$

The output a.c. power is given by:

$$P_{ac} = V_{orms} I_{orms} = \frac{V_{orms}^2}{R} = \frac{(0.5V_m)^2}{R} \quad (3.6)$$

The *PRV* of the thyristor for this configuration is V_m .

To find the power factor of the circuit, the current drawn from the source i_s is the same as the load current. Hence,

$$P_{ac} = V_{orms} I_{orms}$$

$$PF = \frac{P_{ac}}{S} \quad , \quad \text{but} \quad S = V_s I_{orms}$$

$$\text{Hence} \quad PF = \frac{V_{orms} I_{orms}}{V_s I_{orms}} = \frac{V_{orms}}{V_s}$$

$$= \frac{V_m}{2V_s} \sqrt{\frac{1}{\pi} (\pi - \alpha + \frac{1}{2} \sin 2\alpha)}$$

$$= \frac{\sqrt{2}V_s}{2V_s} \sqrt{\frac{1}{\pi} (\pi - \alpha + \frac{1}{2} \sin 2\alpha)}$$

$$\therefore PF = \sqrt{\frac{1}{2\pi} (\pi - \alpha + \frac{1}{2} \sin 2\alpha)} \quad (3.7)$$

3.2.2 Single-Phase, Half-Wave, Controlled Rectifier Loaded with Series Resistive – Inductive Load

Fig.3.2 shows a single-phase half-wave controlled rectifier with R - L load. This circuit is the same as that of Fig.3.1(a) except that the load consists of a resistor and inductor connected in series. At $\omega t = \alpha$, the voltage across the load will be the instantaneous value of the supply voltage at this firing angle. At this instant, because of the existence of the inductor, the current will increase slowly in the same manner as the diode case discussed in Chapter Two. The load voltage and current waveforms are as depicted in Fig.3.3.

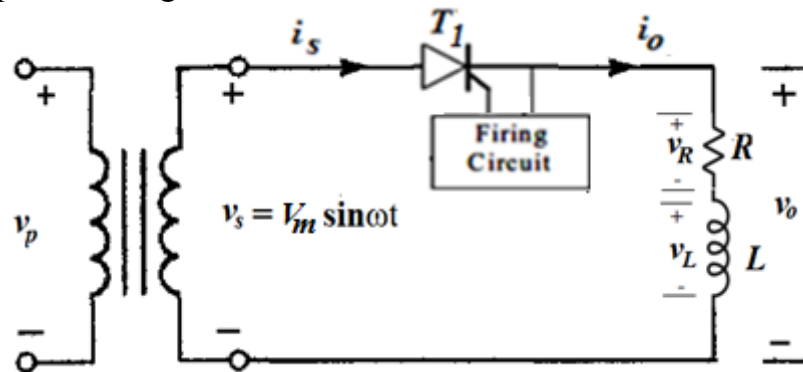


Fig.3.2 Single-phase half-wave controlled rectifier with series R - L load.

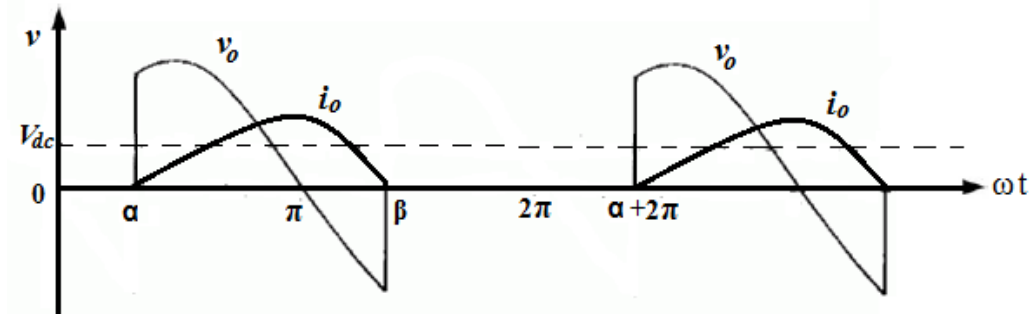


Fig.3.3 Waveforms for the single-phase half-wave controlled rectifier with R - L load.

The average value of the load voltage V_{dc} can be calculated as follows:

$$\begin{aligned}
 V_{dc} &= \frac{1}{2\pi} \int_{\alpha}^{\beta} v_s(\omega t) d\omega t = \frac{1}{2\pi} \int_{\alpha}^{\beta} V_m \sin \omega t d\omega t \\
 &= \frac{V_m}{2\pi} (-\cos \beta + \cos \alpha) \\
 &= \frac{V_m}{2\pi} (\cos \alpha - \cos \beta)
 \end{aligned} \tag{3.8}$$

Consequently, the average value of the load current I_{dc} is

$$I_{dc} = \frac{V_{dc}}{R} = \frac{V_m}{2\pi R} (\cos \alpha - \cos \beta) \tag{3.9}$$

The output d.c. power is given by:

$$P_{dc} = V_{dc} I_{dc} = \frac{V_{dc}^2}{R} \tag{3.10}$$

The *rms* value of the load voltage V_{orms} can be calculated as follows:

$$\begin{aligned}
 V_{orms} &= \sqrt{\frac{1}{2\pi} \int_{\alpha}^{\beta} v_s^2(\omega t) d\omega t} \\
 &= \sqrt{\frac{1}{2\pi} \int_{\alpha}^{\beta} (V_m \sin \omega t)^2 d\omega t} = \sqrt{\frac{(V_m)^2}{2\pi} \int_{\alpha}^{\beta} \frac{1}{2} (1 - \cos 2\omega t) d\omega t}
 \end{aligned}$$

$$= \frac{V_m}{2} \sqrt{\frac{1}{\pi} [(\beta - \alpha) - \frac{1}{2}(\sin 2\beta - \sin 2\alpha)]} \quad (3.11)$$

Therefore, the *rms* value of the load current I_{orms} is :

$$I_{orms} = \frac{V_{orms}}{R} = \frac{V_m}{2R} \sqrt{\frac{1}{\pi} [(\beta - \alpha) - \frac{1}{2}(\sin 2\beta - \sin 2\alpha)]} \quad (3.12)$$

The output a.c. power is given by:

$$P_{ac} = V_{orms} I_{orms} \quad (3.13)$$

The *PRV* of the thyristor for this configuration is V_m . To find the power factor of the circuit, the current drawn from the source i_s is the same as the load current. Hence,

$$S = V_s I_{orms} \quad (3.14)$$

$$\text{But } PF = \frac{P_{ac}}{S}$$

$$\text{Hence } PF = \frac{V_{orms} I_{orms}}{V_s I_{orms}} = \frac{V_{orms}}{V_s}$$

$$= \frac{V_m}{2V_s} \sqrt{\frac{1}{\pi} [(\beta - \alpha) - \frac{1}{2}(\sin 2\beta - \sin 2\alpha)]}$$

$$\text{Since } V_m/2V_s = \sqrt{2}V_s/2V_s$$

$$\therefore PF = \sqrt{\frac{1}{\sqrt{2}\pi} [(\beta - \alpha) - \frac{1}{2}(\sin 2\beta - \sin 2\alpha)]} \quad (3.15)$$

3.2.3 The Freewheeling Diode in Single-Phase Controlled Rectification

The freewheeling diode (FWD) is connected in the circuit across an *R-L* load in such a way as to provide an alternative path for the decaying load current so that the thyristor current is allowed to become zero and the thyristor is allowed to turn off.

Consider the half-wave rectifier of Fig.3.4(a) where FWD is the freewheeling diode. If switch *S* is opened, i.e. the converter operates

without FWD, then the output voltage and current waveforms are as shown in Fig.3.4(b). If S is closed, then the supply voltage is positive, from α to π , FWD is in reverse and passes approximately no current (open circuited), so that source and load current are equal ($i_s = i_o$) as shown in Fig.3.4(d).

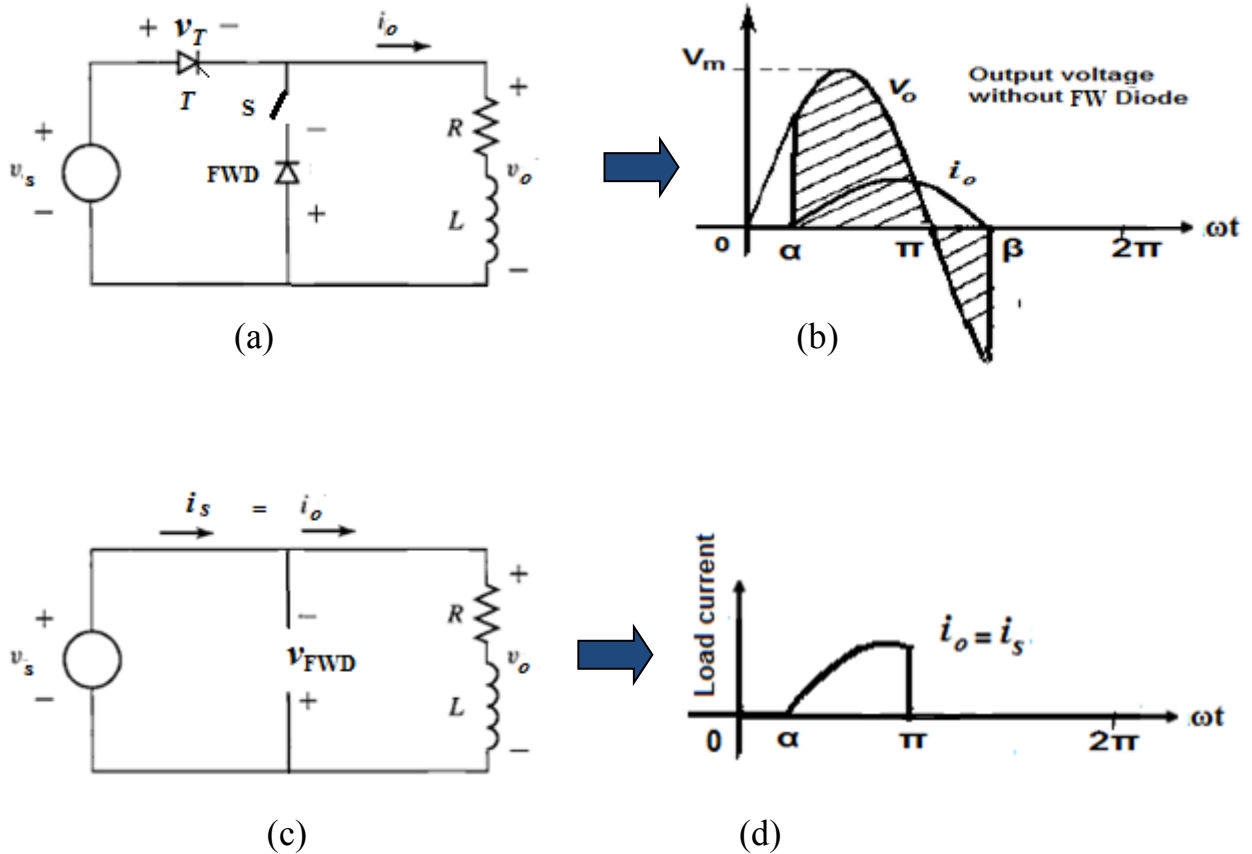


Fig.3.4 Converter operation : (a) Without freewheeling diode, (b) Voltage and current waveforms without freewheeling diode, (c) Operation with freewheeling diode, (d) Supply and load Currents during the positive half-cycle.

During the negative half-cycle of the supply, the load current i_o flows through the low resistance path provided by FWD rather than against the negative supply voltage, so that $i_{FWD} = i_o$, and $i_s = 0$. Hence the thyristor T is allowed to switch off. In this part of the half-cycle, the current is driven by the energy stored in L ; it decays according to the time constant of the circuit (R , L , and FWD), Fig.3.5; v_o is very small and negative, being equal to the voltage drop across FWD. Therefore, the output voltage waveform will be as depicted in Fig.3.6 which is exactly similar to the case of resistive load.

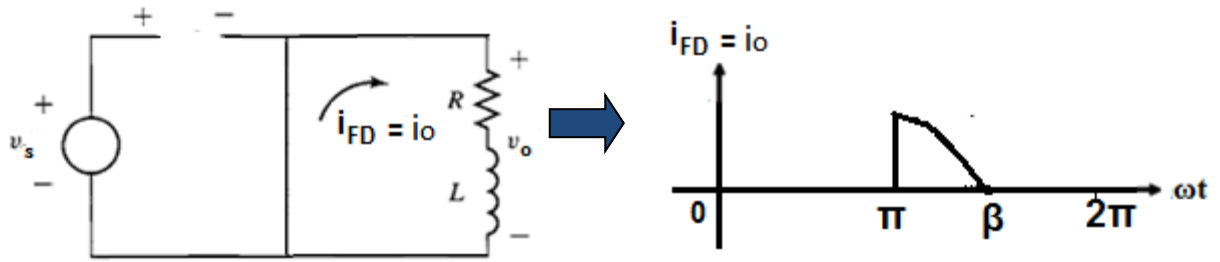


Fig.3.5 Load current during the supply negative half-cycle.

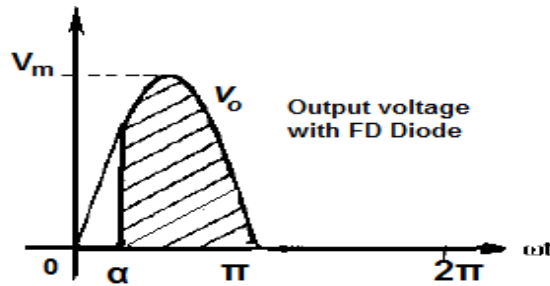


Fig.3.6 Output voltage waveform for $R-L$ load with freewheeling diode.

To derive an expression for the average value of the output voltage for single-phase half-wave controlled rectifier with $R-L$ load and freewheeling diode, referring to the voltage waveform shown in Fig.3.6 one can write,

$$\begin{aligned} V_{dc} &= \frac{1}{2\pi} \int_{\alpha}^{\pi} v_s(\omega t) d\omega t = \frac{1}{2\pi} \int_{\alpha}^{\pi} V_m \sin \omega t d\omega t \\ &= \frac{V_m}{2\pi} (-\cos \pi + \cos \alpha) = \frac{V_m}{2\pi} (1 + \cos \alpha) \end{aligned} \quad (3.16)$$

which is the same expression as that of resistive load case given in Eq.(3.1).

Example 3.1

The single-phase half-wave controlled rectifier shown in Fig.3.2(a) supplies a resistive load draws an average current of 1.62 A. If the converter is operated from a 240 V, 50 Hz supply and if the average value of the output voltage is 81V, calculate the following:

- The firing angle α .
- Load resistance .

- (c) The *rms* load voltage.
- (d) The *rms* load current.
- (e) DC power.
- (f) The ripple factor .

Solution

(a) For single-phase half-wave controlled rectifier with resistive load; the average value of the output voltage is calculated from Eq.(3.1) as

$$V_{dc} = \frac{V_m}{2\pi} (1 + \cos\alpha)$$

$$81 = \frac{240 \times \sqrt{2}}{2\pi} (1 + \cos\alpha) \quad \rightarrow \quad \text{Hence } \alpha = 60^\circ.$$

$$(b) \quad R = \frac{V_{dc}}{I_{dc}} = \frac{81}{1.62} = 50 \, \Omega$$

(c) The *rms* load voltage is calculated using Eq.(3.4) as

$$\begin{aligned} V_{orms} &= \frac{V_m}{2} \sqrt{\frac{1}{\pi} (\pi - \alpha + \frac{1}{2} \sin 2\alpha)} \\ &= \frac{240 \times \sqrt{2}}{2} \sqrt{\frac{1}{\pi} (\pi - \frac{\pi}{3} + \frac{1}{2} \sin 120^\circ)} = 152.14 \, \text{V} \end{aligned}$$

(d) The *rms* load current

$$I_{orms} = \frac{V_{orms}}{R} = \frac{152.14}{50} = 3.04 \, \text{A}$$

(e) The output d.c. power is given by:

$$P_{dc} = V_{dc} I_{dc} = \frac{V_{dc}^2}{R} = \frac{81^2}{50} = 131.22 \, \text{W}$$

$$(f) \quad \text{Ripple factor} = \frac{\sqrt{V_{orms}^2 - V_{dc}^2}}{V_{dc}} = \frac{\sqrt{152.14^2 - 81^2}}{81} = 1.57$$

Example 3.2

The thyristor shown in Fig.3.7 is triggered by RC phase-shift bridge circuit connected to its cathode and gate through 1:1 center tapped transformer. For the resistor R , the minimum value is $R_{min} = 1 \text{ k}\Omega$ and maximum value $R_{max} = 11 \text{ k}\Omega$. The value of the capacitor $C = 0.4 \mu\text{F}$. Assuming that the SCR fires at the instant when the gate / cathode voltage goes positive, determine the range of firing angles available.

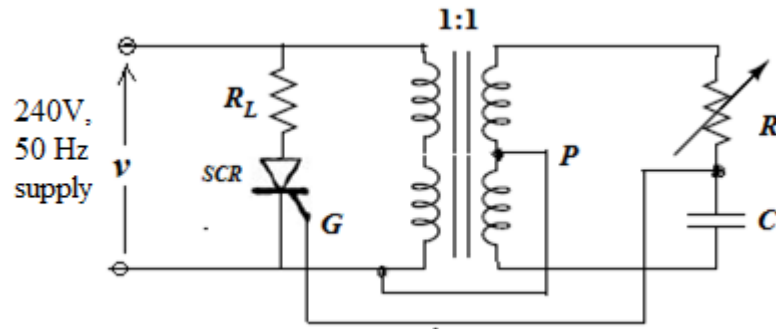


Fig.3.7.

Solution

(a) For $R = R_{min} = 1 \text{ k}\Omega$

Center-tap voltage:

$$V_p = \frac{240 \angle 0^\circ}{2} = 120 \angle 0^\circ \text{ Volts}$$

The impedance of the RC circuit Z_{min} is,

$$\begin{aligned} |Z_{min}| &= \sqrt{R_{min}^2 + \left(\frac{-1}{\omega C}\right)^2} = \sqrt{1000^2 + \left(\frac{-1}{100\pi \times 0.4 \times 10^{-6}}\right)^2} \\ &= \sqrt{10^6 + 7990^2} = 8024.33 \Omega \angle 0^\circ \end{aligned}$$

$$I_{max} = \frac{240 \angle 0^\circ}{|Z_{min}|} = \frac{240 \angle 0^\circ}{8024.33 \angle -83^\circ} = 0.02992 \text{ A} \angle 83^\circ$$

$$V_R = R_{min} I_{max} = 1000 \times 0.02992 = 29.92 \text{ V} \angle 83^\circ$$

Taking voltage V_p as a reference,

$$\begin{aligned} V_{GP} &= 120 \angle 0^\circ - 29.92 V \angle 83^\circ \\ &= 120 - 29.92 \cos 83^\circ - j29.92 \sin 83^\circ \\ &= 116.35 - j29.69 = 120 \angle 14.3^\circ V \quad \therefore \alpha_1 = 14.3^\circ \end{aligned}$$

(b) For $R = R_{max} = 11 \text{ k}\Omega$

$$\begin{aligned} |Z_{max}| &= \sqrt{R_{max}^2 + \left(\frac{-1}{\omega C}\right)^2} = \sqrt{11000^2 + \left(\frac{-1}{100\pi \times 0.4 \times 10^{-6}}\right)^2} \\ &= \sqrt{11^6 + 7990^2} \Omega \end{aligned}$$

$$V_R = I_{min} \cdot R_{max} = \frac{240 \angle 0^\circ}{Z_{max}} = 194 V \angle 36^\circ$$

$$\begin{aligned} V_{GP} &= 120 \angle 0^\circ - 194 V \angle 36^\circ \\ &= 120 - 194 \cos 36^\circ - j194 \sin 36^\circ \\ &= 123 \angle 108.2^\circ \quad \therefore \alpha_2 = 108.2^\circ \end{aligned}$$

Hence the range of angles: $14.3^\circ - 108.2^\circ$

Example 3.3

The circuit of Example 3.2 is used as a half-wave controlled rectifier supplying resistive load with $R_L = 1 \text{ k}\Omega$. For the firing angles obtained in the previous example draw the load current waveforms for both angles and, determine the corresponding power dissipation in the load resistor.

Solution

The load current waveform is shown in Fig.3.8 (a) when $\alpha = 14.3^\circ = 0.249 \text{ rad}$.

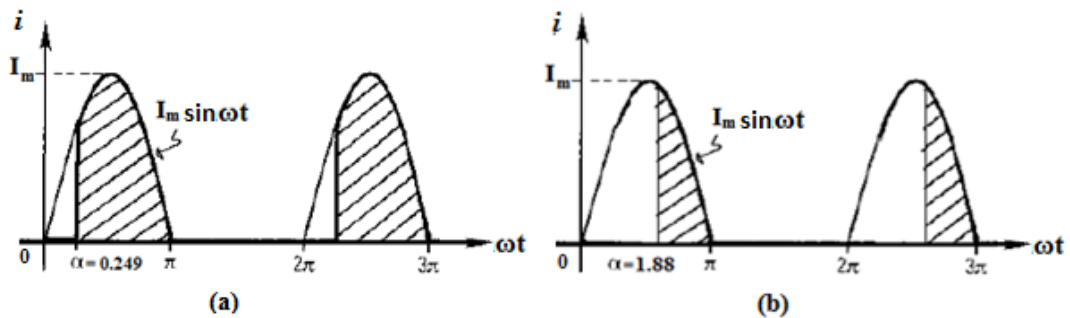


Fig.3.8.

The *rms* value of the load current may be given by,

$$I_{orms}^2 = \frac{V_m^2}{2\pi R_L} \int_{0.249}^{\pi} \sin^2 \omega t d\omega t$$

$$I_{orms}^2 = \frac{V_m^2}{2\pi R_L} \int_{0.249}^{\pi} \frac{1}{2} (1 - \cos 2\omega t) d\omega t$$

$$= \frac{V_m^2}{2\pi R_L} \times \frac{1}{2} (1 - \sin 2\omega t) \Big|_{0.249}^{\pi} = 0.02863 \text{ A}$$

$$P = I_{orms}^2 \cdot R_L = I_{orms}^2 \times 1000 = 28.63 \text{ W}$$

Similarly for $\alpha_2 = 108.2^\circ$ (Fig.3.8(b)) , $P = 8.64 \text{ W}$

Example 3.4

A single-phase half-wave controlled rectifier shown in Fig.3.2 supplied from 230V a.c. supply is operating at $\alpha = 60^\circ$. If the load resistor is 10Ω , determine:

- The power absorbed by the load (P_{dc}).
- The power drawn from the supply (P_{ac}).
- The power factor at the a.c. source.

Solution

(a) The d.c. power absorbed by the load (P_{dc}) :

$$P_{dc} = V_{dc} I_{dc} = \frac{V_{dc}^2}{R}$$

$$V_{dc} = \frac{V_m}{2\pi} (1 + \cos \alpha) = \frac{\sqrt{2} \times 230}{2\pi} (1 + \cos 60^\circ) = 77.68 \text{ V}$$

$$P_{dc} = \frac{77.68^2}{10} = 603.4 \text{ W}$$

(b) The power drawn from the supply P_{ac} :

$$P_{ac} = V_{orms} I_{orms} = \frac{V_{orms}^2}{R} = \frac{(0.5V_m)^2}{R} = \frac{1}{4} \times \frac{(\sqrt{2} \times 230)^2}{10}$$

$$= 2644.2 \text{ W}$$

(c) The power factor at the a.c. source.

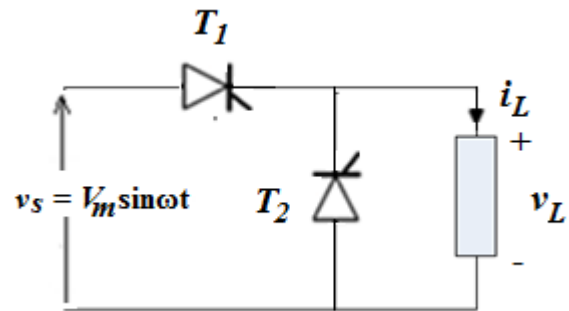
The power factor can be calculated from Eq.(3.7) as

$$PF = \sqrt{\frac{1}{2\pi} (\pi - \alpha + \frac{1}{2} \sin 2\alpha)} = \sqrt{\frac{1}{2\pi} (\pi - \frac{\pi}{3} + \frac{1}{2} \sin 120^\circ)} \\ = 0.633 \text{ lagging}$$

Example 3.5

For the single-phase half-wave controlled rectifier shown in Fig.3.9, thyristor T_1 is operating at $\alpha_1 = 80^\circ$. Thyristor T_2 is connected across the load and operating with a delay angle α_2 of 40° . Assume the load is highly inductive such that I_L is continuous. Plot waveforms for the instantaneous values of v_L , i_{T1} , i_{T2} , i_L , v_{T1} and v_{T2} . Derive an expression for the average load voltage V_{dc} as a function of α_1 and α_2 (with $\alpha_1 < \alpha_2$).

Fig.3.9.



Solution

The voltage and current waveforms are shown in Fig.3.10.

The average d.c. voltage may be obtained as

$$V_{dc} = \frac{1}{2\pi} \int_{\alpha_1}^{\alpha_2} V_m \sin \omega t \, d\omega t = \frac{V_m}{2\pi} \int_{\alpha_1}^{\alpha_2} \sin \omega t \, d\omega t \\ = \frac{V_m}{2\pi} \left\{ -\cos \omega t \right\}_{\alpha_1}^{\alpha_2} = \frac{V_m}{2\pi} \{ \cos \alpha_1 - \cos \alpha_2 \}$$

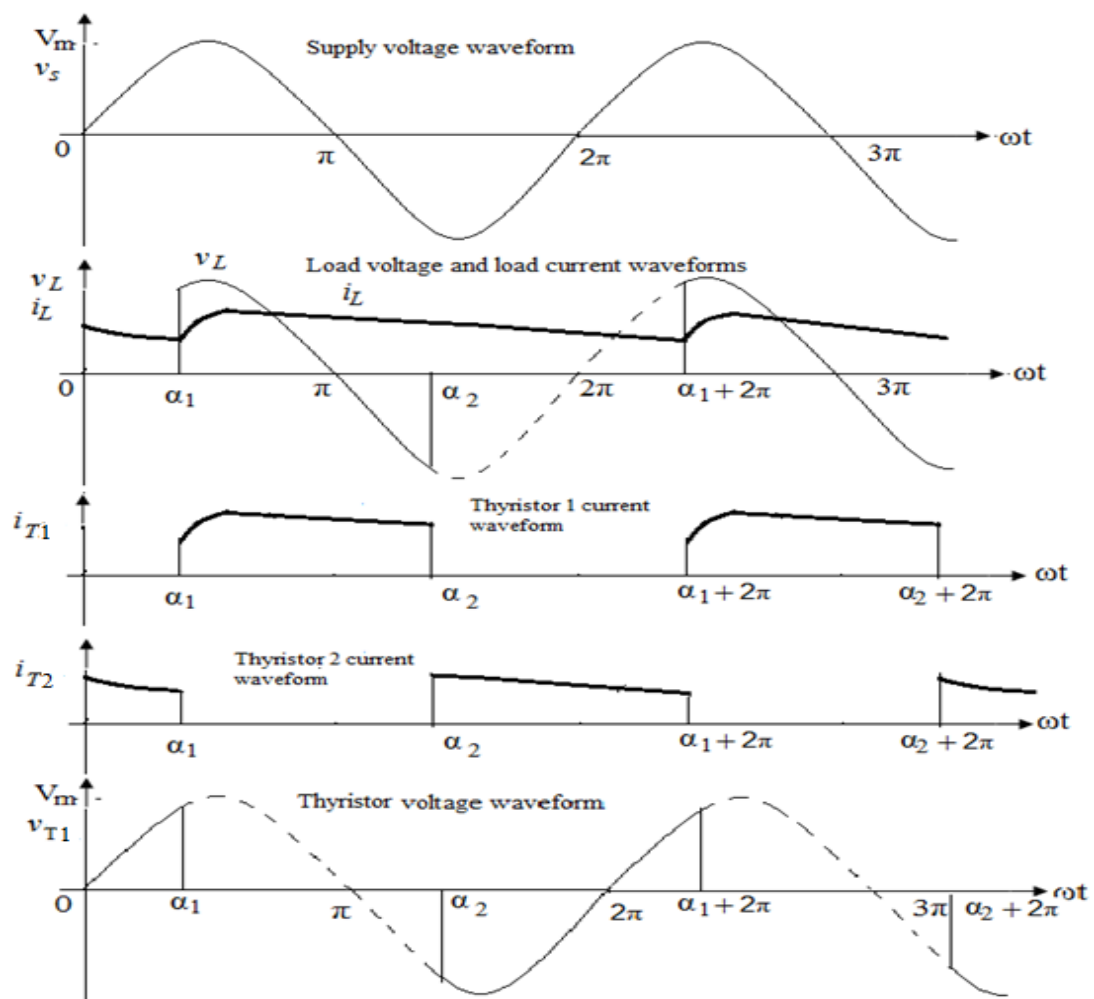
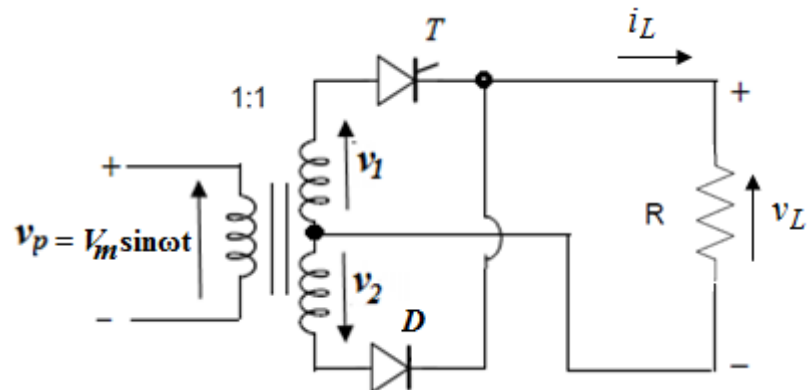


Fig.3.10 The voltage and current waveforms of Example 3.5.

Example 3.6

The single-phase half-controlled rectifier shown in Fig.3.11 operating at a triggering angle of 60° from a.c. source $v_s = 300 \sin \omega t$. Assuming the load is resistive, express V_{dc} of the load as a function of α , and calculate its value.

Fig.3.11



Solution

$$\begin{aligned}
 V_{dc} &= \frac{1}{2\pi} \left\{ \int_{\alpha}^{\pi} V_m \sin \omega t \, d\omega t + \int_{\pi}^{2\pi} -V_m \sin \omega t \, d\omega t \right\} \\
 &= \frac{V_m}{2\pi} \left[-\cos \omega t \Big|_{\alpha}^{\pi} \right] - \left\{ -\cos \omega t \Big|_{\pi}^{2\pi} \right\} \\
 &= \frac{V_m}{2\pi} \{ 1 + \cos \alpha + 1 + 1 \} = \frac{V_m}{2\pi} \{ 3 + \cos \alpha \}
 \end{aligned}$$

$$\text{For } \alpha = 60^\circ : V_{dc} = \frac{V_m}{2\pi} \{ 3 + \cos 60^\circ \} = 167.2 \text{ V}$$

3.3 SINGLE-PHASE, FULL-WAVE, FULLY-CONTROLLED BRIDGE RECTIFIER (P = 2)

The single-phase, fully-controlled full-wave rectifier bridge is shown in Fig.3.12(a). In this circuit, two thyristors must be triggered simultaneously to permit current to flow. For example, with the instantaneous polarity of the sinusoidal supply voltage indicated in Fig.3.12 (b), T_1 and T_2 must be triggered, while in reverse, T_3 and T_4 must triggered at the same time. The output voltage and current waveforms are shown in Fig.3.12 (b) for the case of resistive load.

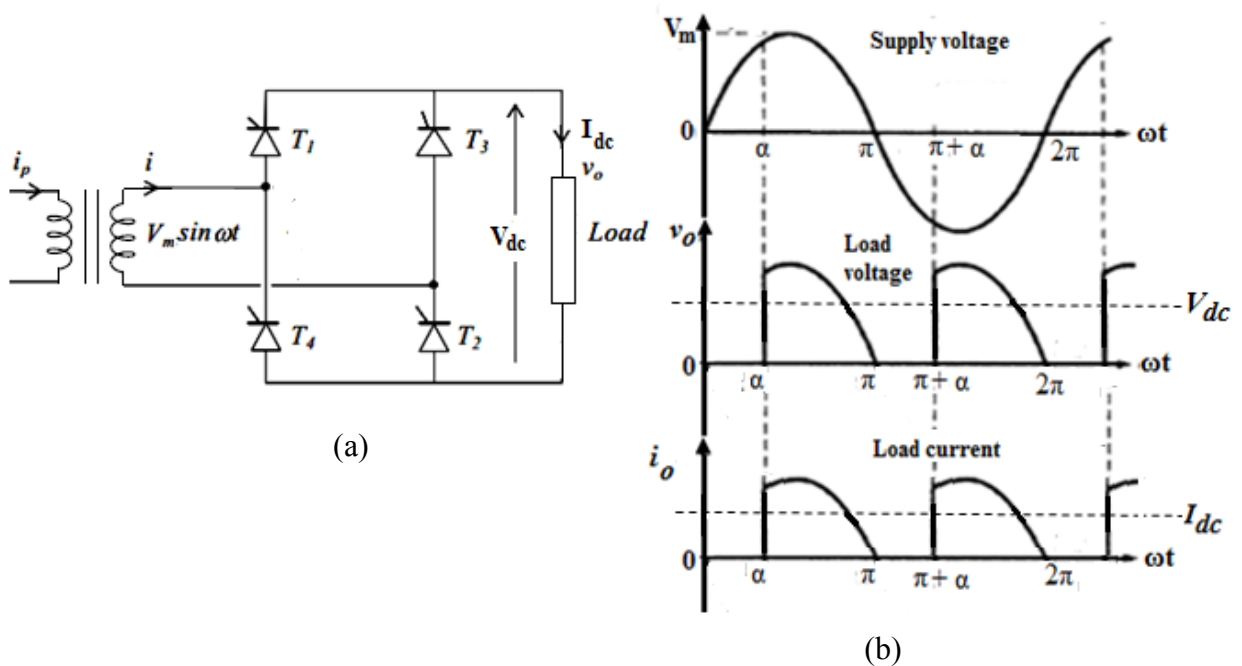


Fig. 3.12 Single-phase fully-controlled full-wave rectifier bridge with R -load, (a) Circuit, (b) Waveforms.

The single-phase supply to the rectifier is usually taken from the secondary of a transformer and not directly from the main and is generally expressed as

$$v_s(\omega t) = V_m \sin \omega t$$

In the following analysis, the rectifier will be considered to operate with three types of passive loads, namely: pure resistive load, R - L load with small L/R ratio (in which the resistance is dominant, R - L load with large L/R ratio (in which ωL is larger than R) and R - L load which is highly inductive ($L \gg R$).

3.3.1 Operation of the Converter with Resistive Load

Referring to Fig.3.12(b), the average d.c. output voltage of the converter with resistive load is given by,

$$V_{dc} = \frac{1}{\pi} \int_{\alpha}^{\pi} V_m \sin \omega t d\omega t = \frac{V_m}{\pi} (1 + \cos \alpha) \quad (3.17)$$

The *rms* value of the load voltage V_{orms} can be calculated as follows:

$$\begin{aligned} V_{orms} &= \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi} v_s^2(\omega t) d\omega t} \\ &= \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi} (V_m \sin \omega t)^2 d\omega t} = \sqrt{\frac{(V_m)^2}{\pi} \int_{\alpha}^{\pi} \frac{1}{2} (1 - \cos 2\omega t) d\omega t} \\ &= \frac{V_m}{\sqrt{2}} \sqrt{\frac{1}{\pi} (\pi - \alpha + \frac{1}{2} \sin 2\alpha)} \end{aligned} \quad (3.18)$$

Therefore, the *rms* value of the load current I_{orms} is

$$I_{orms} = \frac{V_{orms}}{R} = \frac{V_m}{\sqrt{2}R} \sqrt{\frac{1}{\pi} (\pi - \alpha + \frac{1}{2} \sin 2\alpha)} \quad (3.19)$$

3.3.2 Operation of the Converter with R - L load

(i) Case of R - L load with small L/R ratio: Discontinuous load current

In this case the current will be discontinuous as shown in Fig.3.13. For $\alpha \leq \omega t \leq \beta$, the circuit equation is given by

$$V_{dc} = \frac{1}{\pi} \int_{\alpha}^{\beta} V_m \sin \omega t d\omega t = \frac{V_m}{\pi} (\cos \alpha - \cos \beta) \quad (3.20)$$

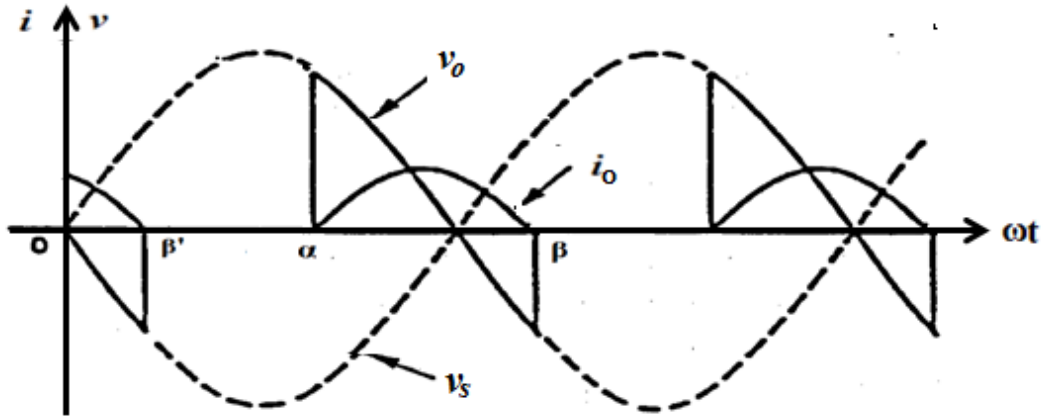


Fig.3.13 Waveforms of the output voltage and current for discontinuous current operation.

(ii) Case of R - L load with large L/R ratio: Continuous load current

Under these conditions, a thyristor is still conducting when another is still forward-biased and is turned on. The first device is instantaneously reverse-biased by the second device which has been turned on. The first device is commutated and load current is instantaneously transferred on the incoming device. In this case the current is continuous as shown in Fig.3.14.

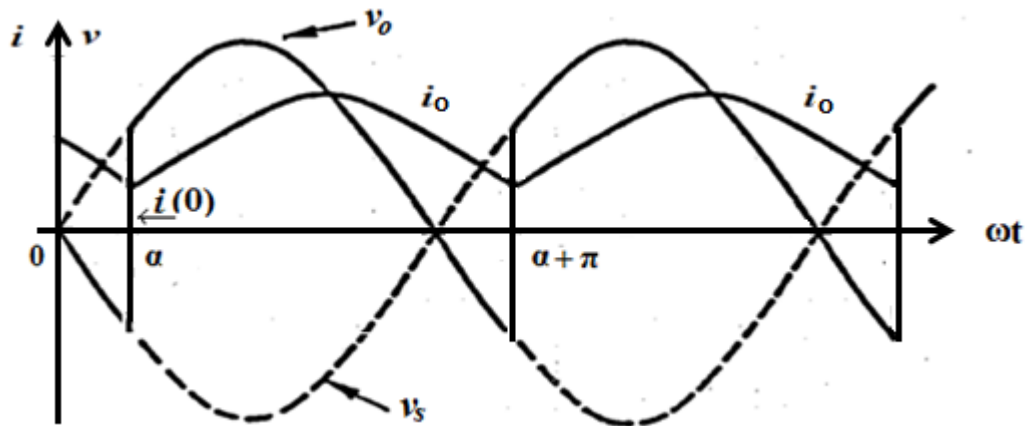


Fig. 3.14 Waveforms of the output voltage and current for continuous current operation.

The average value of the load voltage V_{dc} can be calculated as follows:

$$V_{dc} = \frac{1}{\pi} \int_{\alpha}^{\alpha+\pi} V_m \sin \omega t d\omega t = \frac{2V_m}{\pi} \cos \alpha \quad (3.21)$$

Under all delay angle condition, the average current is given by:

$$I_{dc} = \frac{V_{dc}}{R} \quad (3.22)$$

From the above voltage equation, (3.21), it is clear that, if the firing angle is greater than 90° , the average voltage can be negative because $\cos \alpha$ is negative. Thus if the firing angle is suddenly increased to, say, 160° , a large negative voltage will be applied to the load and the power is fed back to the supply. This process is known as 'INVERSION'. The current remains positive while the voltage is negative. Hence a negative pulse of energy will be produced which is fed back to the supply. In this case the converter is said to operate in two quadrants as shown in Fig.3.15(a). The graph shown in Fig. 3.15(b) gives the relation between the firing angle and the output voltage in percentage for the two modes of operation (continuous and discontinuous) for full-wave single-phase rectifier.

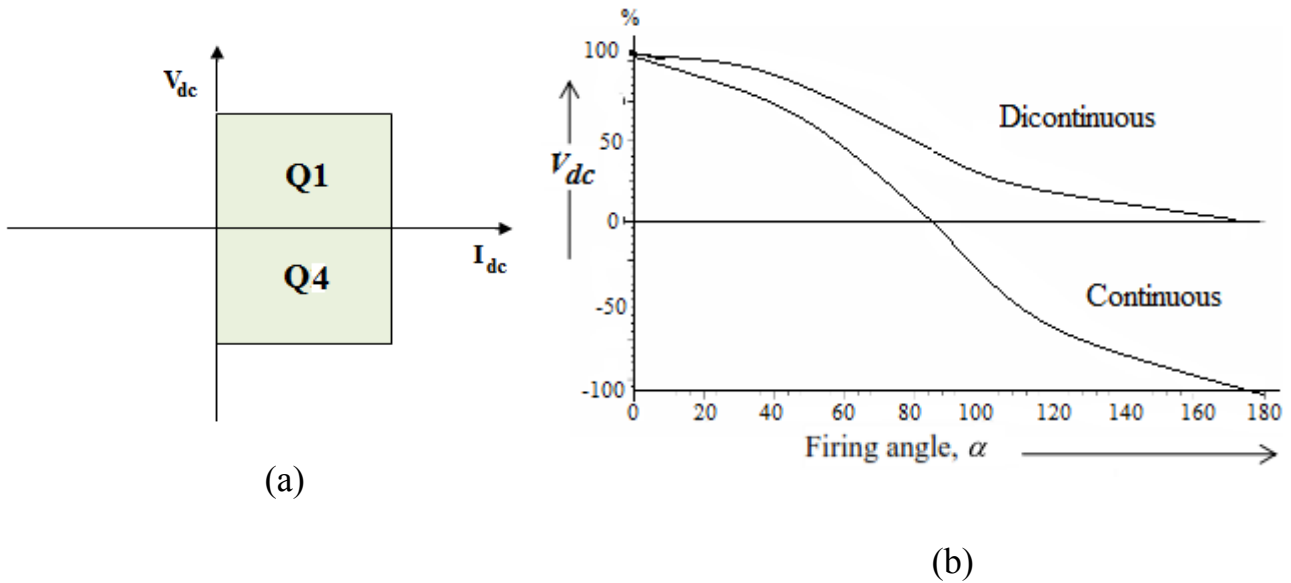


Fig. 3.15 Full-wave fully-controlled single-phase rectifier: (a) Two quadrants operation, (b) The relation between the firing angle and the output voltage in p.u. for the two modes of operation, continuous and discontinuous current.

To find roughly whether the converter is operating in continuous or discontinuous current mode, the following simple rule-of-thumb may be used:

If $\pi + \alpha < \beta$ the current is continuous
 If $\pi + \alpha > \beta$ the current is discontinuous

(iii) Case of highly inductive load ($L \gg R$)

Fig. 3.16 (a) shows the circuit connection for a single-phase, full-wave, controlled rectifier loaded with a highly inductive load. For one total period of operation of this circuit, the corresponding waveforms are shown in Fig. 3.16 (b).

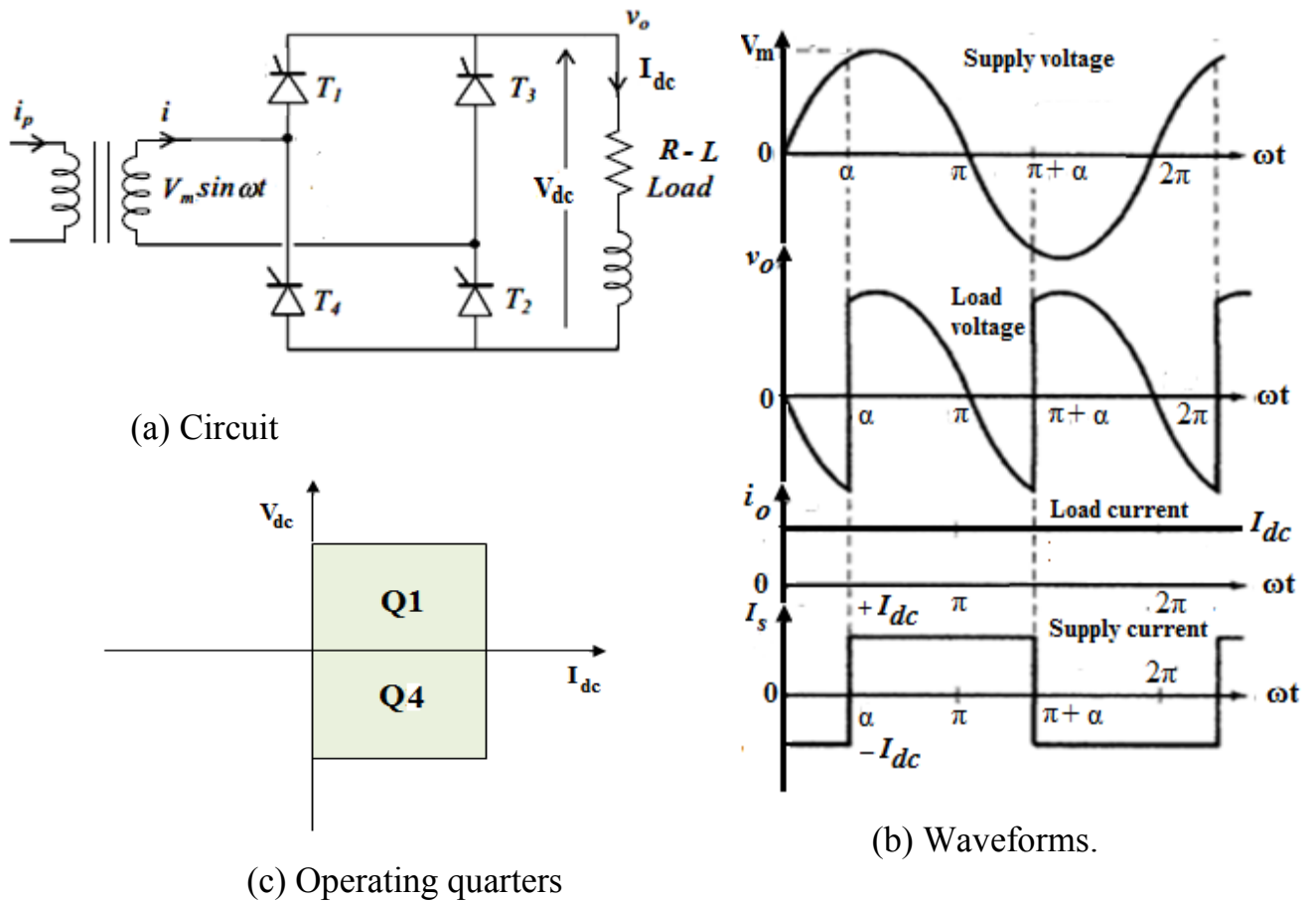


Fig.3.16 Single phase full-wave rectifier loaded with highly inductive load.

The average value of the load voltage V_{dc} can be calculated as follows,

$$V_{dc} = \frac{1}{\pi} \int_{\alpha}^{\alpha+\pi} V_m \sin \omega t d\omega t = \frac{2V_m}{\pi} \cos \alpha \quad (3.23)$$

Which is the same as Eq.(3.21) for continuous current operation mode. Therefore, the average output voltage can vary from $+2V_m/\pi$ to $-2V_m/\pi$ when varying α from π to 0 , respectively. Moreover, since the load voltage for this configuration can be positive or negative while the load current is always positive because the thyristors prevents a reverse current flow. Therefore, this converter operates in the first and the fourth quadrants as shown in Fig.3.16 (c).

The *rms* value of the load voltage V_{orms} can be calculated as follows,

$$\begin{aligned} V_{orms} &= \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi+\alpha} v_s^2(\omega t) d\omega t} = \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi+\alpha} (V_m \sin \omega t)^2 d\omega t} \\ &= \sqrt{\frac{V_m^2}{\pi} \int_{\alpha}^{\pi+\alpha} \frac{1}{2} (1 - \cos 2\omega t) d\omega t} = \frac{V_m}{\sqrt{2}} \end{aligned} \quad (3.24)$$

Since the load current is constant over the studied period, therefore the *rms* value of the load current I_{orms} is : $I_{orms} = I_{dc} = I_a$

The *PRV* for any thyristor in this configuration is (V_m).

(iv) Case of resistive-inductive load in series with d.c. source (**R-L-E**)

Fig.3.17(a) shows the circuit diagram for a single-phase, full-wave, controlled rectifier supplying *RLE* load. When the current is continuous, the output voltage and current waveforms are similar to the case of inductive load with large L/R ratio, Fig.3.14. This means that E has no effect on the waveforms as long as the current remains continuous. Hence the equations for the average and *rms* values of the output voltages are same as that given in Eqs.(3.21) and (3.24).

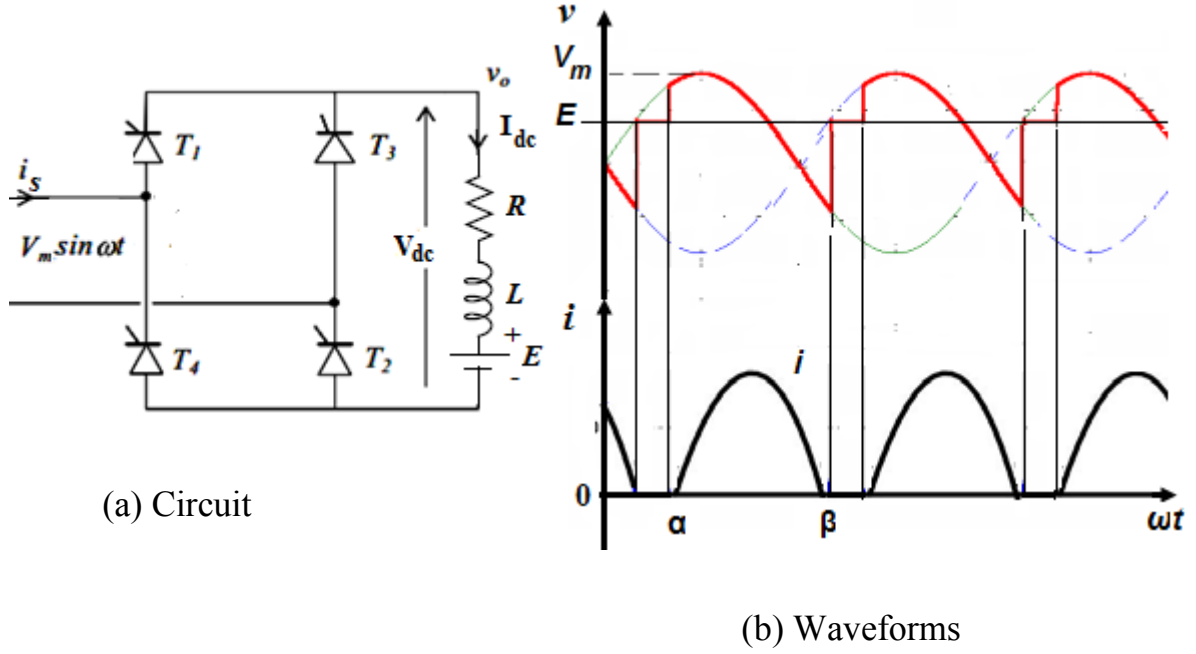
To find the average load current for continuous case, referring to Fig.3.14, by applying KVL, for the interval $\alpha \leq \omega t \leq \alpha + \pi$ which is repeated every π :

$$v_s(\omega t) = V_m \sin \omega t = R i + L \frac{di}{dt} + E \quad (3.25)$$

This is a first order differential equation. The solution of this equation has three parts:

1- Steady-state solution due to source voltage:

$$i_{ss1}(\omega t) = \frac{V_m}{Z} \sin(\omega t - \theta)$$

Fig.3.17 Single-phase full-wave rectifier operating with RLE load.

2- Steady-state solution due to d.c. voltage E : $i_{ss2}(\omega t) = -\frac{E}{R}$

3- Transient solution: $i_{tr}(\omega t) = Ae^{-\frac{t}{\tau}}$
where

$$|Z| = \sqrt{R^2 + (\omega L)^2}$$

$$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

$$\tau = \frac{L}{R} = \text{time constant} \quad \text{and} \quad A = \text{Constant}$$

The complete solution is: $i(\omega t) = i_{ss1}(\omega t) + i_{ss2}(\omega t) + i_{tr}(\omega t)$

$$i(\omega t) = \frac{V_m}{|Z|} \sin(\omega t - \theta) - \frac{E}{R} + Ae^{-\frac{t}{\tau}} \quad (3.26)$$

For continuous current case, the constant A can be found from initial conditions: $i(\omega t = \alpha) = i(0)$,

$$i(\alpha) = i(0) = \frac{V_m}{|Z|} \sin(\alpha - \theta) - \frac{E}{R} + Ae^{-\frac{R}{\omega L}\alpha}$$

$$\therefore A = \left[i(0) - \frac{V_m}{|Z|} \sin(\alpha - \theta) + \frac{E}{R} \right] e^{-\frac{R}{\omega L}\alpha} \quad (3.27)$$

Substitute Eq.(3.27) into Eq.(3.26) and simplify yields

$$i(\omega t) = \frac{V_m}{|Z|} \sin(\omega t - \theta) - \frac{E}{R} + [i(0) - \frac{V_m}{|Z|} \sin(\alpha - \theta) + \frac{E}{R}] e^{\frac{-R}{\omega L}(\omega t - \alpha)} \quad (3.28)$$

Note that, $i(0)$ is the initial value of the load current at $\omega t = \alpha$, $\alpha + \pi$, $2\pi + \alpha, \dots$. This can be obtained by substituting this value into Eq.(3.28) and solving for $i(0)$ we get,

$$i(0) = -\frac{V_m}{|Z|} \sin(\alpha - \theta) \left[\frac{1 + e^{-\frac{\pi R}{\omega L}}}{1 - e^{-\frac{\pi R}{\omega L}}} \right] - \frac{E}{R} \quad (3.29)$$

Once $i(0)$ in Eq.(3.29) is found, the average value of $i(\omega t)$ can be found as

$$\begin{aligned} I_{av} &= \frac{1}{\pi} \int_{\alpha}^{\alpha+\pi} i(\omega t) d\omega t \\ I_{av} &= \frac{1}{\pi} \int_{\alpha}^{\alpha+\pi} \left\{ \frac{V_m}{|Z|} \sin(\omega t - \theta) - \frac{E}{R} + \left[i(0) + \frac{E}{R} - \frac{V_m}{|Z|} \sin(\alpha - \theta) \right] e^{\cot \theta (\alpha - \omega t)} \right\} d\omega t \quad (3.30) \end{aligned}$$

For discontinuous current case, one can find the average value of the current I_{av} by referring to Fig.3.17(b) for the interval $\alpha \leq \omega t \leq \beta$ as

$$\begin{aligned} I_{av} &= \frac{1}{\pi} \int_{\alpha}^{\beta} \left\{ \frac{V_m}{|Z|} \sin(\omega t - \theta) - \frac{E}{R} + \left[i(0) + \frac{E}{R} - \frac{V_m}{|Z|} \sin(\alpha - \theta) \right] e^{\cot \theta (\alpha - \omega t)} \right\} d\omega t \quad (3.31) \end{aligned}$$

where

$$\alpha = \sin^{-1} \frac{E}{V_m}, \quad \beta = \pi - \sin^{-1} \frac{E}{V_m} \quad \text{and} \quad i(0) = 0$$

Equation (3.31) above can also be applied to single-phase half-wave controlled rectifier of Fig.3.1 operating with RLE -load by modifying it as:

$$I_{av} = \frac{1}{2\pi} \int_{\alpha}^{\beta} \left\{ \frac{V_m}{Z} \sin(\omega t - \theta) - \frac{E}{R} + \left[\frac{E}{R} - \frac{V_m}{Z} \sin(\alpha - \theta) \right] e^{\cot\theta(\alpha - \omega t)} \right\} d\omega t \quad (3.32)$$

3.4 SINGLE-PHASE HALF-CONTROLLED (SEMICONVERTER) RECTIFIER

Fig.3.18(a) shows a single-phase half-controlled (semiconverter) rectifier. This configuration consists of a combination of thyristors and diodes to reduce the cost of the full-wave fully-controlled rectifier. With resistive load, the converter operates in the same manner as fully controlled one. The load current and voltage have the same form as presented in Subsection 3.3.1 also apply to this circuit.

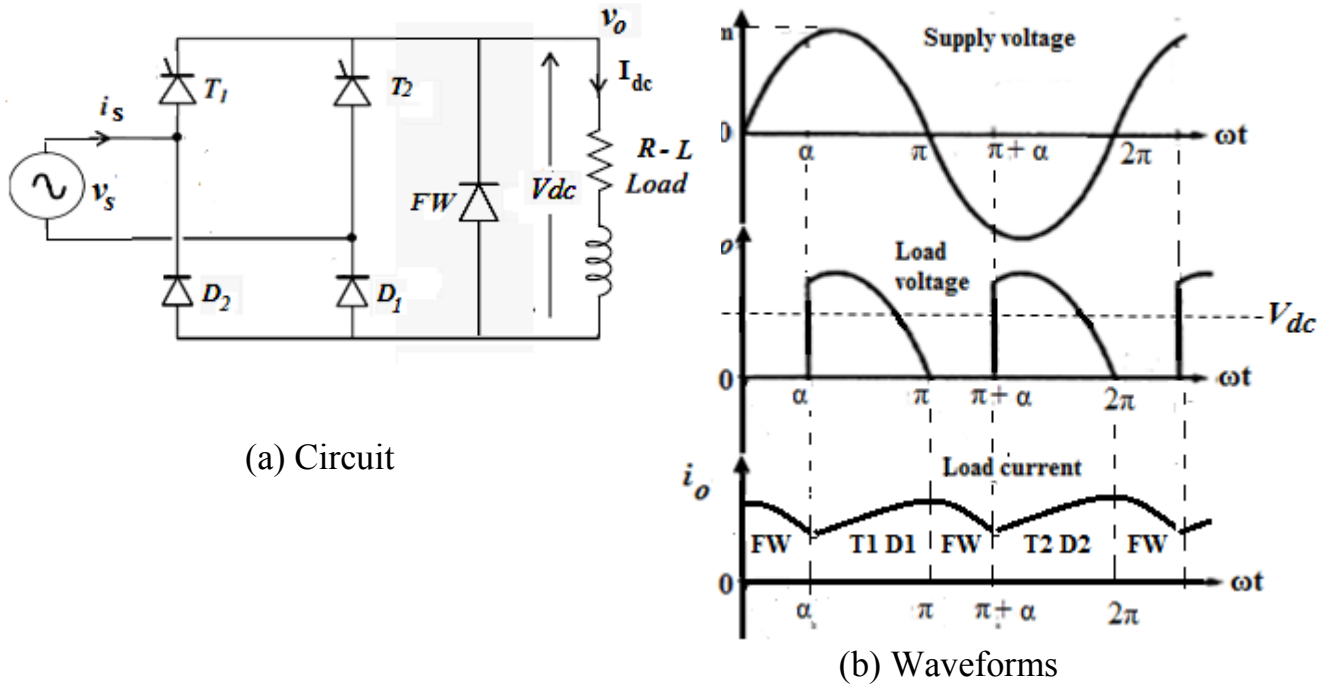


Fig.3.18 Single-phase, half-controlled (Semiconverter): (a) Circuit, (b) Waveforms.

Equations (3.17) to (3.19) for the voltage and current are also applied for the semiconverter with resistive load. However with resistive-inductive load the converter can operate in continuous or discontinuous modes depend on the L/R ratio as discussed in Subsection 3.3.2.

In continuous current mode, freewheeling diode must be used to eliminate any negative voltage occurrence at the load terminals as shown in Fig.3.18 (a). This is because the diode FWD is always activated (forward biased) whenever the load voltage tends to be negative. For one total period of operation of this circuit, the corresponding waveforms are

shown in Fig. 3.18 (b). The average value of the load voltage V_{dc} in case of R - L load can be calculated as follows,

$$V_{dc} = \frac{1}{\pi} \int_{\alpha}^{\pi} V_m \sin \omega t d\omega t = \frac{V_m}{\pi} (1 + \cos \alpha) \quad (3.33)$$

which is the same expression for the case of pure resistive load. Therefore, the average output voltage can vary from 0 to V_m / π when varying α from π to 0 respectively. The average value of the load current I_{dc} is calculated as :

$$I_{dc} = \frac{V_{dc}}{R}$$

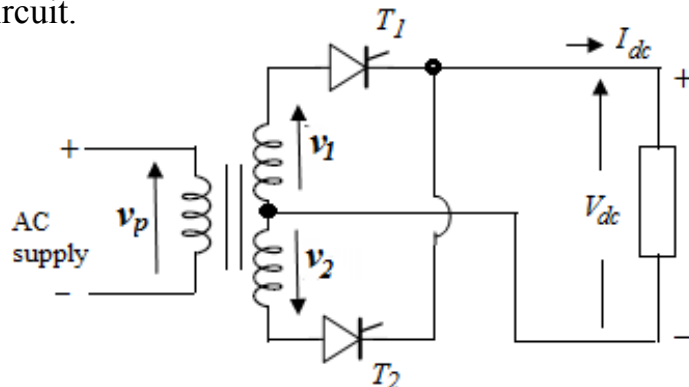
The *rms* value of the load voltage V_{orms} can be calculated as follows:

$$\begin{aligned} V_{orms} &= \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi} v_s^2(\omega t) d\omega t} = \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi} (V_m \sin \omega t)^2 d\omega t} \\ &= \sqrt{\frac{(V_m)^2}{\pi} \int_{\alpha}^{\pi} \frac{1}{2} (1 - \cos 2\omega t) d\omega t} \\ &= \frac{V_m}{\sqrt{2}} \sqrt{\frac{1}{\pi} (\pi - \alpha + \frac{1}{2} \sin 2\alpha)} \end{aligned} \quad (3.34)$$

3.5 BI-PHASE (MID-POINT) CONTROLLED RECTIFIER

Fig.3.19 shows the circuit diagram of a bi-phase (mid-point) controlled rectifier. This circuit acts exactly as the single-phase full-wave rectifier discussed in the previous section. The only difference is that it uses two thyristors T_1 and T_2 only which is considered as an advantage over the bridge type rectifier. However, the voltage rating of each thyristor must be twice as that of the bridge type, i.e. $PRV = 2V_m$ on the secondary side whereas it is only V_m on the bridge type. The major disadvantage of the bi-phase converter is that it needs large center-tapped transformer which adds cost to the circuit.

Fig.3.19



The operation of the converter can be illustrated as follows:

During positive half cycle of v_1 , T_1 is forward biased while T_2 is reversed biased since v_2 is now negative. T_1 conducts from its firing angle α till π and positive voltage appears across the load. From π to 2π , v_2 becomes positive T_1 is reversed biased whereas T_2 is forward biased and positive voltage also appears on the load same as the full-wave bridge rectifier output. Therefore, this converter is considered as an alternative to the bridge converter for producing full-wave rectification. Hence all mathematical analysis and waveforms for the output voltage, current, power and power factor are the same for that of the bridge type converter, so it will not be repeated here.

Example 3.7

A fully-controlled single-phase bridge rectifier is supplied from a 50Hz, 230/100V transformer. The rectifier supplying a highly inductive load of 10Ω resistor. For a firing angle of 45° , determine the rectified voltage, the rectified current and the power factor.

Solution

For highly inductive load, referring to Fig. 3.16(b), the average output voltage V_{dc} is calculated as follows,

$$V_{dc} = \frac{1}{\pi} \int_{\alpha}^{\alpha+\pi} V_m \sin \omega t \, d\omega t = \frac{V_m}{\pi} \left\{ -\cos \omega t \right\}_{\alpha}^{\alpha+\pi} = \frac{2V_m}{\pi} \cos \alpha$$

$$V_m = 100 \times \sqrt{2} = 141.42 \text{ V} \quad \text{since } V_m = \sqrt{2} V_{rms}$$

$$\therefore V_{dc} = \frac{2 \times 141.24}{\pi} \cos 45^\circ = 63.64 \text{ V}$$

$$I_{dc} = \frac{V_{dc}}{R} = \frac{63.64}{10} = 6.364 \text{ A}$$

$$\begin{aligned} PF &= \frac{\text{Average Power}}{V_{orms} \times I_{orms}} = \frac{\left(\frac{2V_m}{\pi} \cos \alpha\right) \times I_{orms}}{V_{orms} \times I_{orms}} \\ &= \frac{2\sqrt{2}}{\pi} \cos \alpha = \frac{2\sqrt{2}}{\pi} \cos 45^\circ = 0.636 \end{aligned}$$

Example 3.8

A single-phase, fully-controlled bridge rectifier supplies a constant current to a highly inductive load. Derive an expression for the average d.c. output voltage from the bridge and also the *rms* value of the fundamental component of the alternating line current assuming (a) that there is no freewheel diode, and (b) that a freewheel diode is used.

Solution:

(a) No freewheeling diode:

The load voltage and current waveforms will be as shown in Fig.3.20 (a) and (b) respectively.

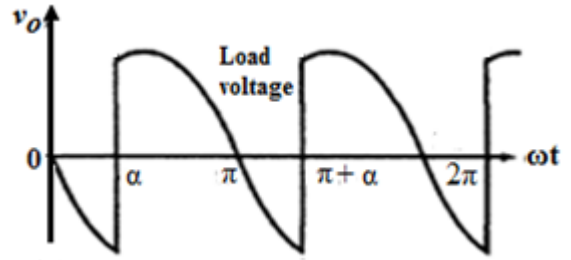
$$V_{dc} = \frac{1}{\pi} \int_{\alpha}^{\alpha+\pi} V_m \sin \omega t \, d\omega t$$

$$= \frac{2V_m}{\pi} \cos \alpha$$

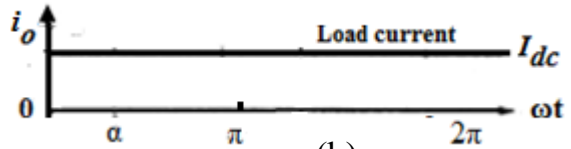
Current seen by the input transformer is as depicted in Fig.3.20 (c) ,

$$I_s = I_{dc} \quad 0 < \omega t < \pi$$

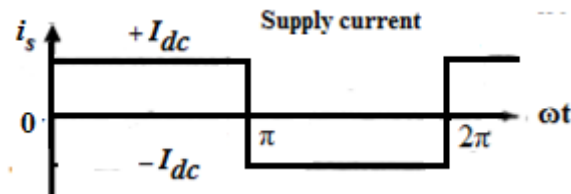
$$I_s = -I_{dc} \quad \pi < \omega t < 2\pi$$



(a)



(b)



(c)

Fig.3.20 Waveforms.

Using Fourier series analysis, the Fourier coefficients are,

$$a_1 = \frac{2}{\pi} \int_0^{\pi} I_{dc} \cos \omega t \, d\omega t = \frac{2I_{dc}}{\pi} \left\{ \sin \omega t \right\}_0^{\pi} = 0$$

$$b_1 = \frac{2}{\pi} \int_0^{\pi} I_{dc} \sin \omega t \, d\omega t - \frac{2I_{dc}}{\pi} \int_{\pi}^{2\pi} \sin \omega t \, d\omega t = \frac{4I_{dc}}{\pi} \left\{ -\cos \omega t \right\}_0^{\pi} = \frac{4I_{dc}}{\pi}$$

The amplitude c_1 of the fundamental component is given by

$$c_1 = \sqrt{a_1^2 + b_1^2} = \frac{4I_{dc}}{\pi}$$

The *rms* value of the fundamental component is

$$V_{o1} = \frac{c_1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \times \frac{4I_{dc}}{\pi} = \frac{2\sqrt{2}}{\pi} I_{dc}$$

(b) With freewheeling diode : In this case the output current will be discontinuous as depicted in Fig.3.21 (b),

$$V_{dc} = \frac{1}{\pi} \int_{\alpha}^{\pi} V_m \sin \omega t \, d\omega t$$

$$V_{dc} = \frac{V_m}{\pi} \left\{ -\cos \omega t \right\}_{\alpha}^{\pi}$$

$$= \frac{V_m}{\pi} (1 + \cos \alpha)$$

$$a_1 = \frac{2}{\pi} \int_{\alpha}^{\pi} I_{dc} \cos \omega t \, d\omega t$$

$$= \frac{2 I_{dc}}{\pi} \left\{ \sin \omega t \right\}_{\alpha}^{\pi}$$

$$= \frac{2 I_{dc}}{\pi} \sin \alpha$$

$$b_1 = \frac{2}{\pi} \int_{\alpha}^{\pi} I_{dc} \sin \omega t \, d\omega t$$

$$= \frac{2 I_{dc}}{\pi} \left\{ -\cos \omega t \right\}_{\alpha}^{\pi} = \frac{2 I_{dc}}{\pi} (1 + \cos \alpha)$$

Hence, $V_{o1} = \frac{1}{\sqrt{2}} \sqrt{a_1^2 + b_1^2} = \frac{2\sqrt{2}}{\pi} I_{dc} \cos \frac{\alpha}{2}$

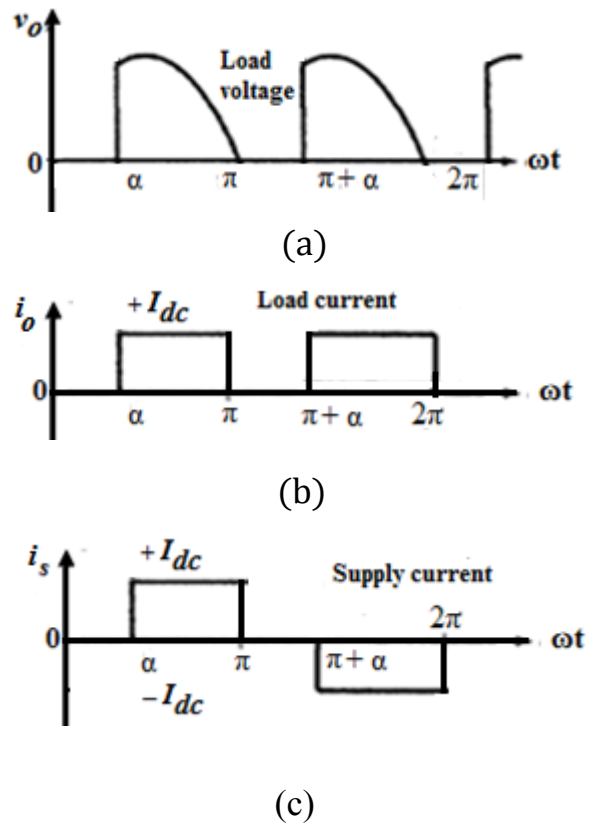


Fig. 3.21.

Example 3.9

For a single-phase, fully-controlled, bridge rectifier derive an expression for the a.c. power dissipated in a resistive load as the triggering angle is varied. If one thyristor remains in the non-conducting state, calculate the firing angle which results in 25% of maximum power dissipation in the load.

Solution

The power dissipated in resistive load is given by,

$$P = I_{orms}^2 \times R$$

$$I_{orms} = \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi} \left(\frac{V_m}{R} \sin \omega t \right)^2 d\omega t}$$

$$I_{orms}^2 \times R = \frac{V_m^2}{2\pi R} \left(\pi - \alpha + \frac{1}{2} \sin 2\alpha \right)$$

$$P_{max} = \frac{dP}{d\alpha} = \frac{V_m^2}{2\pi R} \frac{d \left(\pi - \alpha + \frac{1}{2} \sin 2\alpha \right)}{d\alpha} = 0$$

$$\therefore P_{max} = \frac{V_m^2}{2R}$$

If one thyristor is open circuit

$$P_o = \frac{1}{2} P = \frac{1}{2} \frac{V_m^2}{2\pi R} \left(\pi - \alpha + \frac{1}{2} \sin 2\alpha \right) = 0.25 P_{max}$$

$$0.25 \frac{V_m^2}{2R} = \frac{1}{2} \frac{V_m^2}{2\pi R} \left(\pi - \alpha + \frac{1}{2} \sin 2\alpha \right) = \alpha + \frac{1}{2} \sin 2\alpha = \frac{\pi}{2}$$

$$\text{Since : } \frac{1}{2} \sin 2\alpha \text{ is very small, } \therefore \alpha = \frac{\pi}{2}.$$

Example 3.10

A single-phase fully controlled, full-wave, bridge rectifier has a source of 230 V *rms* at 50 Hz, and is feeding a load $R = 25 \Omega$ and $L = 10 \text{ mH}$. The

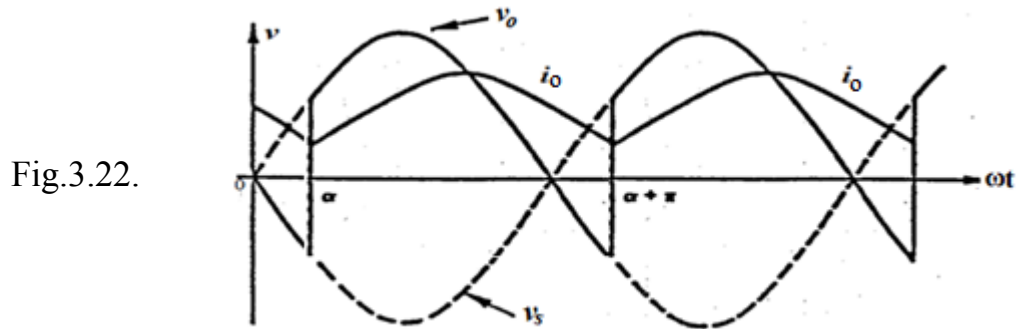
firing angle $\alpha = 45^\circ$ and the current extinction angle $\beta = 230^\circ$. It is required to:

- Sketch the appropriate load voltage and load current waveforms.
- Determine whether the current is continuous or discontinuous.
- Determine the average load voltage and current.
- Determine the *rms* load voltage and current.
- Determine the a.c. and d.c powers absorbed by the load.
- Determine the efficiency of the converter.

Solution

(a) Since $\pi + \alpha = 180^\circ + 45^\circ = 225^\circ < \beta$, then the current is continuous.

(b) The waveforms for v_o , and i_o are as shown in Fig.3.22,



$$(c) \quad V_{dc} = \frac{1}{\pi} \int_{\alpha}^{\alpha+\pi} V_m \sin \omega t \, d\omega t = \frac{2 V_m}{\pi} \cos \alpha$$

$$= \frac{2 \times 230 \times \sqrt{2}}{\pi} \cos 45 = 146.45 \text{ V}$$

$$I_{dc} = \frac{V_{dc}}{R} = \frac{146.45}{25} = 5.85 \text{ A}$$

(d) The *rms* values of the output voltage and current: Using Eq.(3.24),

$$V_{orms} = \frac{V_m}{\sqrt{2}} = \frac{230 \times \sqrt{2}}{\sqrt{2}} = 230 \text{ V}$$

$$I_{orms} = \frac{V_{rms}}{Z_L} = \frac{V_{rms}}{\sqrt{R^2 + (\omega L)^2}} = \frac{230}{\sqrt{25^2 + (2\pi \times 50 \times 10 \times 10^{-3})^2}}$$

$$\therefore I_{orms} = 9.12 \text{ A}$$

(f) The d.c. and a.c. powers are,

$$P_{dc} = V_{dc} I_{dc} = 146.45 \times 5.85 = 856.73 \text{ W}$$

$$P_{ac} = V_{rms} I_{rms} = 230 \times 9.12 = 2097.6 \text{ W}$$

(g) The efficiency of the converter,

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{856.73}{2097.6} = 0.408 = 40.8\%$$

Example 3.11

The circuit of Fig.3.23 consists of identical resistors $R_1 = R_2 = 200 \Omega$, an ideal diode D and an ideal thyristor T supplied from an ideal power supply $v_s = 200 \sin \omega t$.

(a) Sketch, to scale, corresponding waveforms of the supply voltage and the currents in R_1 , R_2 , D and T if the thyristor firing angle is 90° .

(b) What is the average value of the current in R_1 for a supply cycle?

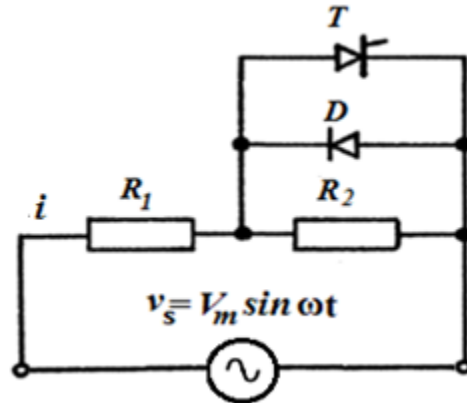


Fig. 3.23.

Solution

(a) The waveforms are as shown in Fig.3.24,

$$(b) \quad i_1 = \frac{V_m}{R_1 + R_2} \sin \omega t = \frac{200}{400} \sin \omega t = 0.5 \sin \omega t$$

$$i_2 = \frac{V_m}{R_1} \sin \omega t = \frac{200}{200} \sin \omega t = \sin \omega t$$

$$\begin{aligned} I_{dc} &= \frac{1}{2\pi} \int_0^{\pi/2} 0.5 \sin \omega t \, d\omega t + \frac{1}{2\pi} \int_{\pi/2}^{2\pi} \sin \omega t \, d\omega t \\ &= \frac{1}{2\pi} \left[-\cos \frac{\pi}{2} + 0.5 - 1 + \cos \frac{\pi}{2} \right] = \frac{-1}{4\pi} = -79.6 \text{ mA} \end{aligned}$$

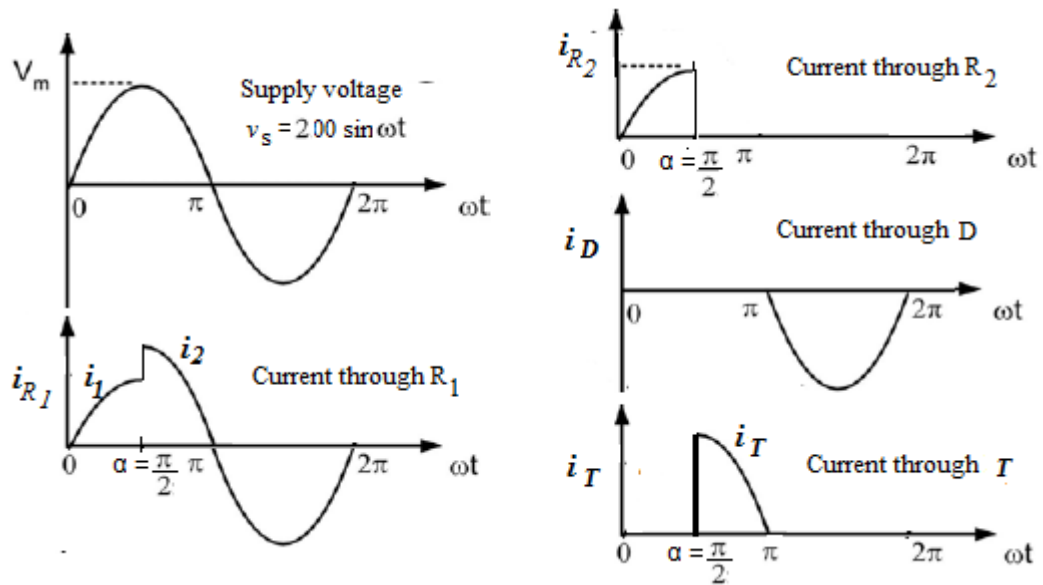


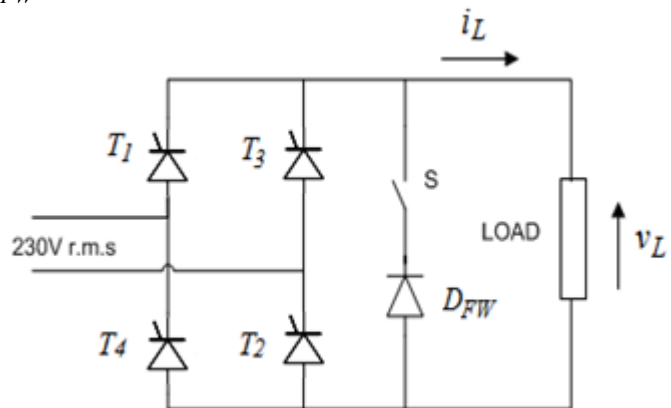
Fig.3.24 Waveforms for example 3.11.

Example 3.12

A single-phase full-wave fully-controlled bridge rectifier is feeding an R - L load with $R = 15 \, \Omega$ and $L = 20 \, \text{mH}$ as shown in Fig.3.25. The *rms* value of the a.c. input voltage is 230 V. The firing angle α is maintained constant at 45° so that the load current has an extinction angle of $\beta = 235^\circ$.

- Specify whether the current is continuous or discontinuous.
- Determine the average load voltage V_{dc} and current I_{dc} .
- Assume that load resistance remains the same; find the voltage V_{dc} and current I_{dc} if a freewheeling diode D_{FW} is used across the load.
- If T_3 is open circuited, find the load voltage and d.c. power while freewheeling diode D_{FW} is still connected and α is the same.

Fig.3.25.



Solution

- The current is continuous or discontinuous is found as follows:

Since $180^\circ + 45^\circ = 225^\circ$, which is $< \beta (=235^\circ)$. Hence , the current is continuous.

- (b) The average load voltage V_{dc} and current I_{dc} for continuous current operation are

$$V_{dc} = \frac{2V_m}{\pi} \cos \alpha$$

$$\therefore V_{dc} = \frac{2 \times 230 \times \sqrt{2}}{\pi} \cos 45^\circ = 146.5 \text{ V}$$

$$I_{dc} = \frac{V_{dc}}{R} = \frac{146.5}{15} = 9.76 \text{ A}$$

- (c) When the freewheeling diode D_{FW} is used across the load, and the load resistance remains the same; the waveform of the voltage will be as depicted in Fig.3.26.

The voltage V_{dc} and current I_{dc} are

$$V_{dc} = \frac{1}{\pi} \int_{\alpha}^{\pi} V_m \sin \omega t d\omega t$$

$$= \frac{V_m}{\pi} \left\{ -\cos \omega t \right\}_{\alpha}^{\pi}$$

$$= \frac{V_m}{\pi} (1 + \cos \alpha)$$

$$= \frac{325.2}{\pi} (1 + \cos 45^\circ) = 176.67 \text{ V}$$

$$I_{dc} = \frac{V_{dc}}{R} = \frac{176.67}{15} = 11.77 \text{ A}$$

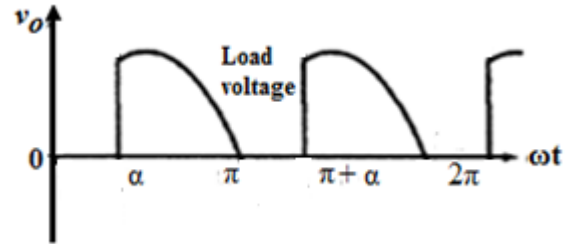


Fig. 3.26.

- (d) If T_3 is open circuited, the circuit and the output voltage will be as in Fig.3.27.

$$V_{dc} = \frac{1}{\pi} \int_{\alpha}^{\pi} V_m \sin \omega t d\omega t = \frac{V_m}{2\pi} \left\{ -\cos \omega t \right\}_{\alpha}^{\pi} = \frac{V_m}{2\pi} (1 + \cos \alpha)$$

$$= \frac{325.2}{2\pi} (1 + \cos 45^\circ) = 88.33 \text{ V}$$

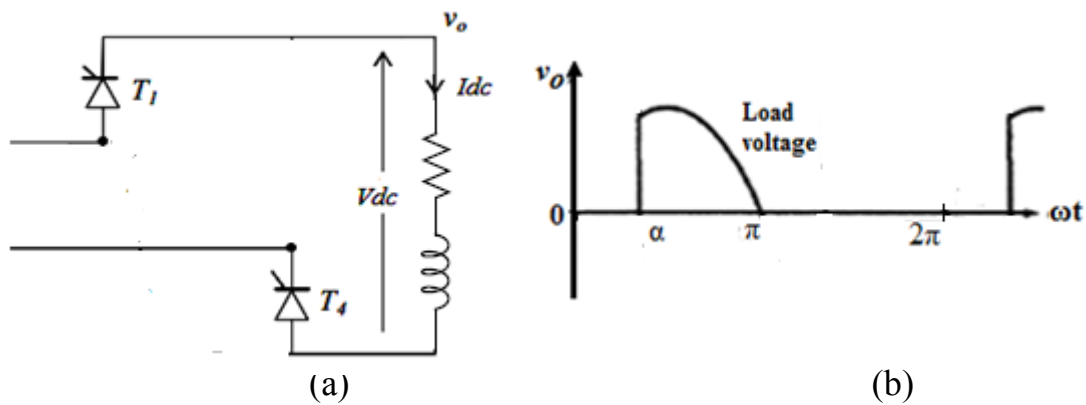


Fig.3.27.

$$I_{dc} = \frac{V_{dc}}{R} = \frac{88.33}{15} = 5.89 \text{ A}$$

$$P_{dc} = V_{dc} \times I_{dc} = 88.33 \times 5.89 = 520.17 \text{ W}$$

Example 3.13

The firing angle of the circuit shown in Fig.3.16 is kept constant to produce an average output voltage of 54 V d.c. and steady load current of 3.6 A. The load has a fixed resistance and infinite inductance. The primary voltage $V_p = 240 \text{ V rms}$ and the transformer has 2:1 voltage ratio.

- Find the triggering angle α and the load resistance R .
- Assume a FWD diode is connected across the load. Find the load current.
- If one thyristor is damaged (open-circuited), find the new load current.
- Specify the required ratings of thyristors and transformer.

Solution

(a) For highly inductive load the d.c. voltage is given by Eq.(3.31) as

$$V_{dc} = \frac{2V_m}{\pi} \cos \alpha$$

$$V_m = \sqrt{2}V_p \times \frac{1}{a} = \sqrt{2} \times 240 \times \frac{1}{2} = 169.7 \text{ V}$$

where a is the transformer voltage ratio.

$$V_{dc} = 54 = \frac{2 \times 169.7}{\pi} \cos \alpha \quad \rightarrow \cos \alpha = 0.5002$$

$$\cos^{-1} 0.5002 \cong 60^\circ$$

The load resistance R is found as

$$R = \frac{V_{dc}}{I_{dc}} = \frac{54}{3.6} = 15\Omega$$

(b) When a freewheel diode is connected across the load, the output voltage waveform will be as depicted in Fig.3.28 and its average value is given by

$$V_{dc} = \frac{1}{\pi} \int_{\alpha}^{\pi} V_m \sin \omega t d\omega t$$

$$V_{dc} = \frac{V_m}{\pi} \left\{ -\cos \omega t \right\}_{\alpha}^{\pi}$$

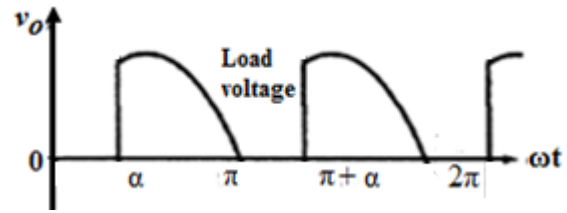


Fig.3.28.

$$= \frac{V_m}{\pi} (1 + \cos \alpha) = \frac{169.7}{\pi} (1 + \cos 60^\circ) = 81 \text{ V}$$

$$I'_{dc} = \frac{V_{dc}}{R} = \frac{81}{15} = 5.4 \text{ A}$$

(c) If one thyristor is damaged, the circuit becomes single-phase, half-wave rectifier, and

$$V_{dc} = \frac{1}{2\pi} \int_{\alpha}^{\pi} V_m \sin \omega t d\omega t$$

$$V_{dc} = \frac{V_m}{2\pi} \left\{ -\cos \omega t \right\}_{\alpha}^{\pi} = \frac{V_m}{2\pi} (1 + \cos \alpha) = \frac{169.7}{2\pi} (1 + 0.5) = 40.5 \text{ V}$$

$$I''_{dc} = \frac{V_{dc}}{R} = \frac{40.53}{15} = 2.7 \text{ A}$$

(d) Thyristors voltage ratings:

$$\text{PRV} = 2 V_m = 2 \times 169.7 = 339.4 \text{ V}$$

$$\text{and PFV} = 2 V_m = 2 \times 169.7 = 339.4 \text{ V}$$

Thyristor average current , since the current is steady (constant) , for highly inductive load ,the current through each thyristor is a rectangular

pulse start from α to $\alpha + \pi$ for a complete cycle of 2π as shown in Fig.3.29, hence its average value is

$$I_{Tav} = \frac{1}{2\pi} \int_{\alpha}^{\alpha+\pi} I_{dc} d\omega t = \frac{I_{dc}}{2\pi} (\alpha + \pi - \alpha) = \frac{I_{dc}}{2} = \frac{3.6}{2} = 1.8 \text{ A}$$

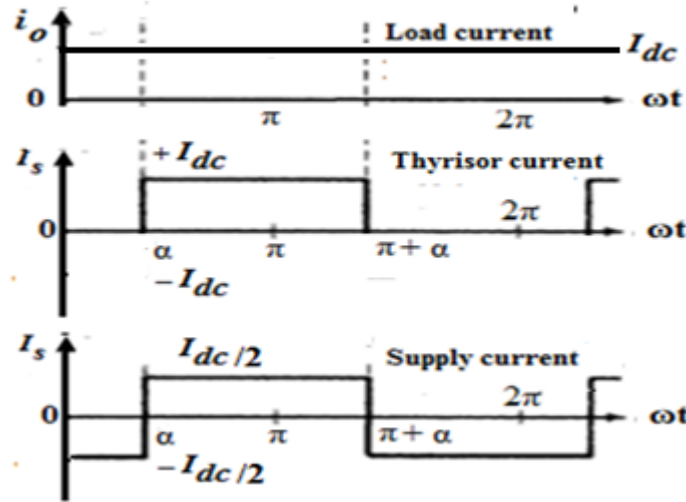


Fig.3.29 Current waveforms.

The *rms* value of the thyristor current is

$$I_{Trms} = \sqrt{\frac{1}{2\pi} \int_{\alpha}^{\alpha+\pi} I_{dc}^2 d\omega t} = \frac{I_{dc}}{\sqrt{2}} = \frac{3.6}{\sqrt{2}} = 2.54 \text{ A}$$

$$(VA)_s = 2 \times \frac{I_{dc}}{\sqrt{2}} \times \frac{V_m}{\sqrt{2}} = I_{dc} V_m = 3.6 \times 169.7 = 610.9$$

$$(VA)_p = \frac{I_{dc}}{2} \times V_{prms} = \frac{3.6}{2} \times 240 = 432$$

The transformer rating is:

$$(VA)_{TR} = \frac{1}{2} ((VA)_s + (VA)_p) = \frac{1}{2} (610.9 + 432) = 521.45$$

Example 3.14

A full-wave, fully-controlled bridge rectifier is fed from 230 V, 50 Hz, single-phase supply. The rectifier is used to charge a 48 V d.c. battery bank through R - L circuit of $R = 0.48 \Omega$ and $L = 0.79 \text{ mH}$. The firing angle is maintained at 41° . Find the required average charging current.

Solution

$$V_m = 230 \times \sqrt{2} = 325.2 \text{ V}$$

To find the average charging current using Eq.(3.28) and Eq.(3.29) as follows :

From equation (3.29),

$$i(0) = -\frac{V_m}{Z} \sin(\alpha - \theta) \left[\frac{1 + e^{-\frac{\pi R}{\omega L}}}{1 - e^{-\frac{\pi R}{\omega L}}} \right] - \frac{E}{R}$$

$$Z = \sqrt{R^2 + (\omega L)^2} = \sqrt{(0.48)^2 + (2\pi \times 50 \times 0.79 \times 10^{-3})^2} \\ = 2.526 \Omega$$

$$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right) = \tan^{-1}\left(\frac{2\pi \times 50 \times 0.79 \times 10^{-3}}{0.48}\right) = \tan^{-1} \frac{2.48}{0.48} \\ = 79^\circ = 0.715 \text{ rad}$$

$$\frac{\pi R}{\omega L} = \frac{3.14 \times 0.48}{2.48} = 0.607$$

$$i(0) = -\frac{325.2}{2.526} \sin(41^\circ - 79^\circ) \left[\frac{1 + e^{-0.607}}{1 - e^{-0.607}} \right] - \frac{48}{0.48} = 169.13 \text{ A}$$

Substituting this value into Eq.(3.28),

$$i(\omega t) = \frac{325.2}{2.526} \sin(\omega t - 79^\circ) - \frac{48}{0.48} + [169.13 \\ - \frac{325.2}{2.526} \sin(41^\circ - 79^\circ) + \frac{48}{0.48}] e^{\frac{-0.48}{2.48}(\omega t - 0.193)}$$

$$i(\omega t) = 128.74 \sin(\omega t - 79^\circ) - 100 + 400e^{-193\omega t}$$

To find the average value of the current using Eq.(3.30),

$$I_{av} = \frac{1}{\pi} \int_{\alpha}^{\alpha+\pi} i(\omega t) d\omega t \\ = \frac{1}{\pi} \int_{0.715}^{0.715+\pi} \{ 128.74 \sin(\omega t - 79^\circ) - 100 + 400e^{-193\omega t} \} d\omega t \\ \cong 220 \text{ A}$$

3.6 POLY-PHASE CONTROLLED RECTIFIERS

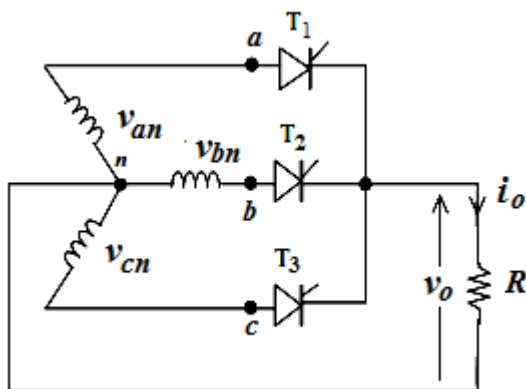
The poly-phase controlled rectifiers are used to convert a.c. input power supply into d.c. variable output voltage across the load. The features of poly-phase controlled rectifiers are:

- Operate from poly-phase a.c. supply voltage.
- They provide higher d.c. output voltage and higher d.c. output power.
- Higher output voltage ripple frequency.
- Filtering requirements are simplified for smoothing output load voltage and load current.

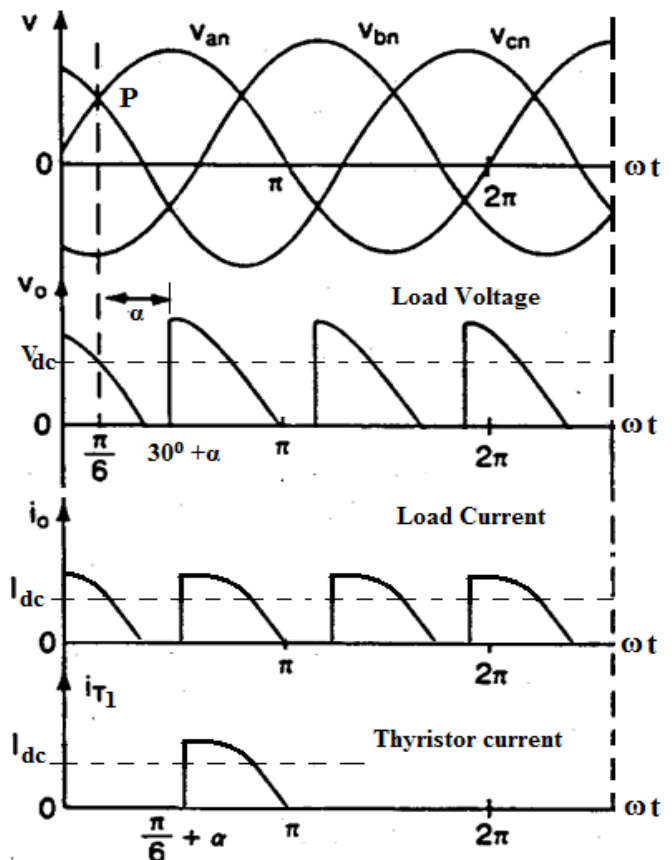
Poly-phase controlled rectifiers are classified into half-wave and full-wave, three-phase, six-phase, twelve-phase depending on the number of input phases which indicates the number of pulses p of the output voltage waveform.

3.6.1 Three-Phase Half-wave Controlled Rectifier ($p = 3$)

The Three-phase half-wave controlled rectifier is shown in Fig.3.30 (a). The waveforms for the supply voltage, output voltage, and load current are shown in Fig.3.30 (b) for case of resistive load.



(a) Circuit.



(b) Waveforms.

Note: α is usually measured from point P in Fig.3.30 (b).

Fig.3.30 Three-phase half-wave controlled rectifier. Case of resistive load.

As for the half-wave 3-phase uncontrolled diode rectifier, the load is connected between the converter positive terminal (cathodes of all thyristors) and the supply neutral. The thyristor with the highest voltage with respect to the neutral, when triggered, conducts. As the voltage of another thyristor becomes the highest, the load current is transferred to that device when triggered, and the previously conducting device is reverse-biased and naturally commutated. This process can be illustrated with the aid of Fig.3.30 (a) and assuming resistive load as follows:

The three-phase half-wave converter can be considered as it combines three single-phase half-wave controlled rectifiers in one single circuit feeding a common load. The thyristor T_1 in series with one of the supply phase windings ' v_{an} ' acts as one half-wave controlled rectifier. The second thyristor T_2 in series with the supply phase winding ' v_{bn} ' acts as the second half-wave controlled rectifier. The third Thyristor T_3 in series with the supply phase winding ' v_{cn} ' acts as the third half-wave controlled rectifier. The 3-phase input supply is applied through the star-connected supply transformer. The common neutral point of the supply is connected to one end of the load while the other end of the load connected to the common cathode point.

- When the thyristor T_1 is triggered at $\omega t = (30^\circ + \alpha)$ the phase voltage v_{an} appears across the load when T_1 conducts. The load current flows through the supply phase winding ' $a - n$ ' and through thyristor T_1 as long as T_1 conducts.
- When thyristor T_2 is triggered at $\omega t = (150^\circ + \alpha)$ T_1 becomes reverse biased and turns-off. The load current flows through the thyristor T_2 and through the supply phase winding ' $b - n$ '. When T_2 conducts the phase voltage v_{bn} appears across the load until the thyristor T_3 is triggered .
- When the thyristor T_3 is triggered at $\omega t = (270^\circ + \alpha)$ T_2 is reversed biased and hence T_2 turns-off. The phase voltage v_{cn} appears across the load when T_3 conducts.
- When T_1 is triggered again at the beginning of the next input cycle the Thyristor T_3 turns off as it is reverse biased naturally as soon as T_1 is triggered . The output voltage which appears across the load, and the load current assuming a constant and ripple free load current for a highly inductive load and the current through the thyristor T_1 .

For a purely resistive load where the load inductance ' $L = 0$ ' and the trigger angle $\alpha > 30^\circ$ the load current appears as discontinuous load current and each thyristor is naturally commutated when the polarity of the corresponding phase supply voltage reverses. The frequency of output ripple frequency for a three-phase half wave converter is $3f_s$, where f_s is the input supply frequency.

Analytical properties of the output voltage waveform of the three-phase half-wave controlled rectifier

(A) Case of resistive load

The waveforms for the supply voltage, output voltage, and load current are shown in Fig.3.29 for the case of resistive load. The average value of the output voltage V_{dc} can be found as:

$$\begin{aligned}\text{Let } v_{an} &= V_m \sin \omega t \\ v_{bn} &= V_m \sin(\omega t - 2\pi/3) \\ v_{cn} &= V_m \sin(\omega t - 4\pi/3)\end{aligned}$$

The average value of the load voltage wave is

$$\begin{aligned}V_{dc} &= \frac{1}{\frac{2\pi}{3}} \int_{30^\circ + \alpha}^{180^\circ} V_m \sin \omega t \, d\omega t = \frac{3V_m}{2\pi} [-\cos \omega t]_{30^\circ + \alpha}^{180^\circ} \\ &= \frac{3V_m}{2\pi} [-(\cos(180^\circ) + \cos(30^\circ + \alpha))] \\ &= \frac{3V_m}{2\pi} [1 + \cos(\alpha + 30^\circ)]\end{aligned}\tag{3.35}$$

The load current I_{dc} is:

$$I_{dc} = \frac{3V_m}{2\pi R} [1 + \cos(\alpha + 30^\circ)]\tag{3.36}$$

In case of resistive load, each thyristor ceases to conduct as soon as its anode voltage goes negative. This causes discontinuous current flow in the rectifier circuit when the angle exceeds 30° as shown in Fig.3.30.

(B) Case of series resistive-inductive ($R-L$) load

When the load is highly inductive it tends to maintain constant output current even with large delay angles, and the thyristor current remains rectangular as shown in Fig.3.31. Consequently, the induced *emf* in the load inductor maintains current flow even when the anode polarity has reversed. This means that energy is being returned from the magnetic field of the inductor through the transformer to the supply, and the circuit is temporarily acting as an inverter. The average value of the load voltage wave is

$$V_{dc} = \frac{1}{\frac{2\pi}{3}} \int_{30^\circ + \alpha}^{30^\circ + \alpha + 120^\circ} V_m \sin \omega t \, d\omega t = \frac{3V_m}{2\pi} [-\cos \omega t]_{30^\circ + \alpha}^{150^\circ + \alpha}$$

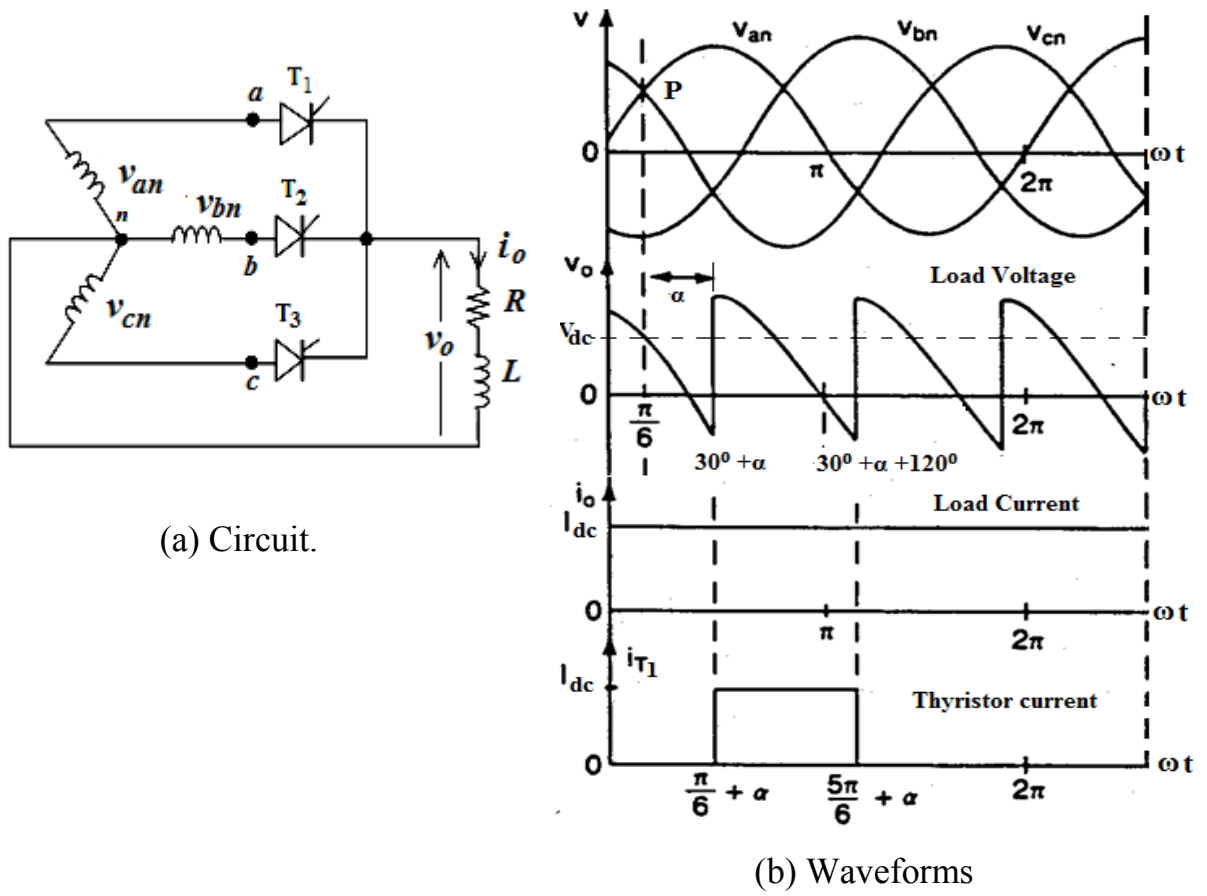


Fig.3.31 Three-phase half-wave controlled rectifier with R - L load.

$$\begin{aligned}
 V_{dc} &= \frac{3V_m}{2\pi} [-(\cos(150^\circ + \alpha) - \cos(30^\circ + \alpha))] \\
 &= \frac{3\sqrt{3}V_m}{2\pi} \cos\alpha
 \end{aligned} \tag{3.37}$$

The load current I_{dc} is:

$$I_{dc} = \frac{3\sqrt{3}V_m}{2\pi R} \cos\alpha \tag{3.38}$$

The operation of the three-phase, half-wave rectifier with different values of α is illustrated in Fig.3.32. It can be seen that this converter can operate either as a rectifier or as an inverter as

For	$0^\circ < \alpha < 90^\circ$	$\Longrightarrow$	Rectification process
	$90^\circ < \alpha < 180^\circ$	$\Longrightarrow$	Inversion process

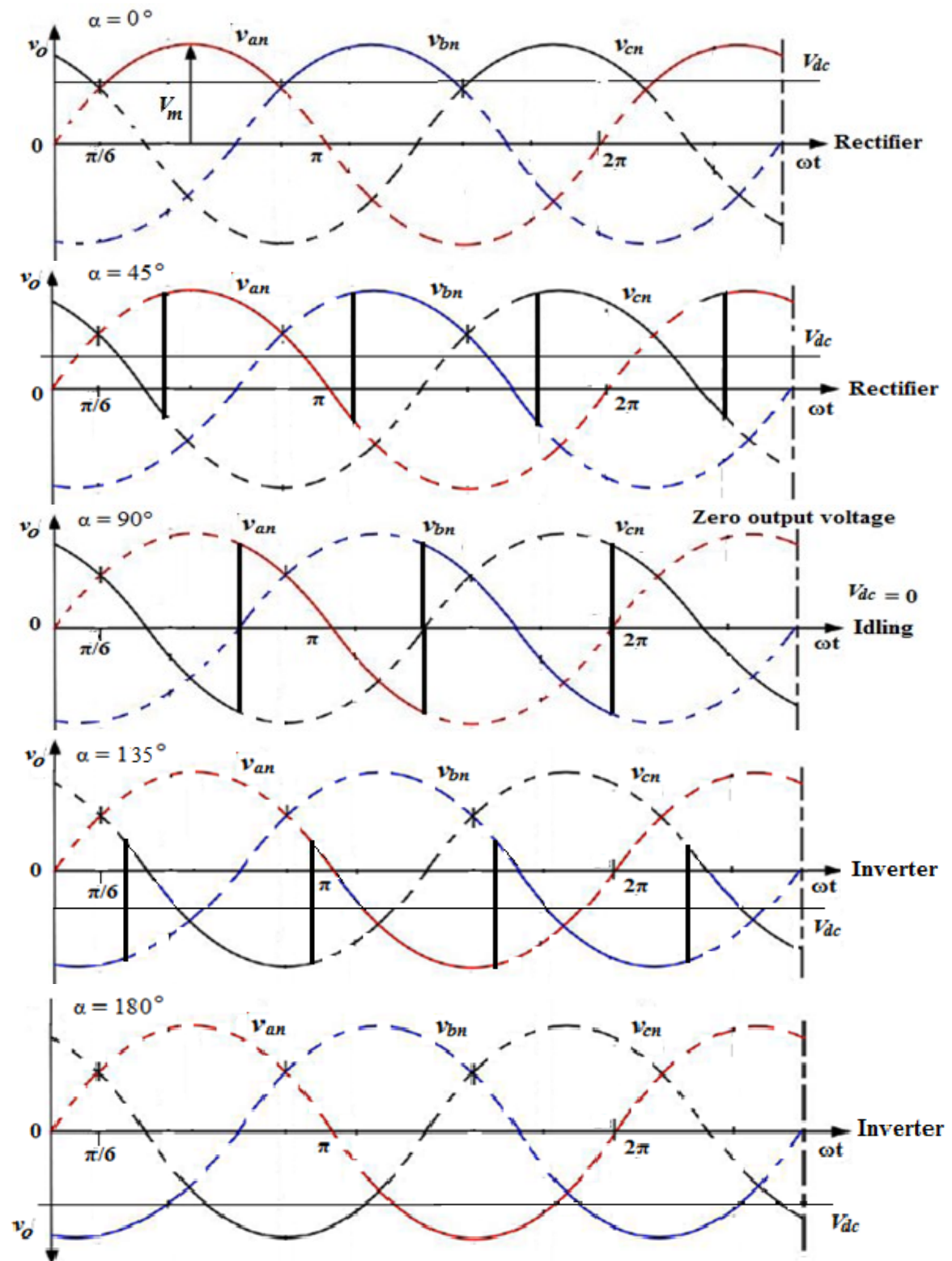


Fig.3.32 Output voltage waveform of the three-phase half-wave rectifier for different values of firing angle α . Case of R - L load.

Notes:

1. The mean output voltage is zero for $\alpha = \pi/2$. The converter is idle (no output).
2. Negative average output voltage occurs when $\alpha > \pi/2$.
3. Power inversion is possible, if a load with an *emf* to assist the current flow.

The three-phase half-wave converter is not normally used in practical converter systems because of the disadvantage that the supply current waveforms contain d.c. components (i.e. the supply current waveforms have an average or d.c. value).

Example 3.15

The load in Fig.3.31 consists of a resistance and a very large inductance. The inductance is so large that the output current I_o can be assumed to be continuous and ripple-free. For $\alpha = 60^\circ$,

- (a) Draw the wave forms of V_o and I_o .
- (b) Determine the average value of the output voltage, if phase voltage $V_{an} = 120$ V.
- (c) Find the average output voltage if a free wheel diode is connected across the load.

Solution

- (a) The waveforms of V_o and I_o are as shown in Fig.3.31(b),
- (b) The average value of the output voltage is given in Eq.(3.37) as,

$$V_{av} = \frac{3\sqrt{3} V_m}{2\pi} \cos \alpha$$

$$\text{For } \alpha = 60^\circ : \quad V_m = \sqrt{2} V_{an} = \sqrt{2} \times 120 \text{ V} = 169.7 \text{ V}$$

$$\therefore V_{dc} = \frac{3\sqrt{3}}{2\pi} \times 169.7 \cos 60^\circ = 70.2 \text{ V}$$

- (c) If a freewheel diode is connected across the load, the circuit and output voltage waveform will be as shown in Fig.3.33.

The average voltage of the waveform of Fig.3.33 (b) can be evaluated as

$$V_{dc} = \frac{1}{\frac{2\pi}{3}} \int_{30^\circ + \alpha}^{\pi} V_m \sin \omega t \, d\omega t = \frac{3V_m}{2\pi} [-\cos \omega t]_{30^\circ + \alpha}^{\pi}$$

$$V_{dc} = \frac{3V_m}{2\pi} (1 + \cos(30^\circ + \alpha))$$

$$\therefore V_{dc} = \frac{3\sqrt{3}}{2\pi} \times 169.7 (1 + \cos(30^\circ + 60^\circ)) = 81 \text{ V}$$

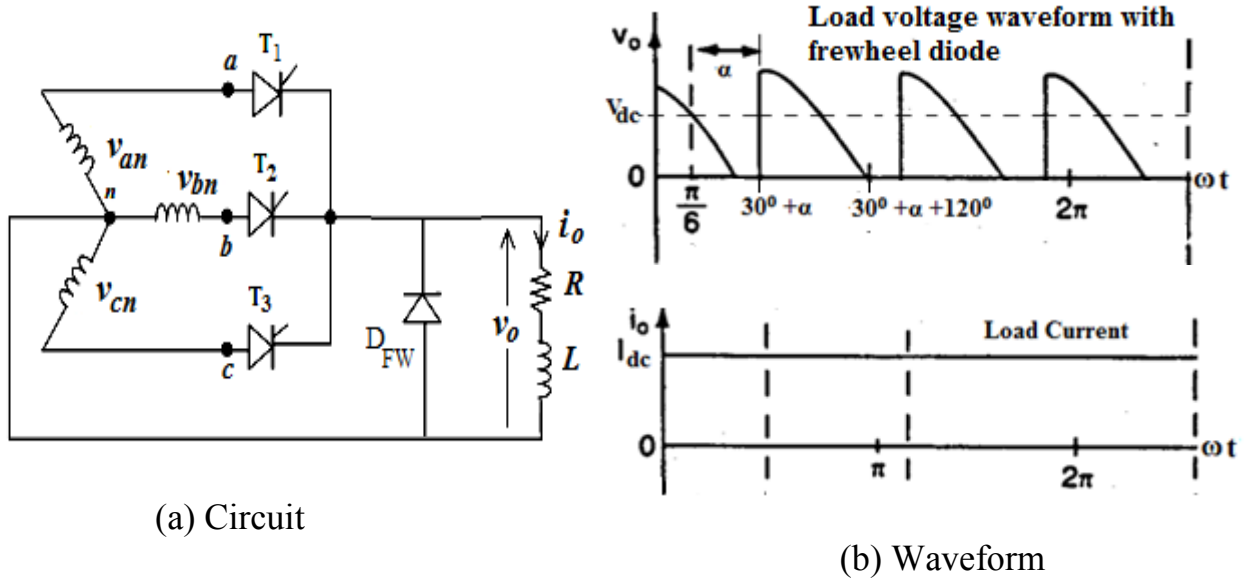


Fig.3.33.

General solution of p -pulse half-wave controlled rectifier

For p -pulse (or p -phase, i.e.: 3-phase, 6-phase, 12-phase etc) half-wave rectifier circuit, a general formula can be obtained as follows:
The output voltage v_o waveform of a p -pulse, fully controlled converter with continuous current operating mode would be as shown in Fig.3.35.

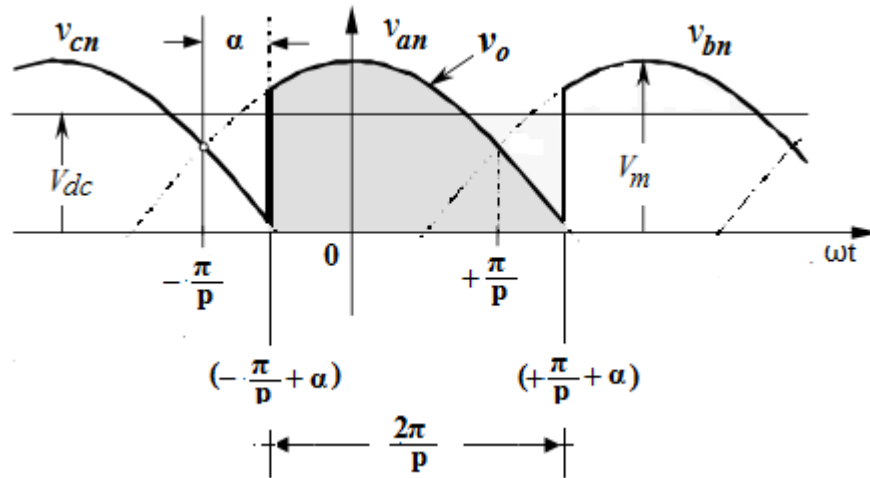


Fig.3.34.

It has been found useful for calculation to express the a.c. voltages on the thyristor side by cosine functions to avoid the mistake in the polarity sign. Hence:

$$v_{an} = V_m \cos (\omega t + 120)$$

$$v_{bn} = V_m \cos \omega t$$

$$v_{cn} = V_m \cos (\omega t - 120)$$

The average d.c. voltage V_{dc} is,

$$\begin{aligned} V_{dc} &= \frac{1}{2\pi} \int_{(-\frac{\pi}{p}+\alpha)}^{(+\frac{\pi}{p}+\alpha)} V_m \cos \omega t d\omega t \\ &= \left(\frac{p}{2\pi}\right) V_m \left[\sin\left(\frac{\pi}{p} + \alpha\right) - \sin\left(-\frac{\pi}{p} + \alpha\right)\right] \\ &= \left(\frac{p}{2\pi}\right) V_m \left[\sin\frac{\pi}{p} \cos \alpha + \sin \alpha \cos \frac{\pi}{p} - \sin\left(-\frac{\pi}{p}\right) \cos \alpha + \cos\left(\frac{-\pi}{p}\right) \sin \alpha\right] \\ &= \frac{p V_m}{2\pi} \left[2 \sin \frac{\pi}{p} \cos \alpha\right] \\ \therefore V_{dc} &= \frac{p}{\pi} V_m \sin \frac{\pi}{p} \cos \alpha \end{aligned} \quad (3.39)$$

or , for a three-phase, half-wave circuit, $p = 3$, hence,

$$V_{dc} = \frac{3}{\pi} V_m \sin \frac{\pi}{3} \cos \alpha = \frac{3\sqrt{3}V_m}{2\pi} \cos \alpha$$

which is the same as Eq.(3.37) obtained previously.

Notes :

- The diode conduction angle is $120^\circ + \alpha$.
- The variation of the average output voltage V_{dc} with the triggering angle α is shown in Fig.3.35 .

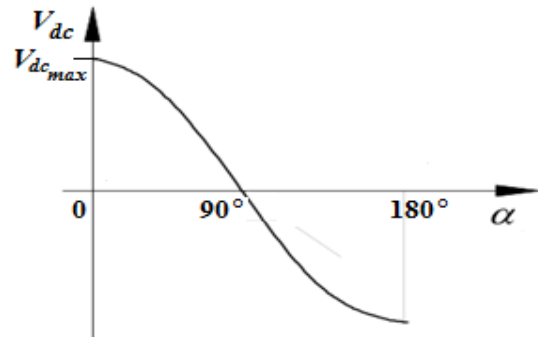


Fig.3.35 variation of the average output voltage V_{dc} with the triggering angle α .

3.6.2 Three-Phase Full-Wave Fully-Controlled Rectifier ($p = 6$)

Three-phase, full-wave converter is a fully-controlled bridge rectifier using six thyristors connected as shown in Fig.3.36(a). All the six thyristors are controlled switches which are turned on at an appropriate time by applying suitable gate trigger signals. The three-phase full-converter is extensively used in industrial power applications up to about 150kW output power level, where two-quadrant operation is required. This circuit is also known as three-phase full wave bridge or as a six-pulse converter. The frequency of output ripple voltage is $6 f_s$ and the filtering requirement is less than that of three-phase half-wave converters.

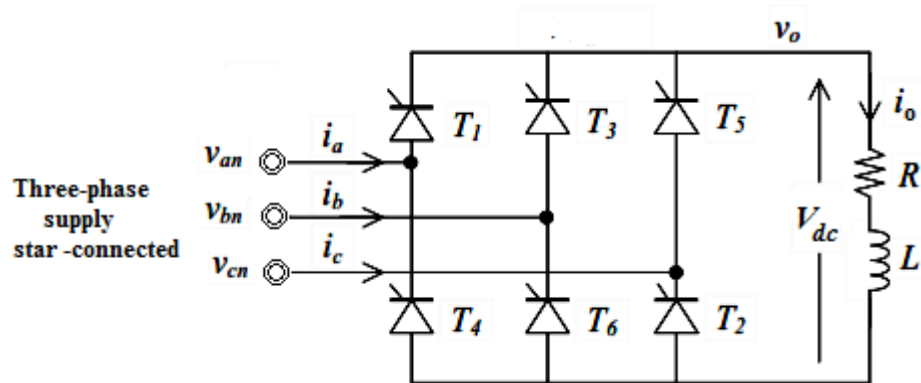


Fig.3.36 Three-phase full-wave fully-controlled rectifier.

In the circuit configuration of the three-phase full-wave controlled rectifier, the thyristor which has the most positive voltage at its anode conducts when triggered, and the thyristor with the most negative voltage at its cathode returns the load current, if triggered. The waveforms are as shown in Fig.3.37.

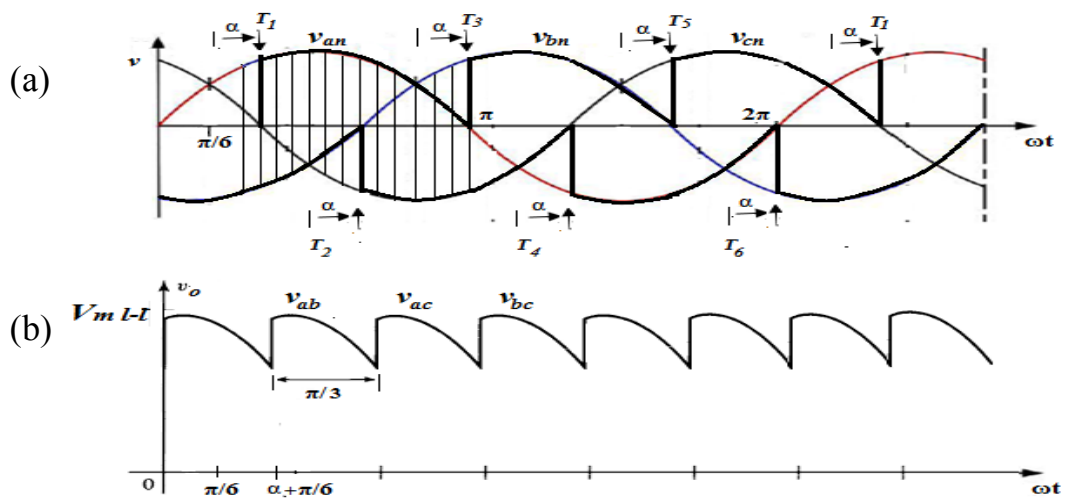


Fig.3.37 Three-phase full-wave fully-controlled rectifier output voltage waveform.

- Commutation of the load current from one thyristor to the next occurs at the firing instant, when the incoming thyristor reverse biases the previously conducting thyristor.
- The output d.c. voltage waveform is determined by the difference of potentials of the positive and negative rails.

Assuming continuous conduction, with highly inductive load, the average d.c. output voltage can be found by referring to Fig.3.37(b) as follows,

$$V_{dc} = \frac{6}{2\pi} \int_{\alpha+\frac{\pi}{6}}^{\alpha+\frac{\pi}{2}} v_s(\omega t) d\omega t$$

where $v_s(\omega t) = v_{ab} = \sqrt{3}V_m \sin(\omega t + 30^\circ)$

$$\text{hence } V_{dc} = \frac{3}{\pi} \int_{\alpha+\frac{\pi}{6}}^{\alpha+\frac{\pi}{2}} \sqrt{3}V_m \sin(\omega t + 30^\circ) d\omega t$$

$$\therefore V_{dc} = \frac{3\sqrt{3}V_m}{\pi} \cos(\alpha) \quad (3.40)$$

The *rms* value of the output voltage waveform is

$$\begin{aligned} V_{o(rms)} &= \left[\frac{6}{2\pi} \int_{\alpha+\frac{\pi}{6}}^{\alpha+\frac{\pi}{2}} v_o^2(\omega t) d\omega t \right]^{\frac{1}{2}} \\ V_{o(rms)} &= \left[\frac{3}{\pi} \int_{\alpha+\frac{\pi}{6}}^{\alpha+\frac{\pi}{2}} 3V_m^2 \sin^2(\omega t + 30^\circ) d\omega t \right]^{\frac{1}{2}} \\ V_{o(rms)} &= \sqrt{3}V_m \left[\frac{1}{2} + \frac{3\sqrt{3}}{4\pi} \cos(2\alpha) \right]^{\frac{1}{2}} \end{aligned} \quad (3.41)$$

Alternatively, V_{dc} can be evaluated from the general p-phase formula, Eq.(3.30), as

$$V_{dc} = \frac{p}{\pi} V_{ml-l} \sin \frac{\pi}{p} \cos \alpha,$$

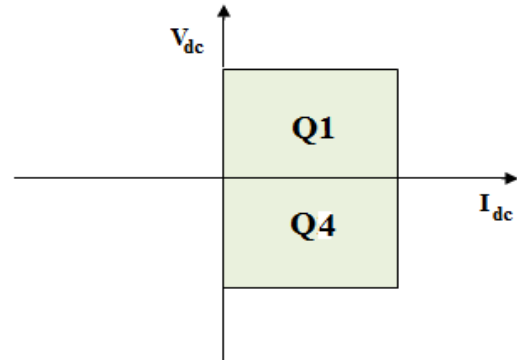
Here $p = 6$, V_{ml-l} = Maximum line-to-line voltage = $\sqrt{3} V_m$, here V_m = maximum line-to-neutral voltage, hence,

$$V_{dc} = \frac{6\sqrt{3}}{\pi} V_m \sin \frac{\pi}{6} \cos \alpha = \frac{3\sqrt{3}V_m}{\pi} \cos(\alpha)$$

which is the same result as in Eq. (3.40).

This converter operates in quadrants Q1 and Q4, developing both positive and negative polarity d.c. output voltage. For firing angles, $0^\circ \leq \alpha \leq 90^\circ$, the converter operates in quadrant-1 (Q1) (giving positive output power, i.e. rectifier operation) and for $90^\circ \leq \alpha \leq 180^\circ$, the operation is in quadrant-4 (giving negative output power, i.e. inverter operation). Operation in quadrant-4 is possible only when the load includes an active d.c. source, able to source power into the a.c. supply circuit. See Fig.3.38.

Fig. 3.38 Quadrants of operation of the three-phase fully-controlled converter.



3.6.3 Three-Phase Full-Wave Half-Controlled Rectifier

This converter is shown in Fig.3.39. It consists of three thyristors and three diodes with freewheeling diode across the load. It gives positive voltage and positive current only (not regenerative converter) i.e. it operates in the first quadrant only.

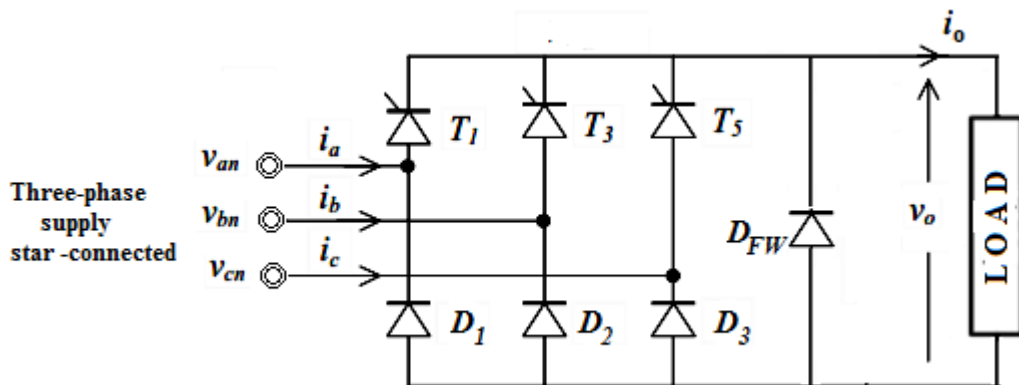


Fig.3.39 Three-phase full-wave, half-controlled rectifier.

In order to understand the principle of operation of this converter, assume that the output (load) current is continuous and ripple free.

- At $\omega t = \pi/6 + \alpha$, T_1 is triggered, hence T_1 and D_3 conduct and voltage $v_o = v_{ac}$ appears across the load.
- At $\omega t = 7\pi/6$ (210°), $v_o = 0$, and from this instant onward v_{an} becomes the most negative phase and diode D_1 becomes forward biased and start conducting. Now since T_1 is still conducting, the

output current i_o will freewheel through T_1 and D_1 if D_{FW} is not connected making $v_o = 0$.

- When T_2 is triggered, current i_o will flow through T_2 and D_1 and voltage v_{ba} appears across the load.
- The process repeated every $2\pi/3$ whenever a thyristor is triggered. The output voltage and current waveforms of the three-phase half-controlled rectifier bridge are as shown in Fig.3.40 for $\alpha = 90^\circ$. The instants of triggering the thyristors and duration of conduction of the diodes are also on the same figure.

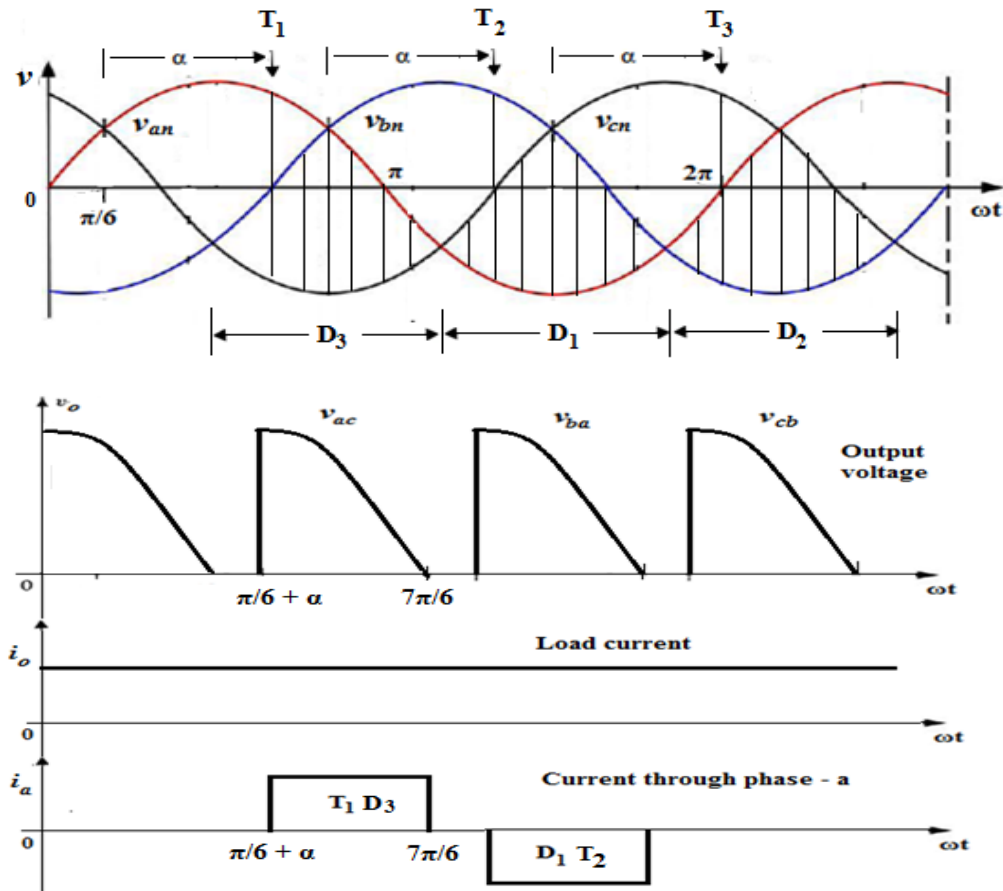


Fig.3.40 Waveforms of three-phase full-wave, half-controlled rectifier.

The output voltage is given by:

For the period $\pi/6 + \alpha$ and $7\pi/6$, the voltage appears across the load is

$$v_o = v_{ac}(\omega t) = \sqrt{3}V_m \sin(\omega t - 30^\circ)$$

$$\begin{aligned}
v_{dc} &= \frac{3}{2\pi} \int_{\alpha+\frac{\pi}{6}}^{\frac{7\pi}{6}} v_{ac}(\omega t) d\omega t \\
&= \frac{3}{2\pi} \int_{\alpha+\frac{\pi}{6}}^{\frac{7\pi}{6}} \sqrt{3}V_m \sin(\omega t - 30^\circ) d\omega t \\
\therefore v_{dc} &= \frac{3\sqrt{3}V_m}{2\pi} (1 + \cos\alpha) \quad (3.42)
\end{aligned}$$

Example 3.16

A three-phase full-wave fully-controlled rectifier supply a highly inductive load with $R = 10 \Omega$ the supply is a three-phase star-connected with 400 V *rms*, calculate:

- The load current when the firing angle $\alpha = 45^\circ$.
- The power drawn from the supply.
- If the current value kept at (a) and α changed to 135° , calculate the power returned to the supply.

Solution

- (a) For $\alpha = 45^\circ$

$$V_m = \frac{\sqrt{2} \times 400}{\sqrt{3}} = 326.6 \text{ V}$$

$$V_{dc} = \frac{3\sqrt{3} V_m}{\pi} \cos \alpha = \frac{3\sqrt{3} \times 326.6}{\pi} \cos 45^\circ = 381.97 \text{ V}$$

$$I_{dc} = \frac{V_{dc}}{R} = \frac{381.97}{10} = 38.197 \text{ A}$$

- (b) The power drawn from the source = the power dissipated at the resistance of the load

$$P_s = P_{load} = V_{dc} I_{dc} = 381.97 \times 38.197 = 14590.21 \text{ W}$$

- (c) For $\alpha = 135^\circ$, $I_o = I_{dc} = 38.197 \text{ A}$

$$V_{dc} = \frac{3\sqrt{3} V_m}{\pi} \cos 135^\circ = \frac{3\sqrt{3} \times 326.6}{\pi} \cos 135^\circ = -381.97 \text{ V}$$

Power return to the source:

$$P_s = V_{dc} \cdot I_{dc} = -381.97 \times 38.197 = -14590.21 \text{ W}$$

Example 3.17

If the converter in Example 3.16 is replaced by a full-wave half-controlled converter, calculate:

- (a) V_{dc} when $\alpha = 45^\circ$
- (b) V_{dc} when $\alpha = 75^\circ$
- (c) V_{dc} when $\alpha = 135^\circ$
- (d) Maximum value of V_{dc}
- (e) The value of α to obtain $I_{dc} = 15 \text{ A}$

Solution

(a) For a three-phase semiconverter, the average d.c. voltage is

$$V_{dc} = \frac{3\sqrt{3} V_m}{2\pi} (1 + \cos \alpha)$$

For $\alpha = 45^\circ$:

$$V_{dc} = \frac{3\sqrt{3} \times 326.6}{2\pi} (1 + \cos 45^\circ) = 461 \text{ V}$$

(b) For $\alpha = 75^\circ$:

$$V_{dc} = \frac{3\sqrt{3} \times 326.6}{2\pi} (1 + \cos 75^\circ) = 340 \text{ V}$$

(c) For $\alpha = 135^\circ$

$$V_{dc} = \frac{3\sqrt{3} \times 169.83}{2\pi} (1 + \cos 135^\circ) = 79.53 \text{ V}$$

(d) Maximum voltage output is when $\alpha = 0^\circ$

$$V_{dc} = \frac{3\sqrt{3} \times 326.6}{2\pi} (1 + \cos 0^\circ) = 540 \text{ V}$$

(e)

$$I_{dc} = \frac{V_{dc}}{R} = 15 \text{ A}$$

$$\text{or } V_{dc} = 15 \times R = 15 \times 10 = 150 \text{ V}$$

$$150 = \frac{3\sqrt{3} \times 326.6}{2\pi} (1 + \cos \alpha)$$

From which $\alpha = 116.4^\circ$

Example 3.18

A d.c. load requires control of V_{dc} from maximum to 1/10 of the maximum value. If a half-controlled three-phase bridge is to be used, determine the range of angles required to trigger the thyristors.

Solution

Maximum V_{dc} is obtained when $\alpha = 0^\circ$, hence for the three-phase half-controlled bridge

$$V_{dcmax} = \frac{3\sqrt{3} V_m}{2\pi} (1 + \cos 0^\circ) = \frac{3\sqrt{3} V_m}{\pi}$$

$$V_{dc} = \frac{V_{dcmax}}{2} (1 + \cos \alpha)$$

$$\frac{V_{dc}}{V_{dcmax}} = \frac{1}{10} = \frac{1}{2} (1 + \cos \alpha)$$

From which

$$\cos \alpha = \frac{1}{5} - 1 = -0.8 \quad \therefore \alpha = 143^\circ$$

The range of α is : $0^\circ \leq \alpha \leq 143^\circ$

Example 3.19

The load in Fig.3.41 consists of a resistance and a very large inductance. The inductance is so large that the output i_o can be assumed to be continuous and ripple-free. The load is supplied from an ideal three-phase half-wave controlled rectifier.

- Sketch a set of waveforms, for thyristor firing angle $\alpha = 45^\circ$, showing: supply voltages (v_{an} , v_{bn} , v_{cn}), load voltage v_o , and load current i_o .
- Derive an expression for the average value V_{dc} of the load voltage.

- (c) Determine the average value of the output voltage and current if the phase voltage peak is $V_{max} = 450$ V and $R = 100 \Omega$ when $\alpha = 0^\circ$, $\alpha = 60^\circ$, and $\alpha = 120^\circ$.
- (d) How is the average load voltage affected if thyristor T_2 , failed to open circuit?

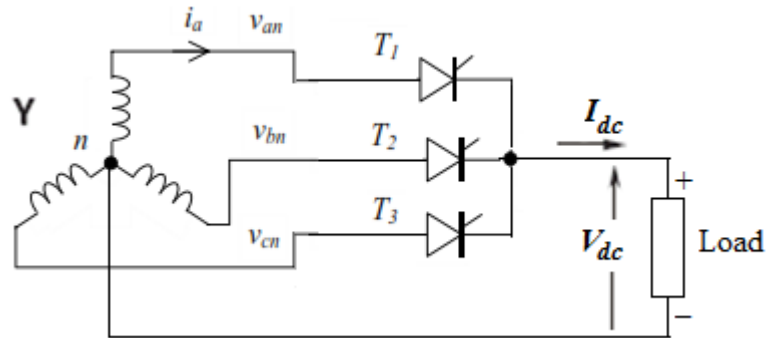
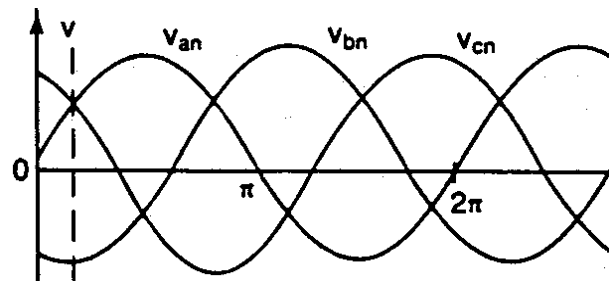


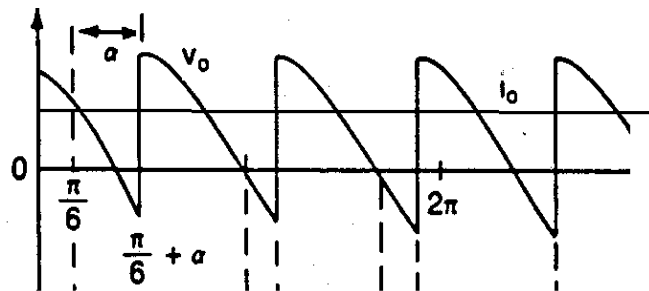
Fig.3.41

Solution

- (a) Waveforms as shown in Fig.3.42.



(a) Supply voltage waveforms



(b) Load voltage and current

Fig.3.42.

- (b) Let $v_{an} = V_m \sin \omega t$,

$$V_{dc} = \frac{1}{2\pi} \int_{30^\circ + \alpha}^{30^\circ + \alpha + 120^\circ} V_{an}(\omega t) d\omega t$$

$$\begin{aligned}
&= \frac{1}{\frac{2\pi}{3}} \int_{30^\circ+\alpha}^{30^\circ+\alpha+120^\circ} V_m \sin \omega t d\omega t \\
&= \frac{1}{\frac{2\pi}{3}} \int_{30^\circ+\alpha}^{30^\circ+\alpha+120^\circ} V_m \sin \omega t d\omega t \\
&= \frac{3V_m}{2\pi} [-\cos(\alpha + 150) + \cos(\alpha + 30)] = \frac{3\sqrt{3} V_m}{2\pi} \cos \alpha
\end{aligned}$$

(c) For $\alpha = 0^\circ$

$$V_{dc} = \frac{3\sqrt{3} \times 450}{2\pi} \cos 0^\circ = 372 \text{ V} \quad \rightarrow \quad I_{dc} = \frac{372}{100} = 3.72 \text{ A}$$

For $\alpha = 60^\circ$

$$V_{dc} = \frac{3\sqrt{3} \times 450}{2\pi} \cos 60^\circ = 189 \text{ V} \quad \rightarrow \quad I_{dc} = \frac{189}{100} = 1.89 \text{ A}$$

For $\alpha = 120^\circ$

$$V_{dc} = \frac{3\sqrt{3} \times 450}{2\pi} \cos(120^\circ) = -322 \text{ V}$$

(d) If T_2 failed to open circuit, phase-b will be eliminated and the voltage and current waveforms will appear as depicted in Fig.3.43 ,

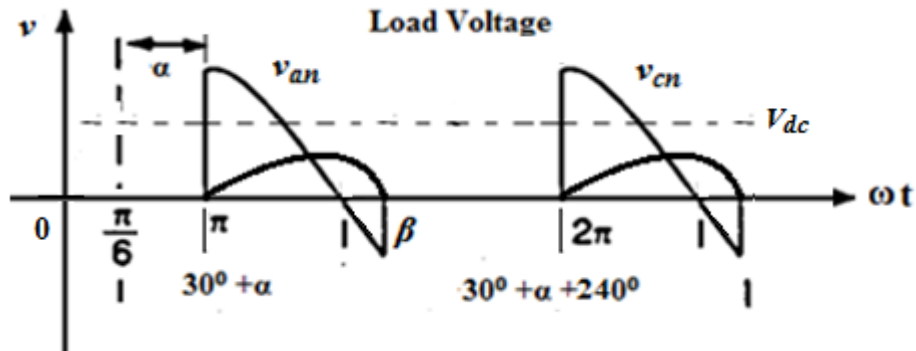


Fig.3.43 Voltage and current waveforms when T_2 fails to open circuit.

The average voltage will be,

$$V_{dc} = \frac{1}{2\pi} \int_{\alpha+30^\circ}^{\beta} V_m \sin \omega t d\omega t + \int_{\alpha+30^\circ+240^\circ}^{\beta+240^\circ} V_m \sin \omega t d\omega t$$

Example 3.20

Sketch the waveform across an R - L load ($L \rightarrow \infty$) for a 3-phase, half-wave converter with delay angle $\alpha = 60^\circ$ (a) without, (b) with freewheeling diode. Drive, in each case, expressions for the average output d.c. voltage and deduce waveforms of transformer secondary current.

Solution

a) Without freewheeling diode, the load voltage is as shown in Fig.3.44.

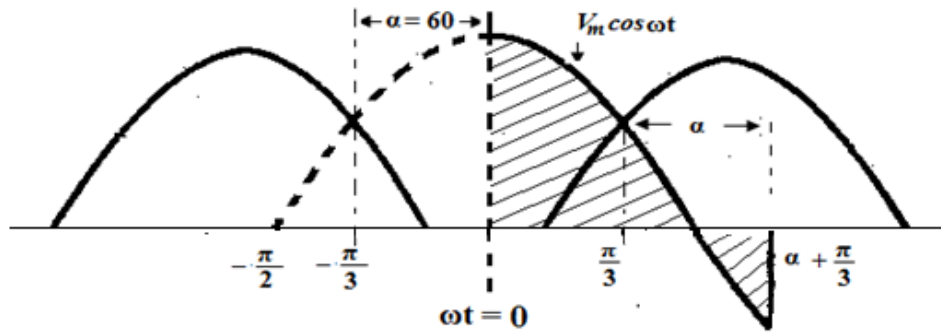


Fig.3.44

$$\begin{aligned} V_{dc} &= \frac{3}{2\pi} \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \sqrt{2} V_s \cos \omega t d\omega t = \frac{3}{2\pi} [\sqrt{2} V_s \sin \omega t]_{-\frac{\pi}{3} + \alpha}^{\frac{\pi}{3} + \alpha} \\ &= \frac{3\sqrt{2}V_s}{2\pi} \left[\sin \left(\frac{\pi}{3} + \alpha \right) - \sin \left(-\frac{\pi}{3} + \alpha \right) \right] \\ &= \frac{3\sqrt{2} V_s}{2\pi} \left[\sin \frac{\pi}{3} \cos \alpha + \cos \frac{\pi}{3} \sin \alpha \right] \\ &= \sin \left(-\frac{\pi}{3} \right) \cos \alpha - \cos \left(-\frac{\pi}{3} \right) \sin \alpha \\ &= \frac{3\sqrt{2}V_s}{2\pi} 2 \sin \frac{\pi}{3} \cos \alpha = \sqrt{2} V_s \left(\frac{3}{\pi} \right) \sin \frac{\pi}{3} \cos \alpha \end{aligned}$$

With freewheeling diode, the negative voltage areas are eliminated

$$\begin{aligned}
 &= \frac{3}{2\pi} \int_{-\frac{\pi}{3}+\alpha}^{\frac{\pi}{2}} \sqrt{2} V_s \cos \omega t \, d\omega t = \frac{3}{2\pi} [\sqrt{2} V_s \sin \omega t]_{-\frac{\pi}{3}+\alpha}^{\frac{\pi}{2}} \\
 &= \frac{3}{2\pi} \sqrt{2} V_s \left[\sin \frac{\pi}{2} - \sin \left(-\frac{\pi}{3} + \alpha \right) \right] \\
 &= \frac{3}{2\pi} \sqrt{2} V_s \left[\sin \frac{\pi}{3} \cos \alpha = \cos \frac{\pi}{3} \sin \alpha + \sin \frac{\pi}{2} \right] \\
 &= \frac{3}{2\pi} \sqrt{2} V_s \left[1 - \sin \left(\alpha - \frac{\pi}{3} \right) \right]
 \end{aligned}$$

Example 3.21

Show that for a p-pulse converter with purely resistive load, where the delay angle α is restricted so that the load current is continuous, the *rms* value of d.c. voltage is given by:

$$V_{orms} = \frac{V_m}{\sqrt{2}} \sqrt{1 + \frac{P}{2\pi} \cos 2\alpha \sin \frac{2\pi}{P}}$$

where V_m is the peak value of the transformer secondary voltage secondary voltage per phase.

Solution

From Fig. 3.45, the *rms* value of the output voltage is

$$\begin{aligned}
 V_{orms}^2 &= \frac{P}{2\pi} \int_{-\frac{\pi}{P}+\alpha}^{\frac{\pi}{P}+\alpha} (V_m)^2 \cos^2 \omega t \, d\omega t \\
 &= \frac{P}{2\pi} V_m^2 \int_{-\frac{\pi}{P}+\alpha}^{\frac{\pi}{P}+\alpha} \left(\frac{1}{2} + \frac{\cos 2\omega t}{2} \right) d\omega t
 \end{aligned}$$

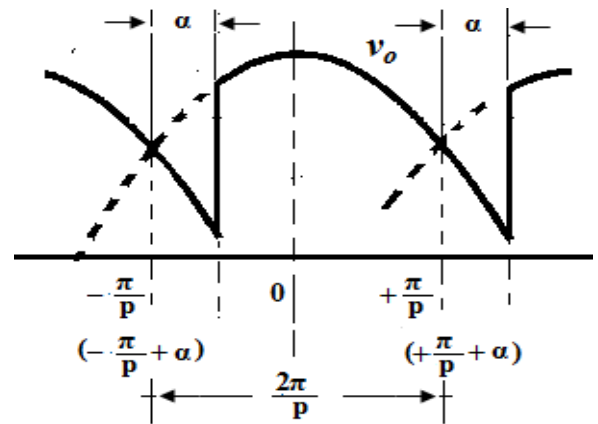


Fig. 3.45.

$$\begin{aligned}
&= \frac{P}{2\pi} V_m^2 \left[\frac{1}{2} \omega t + \frac{\sin 2\omega t}{4} \right] \left\{ \begin{array}{l} \frac{\pi}{P} + \alpha \\ -\frac{\pi}{P} + \alpha \end{array} \right. \\
&= \frac{P}{2\pi} V_m^2 \left\{ \begin{array}{l} \frac{1}{2} \frac{\pi}{P} + \frac{1}{2} \alpha + \frac{1}{4} \left[\sin 2\alpha \cos \frac{2\pi}{P} + \cos 2\alpha \sin \frac{2\pi}{P} \right] - \frac{1}{2} \alpha - \frac{1}{2} \frac{\pi}{P} \\ -\frac{1}{4} \left[\sin 2\alpha \cos \frac{2\pi}{P} - \cos 2\alpha \sin \frac{2\pi}{P} \right] \end{array} \right\} \\
&= \frac{P}{2\pi} V_m^2 \left[\frac{\pi}{P} + \frac{1}{2} \cos 2\alpha \sin \frac{2\pi}{P} \right] \\
\therefore V_{orms} &= \frac{V_m}{\sqrt{2}} \sqrt{1 + \frac{P}{2\pi} \cos 2\alpha \sin \left(\frac{2\pi}{P} \right)}
\end{aligned}$$

Example 3.22

Show that for a P -pulse converter with a purely resistive load, where the delay angle α is such that the load current is discontinuous, the *rms* value of the output voltage is given by

$$V_{orms} = V_s \sqrt{\frac{P}{4} - \frac{1}{2} \frac{\alpha}{\pi} P + \frac{1}{2} - \frac{1}{4} \frac{P}{\pi} \sin 2 \left(\alpha - \frac{\pi}{P} \right)}$$

where V_s is the *rms* value of the transformer secondary voltage per phase.

Solution:

The *rms* value of the output voltage waveform for a p -pulse converter is

$$\begin{aligned}
V_{orms}^2 &= \frac{P}{2\pi} \int_{-\frac{\pi}{P} + \alpha}^{\frac{\pi}{2}} (\sqrt{2} V_s)^2 \cos^2 \omega t d\omega t \\
&= \frac{P}{\pi} V_s^2 \left[\frac{1}{2} \omega t + \frac{\sin 2\omega t}{4} \right] - \frac{\pi}{P} + \alpha
\end{aligned}$$

$$= \frac{P}{\pi} V_s^2 \left[\frac{\pi}{4} - \frac{\alpha}{2} + \frac{\pi}{2p} - \frac{1}{4} \left(\sin 2\alpha \cos^2 \frac{2\pi}{p} - \cos 2\alpha \sin \frac{2\pi}{p} \right) \right]$$

$$\therefore V_{orms} = V_s \sqrt{\frac{P}{4} - \frac{1}{2} \frac{\alpha}{\pi} P + \frac{1}{2} - \frac{1}{4} \frac{P}{\pi} \sin 2 \left(\alpha - \frac{\pi}{p} \right)}$$

3.7 OVERLAP AND HARMONIC CONSIDERATIONS IN CONTROLLED AC- DC CONVERTORS

3.7.1 Principle of Overlap During Commutation

Overlap is the phenomenon due to the effect of source inductance on the a.c. side. The current commutation is delayed due to the source inductance which is normally the leakage reactance of a transformer (as $X \gg R$ for a transformer, the source resistance is usually neglected). Current commutation with diodes starts when the voltage of the incoming diode exceeds that of the outgoing diode. With thyristors, the commutation process starts by triggering the incoming thyristor while its voltage is higher than that of the outgoing one. Due to the inductive reactances of the source (including the reactances of the feeders, up to the rectifying element), certain time is required until the current transfer is completed. Note that, load inductance is not involved in this transfer.

In the following analyses, the source is represented by the equivalent phase reactances or inductances and emfs (v_1 and v_2). The current of the outgoing thyristor T_1 at the instant of triggering T_2 is I . The current of the incoming thyristor i_2 cannot jump immediately to I and T_1 cannot cease conduction. After triggering T_2 , both T_1 and T_2 are conducting, and i_2 starts increasing from zero up to I while i_1 starts decreasing from I down to zero, such that $i_1 + i_2 = I$, as depicted in Fig. 3.46. The period of this transition is called the overlap period of current during commutation. The angular period μ is known as the overlap angle or the commutation angle.

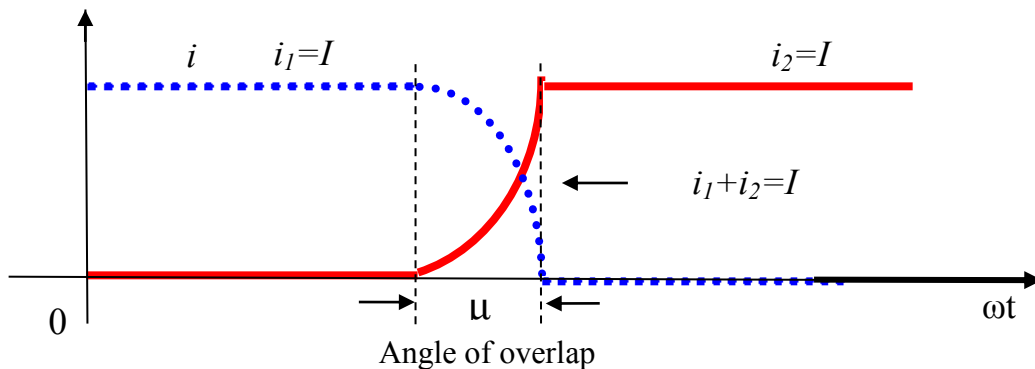


Fig.3.46 Overlap angle during commutation.

Effect of source inductance in single-phase fully-controlled converters

To generalize the problem, the following analyses are based on controlled rectifiers, and for uncontrolled rectifiers the final result and relation are simply obtained by setting $\alpha = 0$. Hence let us first consider the general case when two thyristors T_1 and T_2 encountering commutation process as depicted in Fig.3.47. If we assume that the load is highly inductive such that the load current is continuous and ripple free, then

Initial condition: $i_1 = I, i_2 = 0$

Final condition: $i_1 = 0, i_2 = I$

During overlap period: $i_1 + i_2 = I$

$$\text{or } \frac{di_1}{dt} + \frac{di_2}{dt} = 0$$

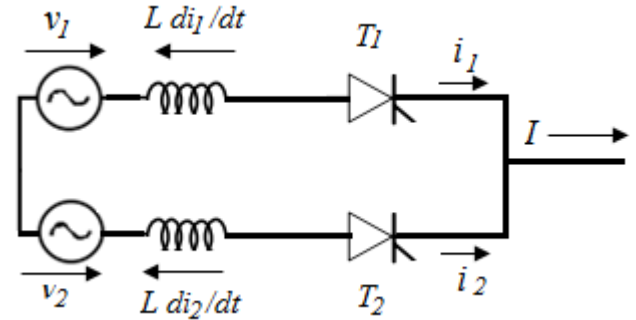


Fig.3.47.

and around the loop shown in Fig.3.47:

$$v_2 - v_1 = L \frac{di_2}{dt} - L \frac{di_1}{dt} = 2L \frac{di_2}{dt} \quad (3.43)$$

Equation (3.43) above will be used as basic relation for the following analysis for the overlap effect in rectifier circuit.

(A) Bi-phase (2-pulse) Converter

For the bi-phase controlled rectifier circuit shown in Fig.3.48,

Let $v_1 = -V_m \sin \omega t$, and $v_2 = V_m \sin \omega t$

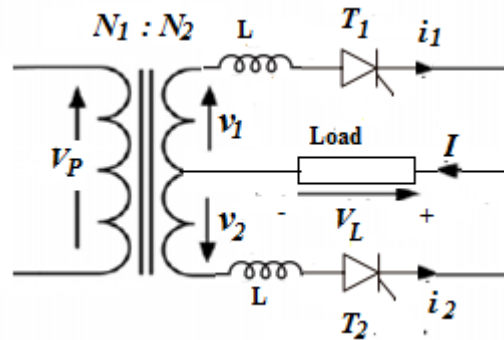


Fig.3.48.

During overlap period, the voltage waveforms will be as depicted in Fig.3.49, where,

$$v_L = \frac{v_1 + v_2}{2} = 0$$

$v_2 - v_1 = 2V_m \sin \omega t$; the forcing voltage, hence from Eq.(3.43):

$$L \frac{di_2}{dt} = V_m \sin \omega t$$

Integrating both sides:

$$\begin{aligned} \int_0^{i_2} di_2 &= \frac{1}{L} \int_{t=\frac{\alpha}{\omega}}^t V_m \sin \omega t d\omega t \\ &= \frac{V_m}{\omega L} \int_{\omega t=\alpha}^{\omega t} \sin \omega t d\omega t \\ i_2 &= \frac{V_m}{\omega L} (-\cos \omega t) \Big|_{\alpha/\omega}^t \\ &= \frac{V_m}{\omega L} (\cos \alpha - \cos \omega t) \end{aligned}$$

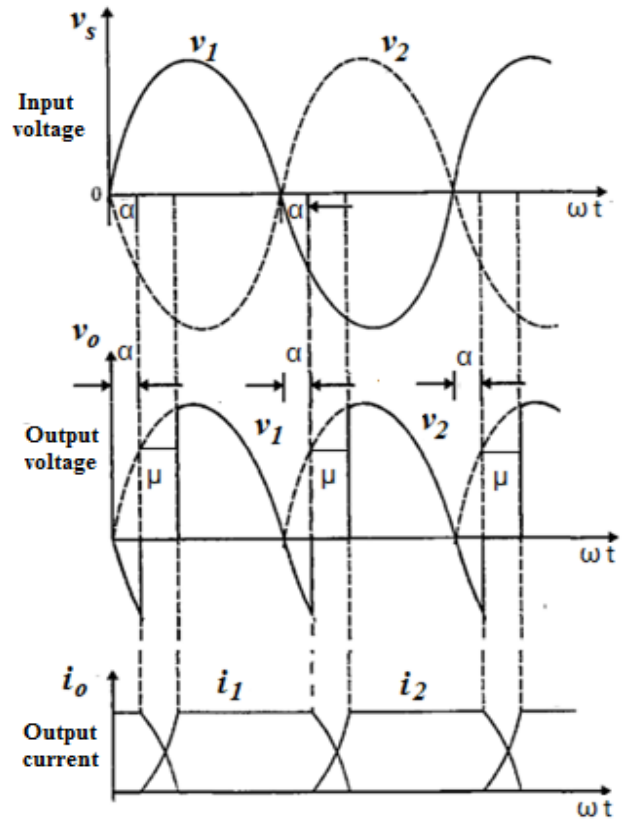


Fig.3.49 Waveforms.

$$i_1 = I - i_2 = I - \frac{V_m}{\omega L} (\cos \alpha - \cos \omega t)$$

From the final conditions: At $\omega t = \alpha + \mu$, $i_2 = I, i_1 = 0$

Therefore,

$$\cos \alpha - \cos(\alpha + \mu) = \frac{\omega L I}{V_m} \quad (3.44)$$

The average output voltage V_{dc} is:

$$\begin{aligned} V_{dc} &= \frac{1}{\pi} \left[\int_{\alpha}^{\alpha+\mu} \left(\frac{v_1 + v_2}{2} \right) d\omega t + \int_{\alpha+\mu}^{\pi+\alpha} v_2 d\omega t \right] \\ &= \frac{1}{\pi} \left[\int_{\alpha+\mu}^{\pi+\alpha} V_m \sin \omega t d\omega t \right] \\ &= \frac{V_m}{\pi} (\cos \alpha + \cos(\alpha + \mu)) \end{aligned}$$

Using Eq.(3.44) V_{dc} can be expressed as

$$V_{dc} = \frac{2V_m}{\pi} \cos\alpha - \frac{\omega LI}{\pi} = \frac{2V_m}{\pi} \cos(\alpha + \mu) + \frac{\omega LI}{\pi} \quad (3.45)$$

This is the output d.c. voltage of the converter considering the effect of the source inductance L which causes overlap. The effect of overlap is, then equivalent to voltage loss of $(\omega LI / \pi)$. This lost voltage can also be determined by calculating the area A which occurs every half cycle or every $\frac{\pi}{\omega}$ second. Thus referring to Fig.3.50, A can be determined as:

$$\begin{aligned} A &= \int_{t=\frac{\alpha}{\omega}}^{(\alpha+\mu)/\omega} \left(v_2 - \frac{v_1 + v_2}{2} \right) dt \\ &= \int_{t=\frac{\alpha}{\omega}}^{(\alpha+\mu)/\omega} \left(\frac{v_2 - v_1}{2} \right) dt \end{aligned}$$

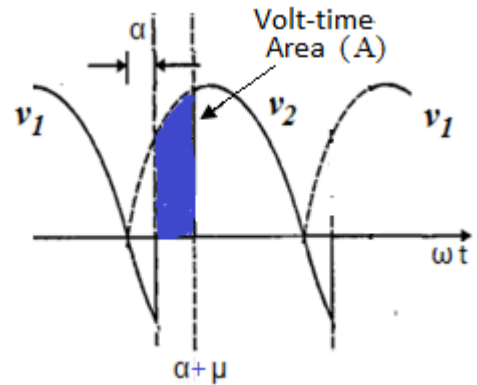


Fig.3.50 Volt-time area.

since $\frac{v_1 + v_2}{2} = L \frac{di_2}{dt}$, then

$$A = \int_{t=\frac{\alpha}{\omega}}^{(\alpha+\mu)/\omega} L di_2 = L(I - 0) = LI$$

The average lost voltage during commutation is,

$$V_{dc(lost)} = \frac{A}{\pi/\omega} = \frac{LI}{\pi/\omega} = \frac{\omega LI}{\pi} \quad (3.46)$$

The net output d.c. voltage for the bi-phase converter will be:

$$V_{dc} = V_{dc}(\text{without } L) - V_{dc(lost)} = \frac{2V_m}{\pi} \cos\alpha - \frac{\omega LI}{\pi} \quad (3.47)$$

(B) Single-phase full-wave bridge rectifier

For the single-phase full-wave bridge rectifier, the equivalent circuit during overlap is shown in Fig.3.51. The supply current i_s is common between T_1 and T_3 . Hence, by KVL;

$$v_s = V_m \sin \omega t = L \frac{di_s(t)}{dt}$$

Initial condition: $i_s = I$,

Final condition: $i_s = -I$

$$V_m \sin \omega t dt = L di_s(t)$$

Integrating both the sides :

$$\int_{\alpha}^{\alpha+\mu} V_m \sin \omega t dt = \int_I^{-I} L di_s(t) = -2LI$$

$$= \frac{V_m}{\omega} (-\cos \alpha + \cos(\alpha + \mu)) = -2LI$$

$$\cos \alpha - \cos(\alpha + \mu) = \frac{2\omega LI}{V_m} = V_{dc(lost)}$$

Therefore, the net output d.c. voltage for the single-phase converter will be:

$$V_{dc} = V_{dc}(\text{without } L) - V_{dc(lost)} = \frac{2V_m}{\pi} \cos \alpha - \frac{2\omega LI}{\pi} \quad (3.48)$$

(C) Three-phase half-wave rectifier

For the three-phase half-wave controlled rectifier circuit shown in Fig.3.30, the output voltage waveform considering overlap angle is depicted in Fig.3.52. Now let :

$$v_1 = V_m \sin \omega t, \quad v_2 = V_m \sin(\omega t + 120^\circ)$$

$$\text{and } v_3 = V_m \sin(\omega t - 120^\circ)$$

Hence,

$$\begin{aligned} v_1 + v_2 &= V_m \sin(\omega t + 60^\circ) \\ &= V_m \cos(\omega t + 60^\circ - 90^\circ) = V_m \cos(\omega t - 30^\circ) \end{aligned}$$

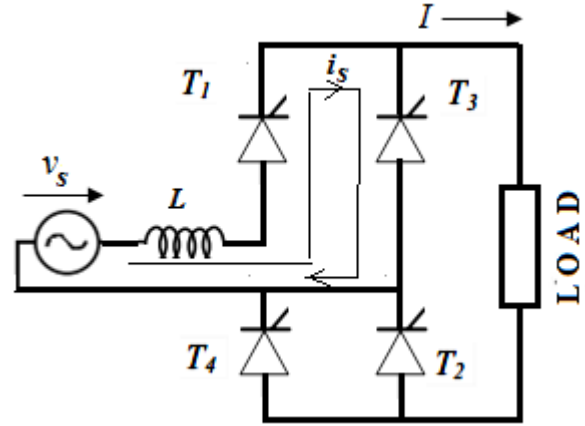


Fig.3.51.

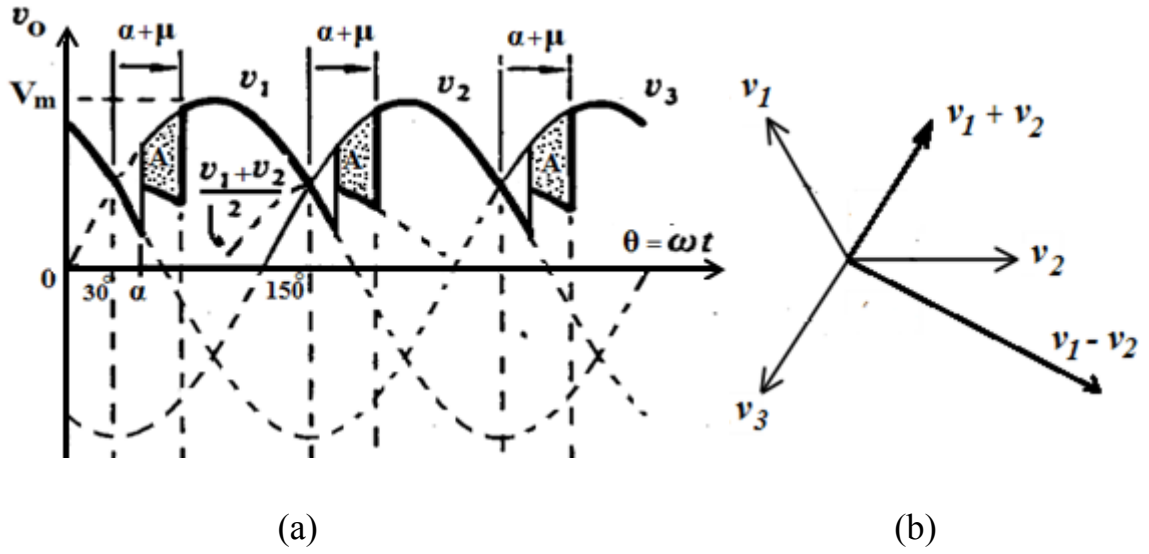


Fig.3.52 Three-phase half-wave controlled rectifier considering overlap:
(a) Output voltage waveform, (b) Voltage phasor diagram.

$$v_1 - v_2 = \sqrt{3}V_m \sin \omega t$$

Using Eq.(3.43) gives :

$$L \frac{di_2}{dt} = \frac{\sqrt{3}}{2} V_m \sin \omega t \quad (3.49)$$

Taking integration of both sides of Eq.(3.49) yields

$$\int_0^{i_2} di_2 = \int_{t=\frac{\alpha}{\omega}}^t \frac{\sqrt{3}}{2L} V_m \sin \omega t dt$$

$$i_2 = \frac{\sqrt{3}V_m}{2\omega L} (\cos \alpha - \cos \omega t) \quad (3.50)$$

Now, at $\omega t = \alpha + \mu$, $i_2 = I = (I_{dc})$ and $i_1 = 0$

$$\cos \alpha - \cos(\alpha + \mu) = \frac{2\omega LI}{\sqrt{3}V_m} \quad (3.51)$$

$$V_{dc} = \frac{1}{T} \int_0^T V_L dt = \frac{1}{2\pi} \left[\int_{30^\circ+\alpha}^{30^\circ+\alpha+\mu} \left(\frac{v_1 + v_2}{2} \right) d\omega t + \int_{30^\circ+\alpha+\mu}^{150^\circ+\alpha} v_2 d\omega t \right]$$

Using $\phi = \omega t - 30^\circ$ and $d\phi = d\omega t$ gives,

$$V_{dc} = \frac{3}{2\pi} \left[\int_{\phi=\alpha}^{\alpha+\mu} \frac{V_m}{2} \cos\phi \, d\phi + \int_{\omega t=30^\circ+\alpha+\mu}^{150^\circ+\alpha} v_2 \, d\omega t \right]$$

$$= \frac{3\sqrt{3}}{4\pi} V_m (\cos\alpha + \cos(\alpha + \mu))$$

With the aid of Eq.(3.51):

$$V_{dc} = \frac{3\sqrt{3}}{2\pi} V_m \cos\alpha - \frac{3\omega LI}{2\pi}$$

$$= \frac{3\sqrt{3}}{2\pi} V_m \cos(\alpha + \mu) + \frac{3\omega LI}{2\pi} \quad (3.52)$$

(D) P-pulse converter

For p-pulse converter the voltage waveforms and the overlap angles are depicted in Fig.3.53. Let the voltage at the anode of the incoming thyristor be $v_2 = V_m \cos \theta$. Since the phase voltages are displaced by $\frac{2\pi}{p}$ radians, then, the voltage at the anode of the outgoing thyristor v_1 is given by:

$$v_1 = V_m \cos\left(\theta + \frac{2\pi}{p}\right)$$

$$v_2 - v_1 = V_m \left[\cos\theta - \cos\left(\theta + \frac{2\pi}{p}\right) \right] = 2V_m \sin\frac{\pi}{p} \sin\left(\theta + \frac{\pi}{p}\right)$$

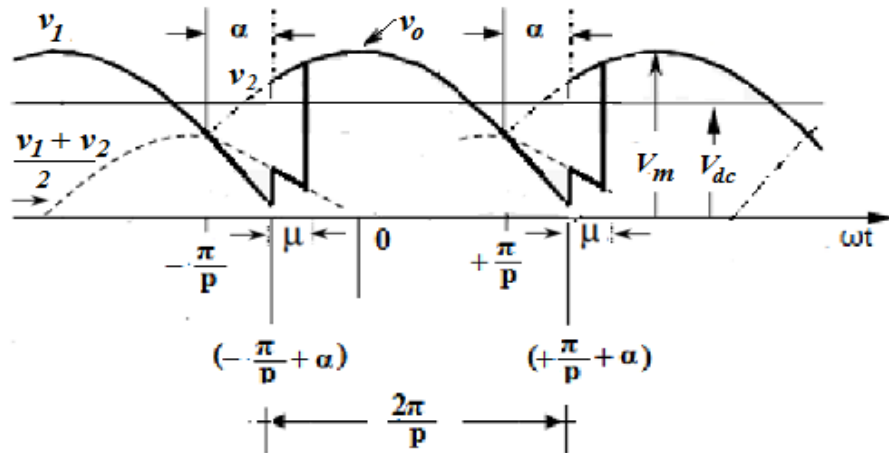


Fig.3.53

Using $\phi = \theta + \frac{\pi}{p} = \omega t + \frac{\pi}{p}$; $d\phi = \omega dt$

From Eq.(3.43) :

$$\frac{v_2 - v_1}{2} = V_m \sin \frac{\pi}{p} \sin \phi = L \frac{di_2}{dt}$$

The overlap starts with $\phi = \alpha$ with $i_2 = 0$ and it lasts at $\phi = \alpha + \mu$ with $i_2 = I$. Therefore, the average d.c. output voltage

$$L \int_0^{i_2} di_2 = \int_{\alpha}^{\alpha+\mu} V_m \sin \frac{\pi}{p} \sin \phi \frac{d\phi}{\omega}$$

$$\omega LI = V_m \sin \frac{\pi}{p} [-\cos \phi]_{\alpha}^{\alpha+\mu}$$

$$\cos \alpha - \cos(\alpha + \mu) = \frac{\omega L I}{V_m \sin \frac{\pi}{p}} \quad (3.53)$$

And the average d.c. output voltage can be evaluated as,

$$\begin{aligned} V_{dc} &= \frac{p}{2\pi} \left[\int_{-\frac{\pi}{p}+\alpha}^{-\frac{\pi}{p}+\alpha+\mu} \left(\frac{v_2 - v_1}{2} \right) d\phi + \int_{-\frac{\pi}{p}+\alpha+\mu}^{\frac{\pi}{p}+\alpha} v_2 d\omega t \right] \\ &= \frac{p}{2\pi} \left[\int_{\alpha}^{\alpha+\mu} V_m \cos \frac{\pi}{p} \cos \phi d\phi + \int_{-\frac{\pi}{p}+\alpha+\mu}^{\frac{\pi}{p}+\alpha} V_m \cos \theta d\omega t \right] \\ &= \frac{p}{2\pi} V_m \sin \frac{\pi}{p} [\cos \alpha + \cos(\alpha + \mu)] \end{aligned}$$

$$\therefore V_{dc} = \frac{p}{\pi} V_m \sin \frac{\pi}{p} \cos \alpha - \frac{p\omega L I}{2\pi} \quad (3.54)$$

$$\text{or } V_{dc} = \frac{p}{\pi} V_m \sin \frac{\pi}{p} \cos(\alpha + \mu) - \frac{p\omega L I}{2\pi} \quad (3.55)$$

Alternatively, one can use the lost voltage due to commutation to derive V_{dc} . $A=LI$ (as derived before) and its average value $V_{dc(lost)}$ for p -pulse converter is thus,

$$V_{dc(lost)} = \frac{A}{\left(\frac{2\pi}{p\omega}\right)} = \frac{p\omega L I}{2\pi}$$

Therefore the net output d.c. voltage for the p -pulse converter will be:

$$\begin{aligned} V_{dc} &= V_{dc}(\text{without } L) - V_{dc(lost)} \\ &= \frac{p}{\pi} V_m \sin \frac{\pi}{p} \cos \alpha - \frac{p\omega L I}{2\pi} \end{aligned} \quad (3.56)$$

Notes:

1. $\omega L = X$ is the supply reactance per phase (up to the anode of the thyristor) and $\frac{\omega L}{2\pi} = f L$ where f is the frequency of the supply.
2. For a given frequency and input voltage (V_m) the overlap angle μ depends upon:
 - α : the triggering angle.
 - L : the leakage inductance of the supply
 - I : the load current or the d.c. side current
3. Minimum overlap angle occurs when the triggering happens near the maximum voltage differences between the phases (v_2-v_1).
4. With a three-phase bridge, the overlap occurs every 60° and the lost voltage is $\frac{A}{\left(\frac{2\pi}{6\omega}\right)}$ or $\frac{3\omega L I}{\pi}$. With a single-phase bridge rectifier, the overlap is equivalent to that taking place every 90° or $\pi/2$ and the lost voltage is $\frac{A}{\left(\frac{2\pi}{4}\right)}$ or $\frac{2\omega L}{I}$. Hence, the output of bridge rectifier is given by (let $I = I_{dc}$)

For bi-phase rectifier

$$V_{dc} = \frac{2 V_m}{\pi} \cos \alpha - \frac{\omega L}{\pi} I_{dc}$$

For single-phase bridge

$$V_{dc} = \frac{2 V_m}{\pi} \cos \alpha - \frac{2\omega L}{\pi} I_{dc}$$

For three-phase bridge rectifier

$$V_{dc} = \frac{3\sqrt{3} V_m}{2\pi} \cos\alpha - \frac{3\omega L}{2\pi} I_{dc}$$

For a bi-phase rectifier

$$\cos\alpha - \cos(\alpha + \mu) = \frac{\omega L}{V_m} I_{dc}$$

For a single-phase bridge rectifier

$$\cos\alpha - \cos(\alpha + \mu) = \frac{2\omega L}{V_m} I_{dc}$$

which is double of a bi-phase rectifier.

For a three-phase half-wave rectifier

$$\cos\alpha - \cos(\alpha + \mu) = \frac{2\omega L}{\sqrt{3} V_m} I_{dc}$$

For a three-phase bridge rectifier

$$\cos\alpha - \cos(\alpha + \mu) = \frac{2\omega L}{\sqrt{3} V_m} I_{dc}$$

This is the same as a three-phase half-wave rectifier but of double occurrence.

For p-pulse converter

$$\cos\alpha - \cos(\alpha + \mu) = \frac{\omega L}{V_m \sin \frac{\pi}{p}} I_{dc}$$

Example 3.23

A single-phase mid-point (bi-phase) converter supplied from 240/120 V, 50Hz transformer is connected to load of 15 Ω resistance and infinite inductance. If the secondary winding of the transformer has a line inductance of 20 mH. Determine the load voltage and current at a firing angle of 60°. Find the overlap angle.

Solution

For bi-phase or mid-point rectifier

$$V_{dc} = \frac{2 V_m}{\pi} \cos \alpha - \frac{\omega L I_{dc}}{\pi}$$

$$\text{or } R I_{dc} = \frac{2 V_m}{\pi} \cos \alpha - \frac{\omega L}{\pi} I_{dc}$$

$$I_{dc} \left[R + \frac{\omega L}{\pi} \right] = \frac{2 V_m}{\pi} \cos \alpha$$

$$I_{dc} = \frac{\frac{2 V_m}{\pi} \cos \alpha}{\left[R + \frac{\omega L}{\pi} \right]} = \frac{\frac{2 \times 120 \times \sqrt{2}}{\pi}}{15 + \frac{100\pi \times 20}{1000\pi}} = 3.17 \text{ A}$$

$$V_{dc} = R I_{dc} = 15 \times 3.17 = 47.6 \text{ V}$$

For bi-phase rectifier

$$\cos \alpha - \cos(\alpha + \mu) = \frac{\omega L}{V_m} I_{dc}$$

$$\begin{aligned} \cos(\alpha + \mu) &= -\frac{\omega L}{V_m} I_{dc} + \cos \alpha \\ &= -\frac{2\pi \times 50 \times 20 \times 10^{-3}}{169.7} \times 3.17 + \cos 60^\circ \end{aligned}$$

$$\cos(\alpha + \mu) = -0.1173 + 0.5 = 0.3886$$

$$\alpha + \mu = 67.5^\circ \quad \therefore \mu = 67.5^\circ - \alpha = 67.5^\circ - 60^\circ = 7.5^\circ$$

Example 3.24

A three-phase, half-wave controlled rectifier shown in Fig.3.54 is supplied by 150 V per phase, at 50 Hz. The source inductance and resistance are 1.2 mH and 0.07 Ω per phase respectively. Assuming a thyristor voltage drop of 1.5 V and a continuous load current of 30 A, determine the mean load voltage at firing angles of 0°, 30° and 60°.

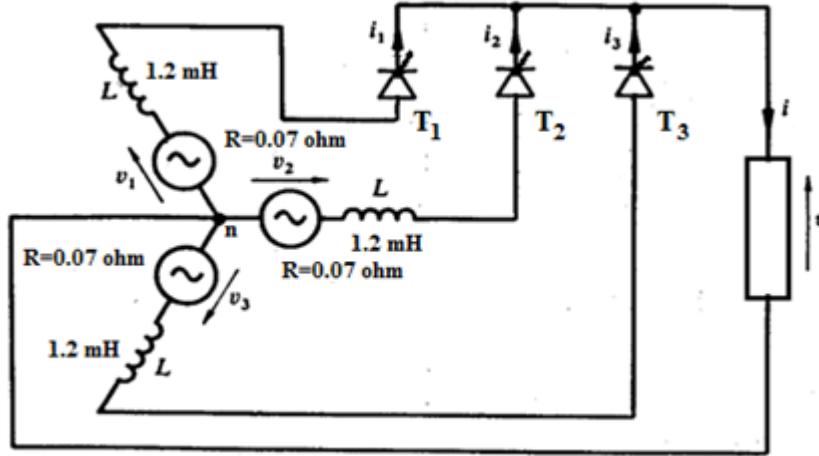


Fig.3.54.

Solution

At all time there is a voltage drop across the thyristor which is 1.5 V, also there is a constant voltage drop across the source resistance which is equal to: $30 \times 0.07 = 2.1$ V.

$$V_{dc} = \frac{3\sqrt{3}}{2\pi} V_m \cos \alpha - \frac{3\omega L}{2\pi} I_{dc}$$

$$\frac{3\omega L}{2\pi} I_{dc} = \frac{3 \times 2\pi \times 50 \times 1.2 \times 10^{-3}}{2\pi} = 5.4 \text{ V}$$

$$\frac{3\sqrt{3}}{2\pi} V_m = \frac{3\sqrt{3} \times 150 \times \sqrt{2}}{2\pi} = 175.4$$

at

$$\alpha = 0^\circ \quad V_d = 175.4 \cos 0^\circ - 1.5 - 2.1 - 5.4 = 166 \text{ V}$$

$$\alpha = 30^\circ \quad V_d = 175.4 \cos 30^\circ - 1.5 - 2.1 - 5.4 = 142.9 \text{ V}$$

$$\alpha = 60^\circ \quad V_d = 175.4 \cos 60^\circ - 1.5 - 2.1 - 5.4 = 78.7 \text{ V}$$

Example 3.25

A three-phase fully-controlled thyristor bridge rectifier is supplied from three-phase source of 400 V, 50 Hz with an inductive reactance of 0.25 Ω per phase and negligible resistance. The output is smoothed by an inductor which has a resistance of 0.15 Ω and the load current is 200 A. Determine the mean bridge output voltage, the mean load voltage and the overlap angle when the delay angle is 45° . Ignore any voltage drop in the thyristors.

Solution

From Eq.(3.55), for p-pulse converter :

$$V_{dc} = \frac{p}{\pi} V_m \sin \frac{\pi}{p} \cos \alpha - \frac{p \omega L I}{2\pi}$$

For three-phase fully-controlled thyristor bridge rectifier $p = 6$, hence

$$V_{dc} = \frac{6}{\pi} V_m \sin \frac{\pi}{6} \cos \alpha - \frac{6 \omega L I}{2\pi}$$

$$V_{dc} = \frac{6}{\pi} V_m \frac{\sqrt{3}}{2} \cos \alpha - \frac{3 \omega L I}{\pi}$$

$$V_m = \sqrt{2} \frac{400}{\sqrt{3}} = 326.6 \text{ V} \rightarrow \sqrt{3} V_m = \sqrt{2} \times 400$$

$$V_{dc} = \frac{6}{\pi} \times \sqrt{2} \frac{400}{\sqrt{3}} \times \frac{\sqrt{3}}{2} \cos 45^\circ - \frac{3 \times 0.25 \times 200}{\pi} = 334.28 \text{ V}$$

$$V'_{dc} = 334.28 - 200 \times 0.15 = 304.28 \text{ V}$$

$$V_{dc} = \frac{P V_m}{2\pi} \sin \frac{\pi}{P} (\cos \alpha + \cos(\alpha + \mu))$$

$$334.28 = \frac{6}{2\pi} \times \sqrt{2} \times 400 \times \frac{1}{2} (\cos \alpha + \cos(\alpha + \mu))$$

$$334.28 = 270.2 (\cos \alpha + \cos(\alpha + \mu))$$

$$\cos \alpha + \cos(\alpha + \mu) = 1.237$$

$$\cos(45^\circ + \mu) = 1.237 - 0.707 = 0.530$$

$$45^\circ + \mu = \cos^{-1} 0.530 = 57.98^\circ$$

$$\therefore \mu = 57.98^\circ - 45^\circ = 12.98^\circ$$

3.7.2 Harmonic Considerations in Controlled P-Pulse ac-to-dc Converter – General Solution

The d.c. output voltage waveform of a p-pulse controlled converter may be represented by Fourier series as

$$v(\omega t) = a_o + \sum_{n=2}^{\infty} a_n \cos(n\omega t) + \sum_{n=2}^{\infty} b_n \sin(n\omega t)$$

Referring to the waveform of p-pulse converter shown in Fig. 3.34, the Fourier coefficients can be evaluated as follows:

$$a_o = V_{dc} = \frac{1}{\frac{2\pi}{p}} \int_{(-\frac{\pi}{p}+\alpha)}^{(+\frac{\pi}{p}+\alpha)} V_m \cos \omega t d\omega t$$

$$a_o = \frac{p}{\pi} V_m \sin \frac{\pi}{p} \cos \alpha = V_{dc} \cos \alpha \quad (3.57)$$

The Fourier coefficients of the n^{th} order harmonics are (for $n \neq 1$):

$$a_n = \frac{1}{\frac{2\pi}{p}} \int_{(-\frac{\pi}{p}+\alpha)}^{(+\frac{\pi}{p}+\alpha)} V_m \cos \omega t \cos n\omega t d\omega t$$

$$b_n = \frac{1}{\frac{2\pi}{p}} \int_{(-\frac{\pi}{p}+\alpha)}^{(+\frac{\pi}{p}+\alpha)} V_m \cos \omega t \sin n\omega t d\omega t$$

Evaluating this integration by the same method as that given in chapter two for uncontrolled case we get,

$$a_n = \frac{2V_{dc}}{(n^2 - 1)} [n \sin n\alpha \sin \alpha + \cos n\alpha \cos \alpha] \quad (3.58)$$

$$b_n = \frac{2V_{dc}}{(n^2 - 1)} [n \cos n\alpha \sin \alpha + \sin n\alpha \cos \alpha] \quad (3.59)$$

where :

V_{dc} = d.c. output value for uncontrolled case

$n = cp$ with c any integer

V_m = the maximum value of converter secondary voltage /phase

α = the triggering (delay) angle

The fundamental component coefficients a_1 , b_1 and c_1 (when $n = 1$) are:

$$\begin{aligned}
 a_1 &= \frac{V_m}{\pi} \int_{\frac{-\pi}{p}+\alpha}^{\frac{\pi}{p}+\alpha} \sin(\omega t) \cos(\omega t) d\omega t \\
 &= \frac{V_m}{\pi} \int_{\frac{-\pi}{p}+\alpha}^{\frac{\pi}{p}+\alpha} \frac{1}{2} \sin(2\omega t) d\omega t \\
 &= \frac{V_m}{4\pi} [-\cos(2\omega t)]_{\frac{-\pi}{p}+\alpha}^{\frac{\pi}{p}+\alpha} \\
 &= \frac{V_m}{4\pi} [-\cos(\frac{2\pi}{p} + 2\alpha) + \cos(-(\frac{2\pi}{p} + 2\alpha))] \\
 &= \frac{V_m}{4\pi} [(\cos(\frac{2\pi}{p}) \cos(2\alpha) - \sin(\frac{2\pi}{p}) \sin(2\alpha)) \\
 &\quad + \cos(-\frac{2\pi}{p}) \cos(2\alpha) - \sin(-\frac{2\pi}{p}) \sin(2\alpha)]
 \end{aligned}$$

After further simplification

$$\therefore a_1 = \frac{V_m}{2\pi} \left[\sin\left(\frac{2\pi}{p}\right) \sin(2\alpha) \right] \quad (3.60)$$

$$\begin{aligned}
 b_1 &= \frac{V_m}{\pi} \int_{\frac{-\pi}{p}+\alpha}^{\frac{\pi}{p}+\alpha} \sin(\omega t) \sin(\omega t) d\omega t \\
 &= \frac{V_m}{\pi} \int_{\frac{-\pi}{p}+\alpha}^{\frac{\pi}{p}+\alpha} \sin^2(\omega t) d\omega t = \frac{V_m}{2\pi} \int_{\frac{-\pi}{p}+\alpha}^{\frac{\pi}{p}+\alpha} [1 - \cos(2\omega t)] d\omega t \\
 &= \frac{V_m}{2\pi} [\omega t - \frac{1}{2} \sin(2\omega t)]_{\frac{-\pi}{p}+\alpha}^{\frac{\pi}{p}+\alpha} \\
 &= \frac{V_m}{2\pi} \left[\frac{\pi}{p} + \alpha + \frac{\pi}{p} - \alpha - \frac{\sin(\frac{2\pi}{p} + 2\alpha)}{2} + \frac{\sin(-\frac{2\pi}{p} + 2\alpha)}{2} \right] \\
 &= \frac{V_m}{2\pi} \left[\frac{2\pi}{p} - \frac{1}{2} (\sin(\frac{2\pi}{p} + 2\alpha) - \sin(2\alpha - \frac{2\pi}{p})) \right]
 \end{aligned}$$

$$\begin{aligned}
&= \frac{V_m}{2\pi} \left[\frac{2\pi}{p} - \frac{1}{2} \left(\sin\left(\frac{2\pi}{p}\right) \cos(2\alpha) + \sin(2\alpha) \cos\left(\frac{2\pi}{p}\right) \right. \right. \\
&\quad \left. \left. - \sin(2\alpha) \cos\left(\frac{2\pi}{p}\right) + \sin\left(\frac{2\pi}{p}\right) \cos(2\alpha) \right) \right] \\
\therefore b_1 &= \frac{V_m}{2\pi} \left[\frac{2\pi}{p} - \sin\left(\frac{2\pi}{p}\right) \cos(2\alpha) \right] \quad (3.61)
\end{aligned}$$

The amplitude c_1 of the fundamental component is, $c_1 = \sqrt{a_1^2 + b_1^2}$

3.7.3 Harmonic Amplitude Spectra of the Output Voltage Waveform for P-Pulse Controlled Rectifiers

(i) Single-phase full-wave controlled rectifier (p=2)

Harmonic amplitude spectra of the output voltage waveform for single-phase full-wave rectifier (p = 2) is shown in Fig.3.55 for triggering angle $\alpha = 30^\circ$. It is clear that the zero frequency component (the d.c. value which has an amplitude of 55.1 percent) is not the dominant one in the frequency spectrum. The second order harmonic ($f=100\text{Hz}$) has amplitude of 66.22 percent, whereas the fundamental component ($f=50\text{Hz}$) has amplitude of 50.01 percent. For all higher order harmonic frequency components, the harmonic amplitude c_n indicates that the output voltage waveform contains both even and odd harmonics. The values of the harmonic amplitudes c_n in percentage ($V_m=100\%$) are given in Table 3.1 for convenience.

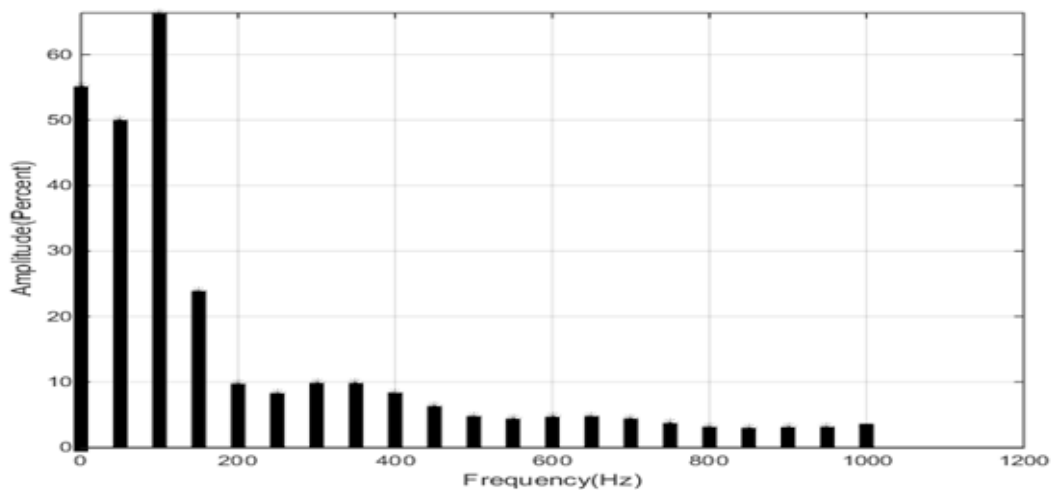


Fig.3.55 Harmonic amplitude spectra for the output voltage waveform of single-phase full-wave rectifier (p = 2) for triggering angle $\alpha = 30^\circ$.

Table 3.1 Harmonic amplitudes for the output voltage waveform of single - phase full-wave rectifier ($p = 2$) for triggering angle $\alpha = 30^\circ$.

n	0	1	2	4	6	8	10	12	16	20
Hz	0	50	100	200	300	400	500	600	800	1000
c(n)	55.10399	50.01005	66.22861	9.716907	9.841967	8.344014	4.75539	4.677029	3.138863	2.971708

(ii) Three -phase half-wave controlled rectifier ($p= 3$)

Harmonic amplitude spectra of the output voltage waveform for three-phase half-wave rectifier ($p = 3$) is shown in Fig.3.56 for triggering angle $\alpha = 30^\circ$.

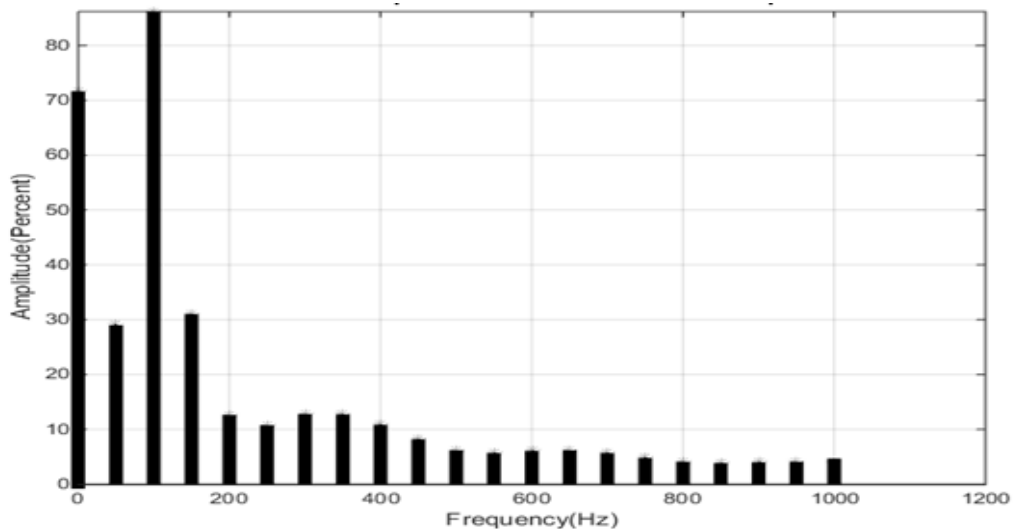


Fig. 3.56 Harmonic amplitude spectra for three-phase half-wave rectifier ($p = 3$) for triggering angle $\alpha = 30^\circ$.

It is clear that the zero frequency component (the d.c. value which has an amplitude of 71.59 percent) is not the dominant one in the frequency spectrum. The second order harmonic ($f = 100$ Hz) has amplitude of 86.05 percent, whereas the fundamental component ($f = 50$ Hz) has amplitude of 29.01 percent. For all higher order harmonic frequency components, the harmonic amplitude c_n indicates that the output voltage waveform contains both even and odd harmonics.

The values of the harmonic amplitudes c_n in percentage ($V_m = 100\%$) are given in Table 3.2 for convenience.

Table 3.2 Harmonic amplitudes for the output voltage waveform three - phase half - wave rectifier ($p = 3$) for triggering angle $\alpha = 30^\circ$.

n	0	1	2	4	6	8	10	12	16	20
Hz	0	50	100	200	300	400	500	600	800	1000
c(n)	71.59961	29.01816	86.05443	12.62571	12.7882	10.84183	6.178937	6.077118	4.078496	3.861302

(iii) Three-phase full-wave fully-controlled rectifier (p= 6)

Harmonic amplitude spectra of the output voltage waveform for three-phase full-wave rectifier (p = 6) is shown in Fig.3.57 for triggering angle $\alpha = 30^\circ$. It is clear that the zero frequency component (the d.c. value which has an amplitude of 82.68 percent) is not the dominant one in the frequency spectrum. The second order harmonic ($f=100\text{Hz}$) has amplitude of 99.37 percent, whereas the fundamental component ($f=50\text{Hz}$) has amplitude of 15.43 percent. For all higher order harmonic frequency components, the harmonic amplitude c_n indicates that the output voltage waveform contains both even and odd harmonics.

The values of the harmonic amplitudes c_n in percentage ($V_m=100\%$) are given in Table 3.3.

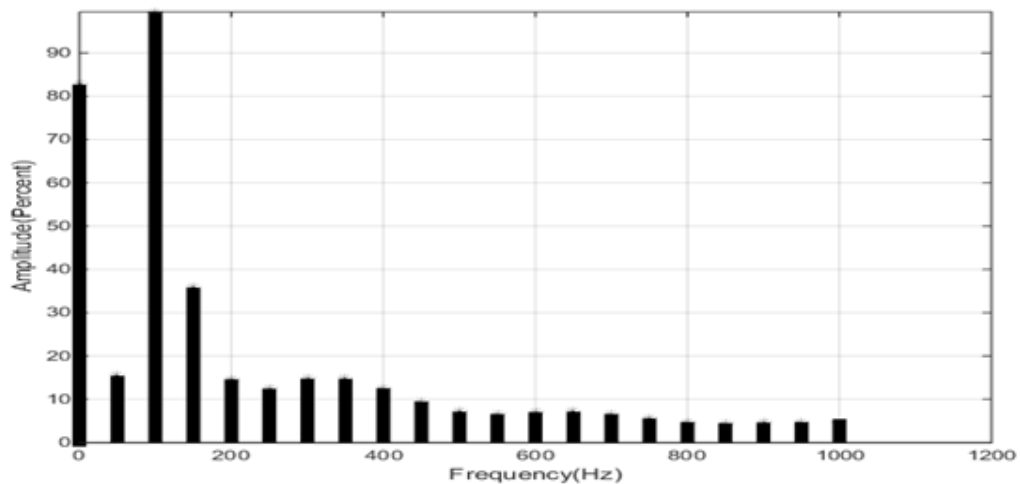


Fig. 3.57 Harmonic amplitude spectra for three-phase full-wave rectifier (p = 6) for triggering angle $\alpha = 30^\circ$.

Table 3.3 Harmonic amplitudes for the output voltage waveform three-phase full-wave rectifier (p = 6) for triggering angle $\alpha = 30^\circ$.

n	0	1	2	4	6	8	10	12	16	20
Hz	0	50	100	200	300	400	500	600	800	1000
c(n)	82.68617	15.43303	99.37919	14.58068	14.76834	12.52059	7.13569	7.018106	4.710015	4.45919

(iv) Poly-phase 12-pulse rectifier (p=12)

Harmonic amplitude spectra of the output voltage waveform for twelve-phase full-wave rectifier (p = 12) is shown in Fig.3.58 for triggering angle $\alpha = 30^\circ$. It is clear that the zero frequency component (the d.c. value which has an amplitude of 82.68 percent) is not the dominant one in the frequency spectrum. The second order harmonic ($f=100\text{Hz}$) has amplitude of 99.37 percent, whereas the fundamental component ($f=50\text{ Hz}$) has

amplitude of 15.43 percent. For all higher order harmonic frequency components, the harmonic amplitude c_n indicates that the output voltage waveform contains both even and odd harmonics. The values of the harmonic amplitudes c_n in percentage ($V_m=100\%$) are given in Table 3.4.

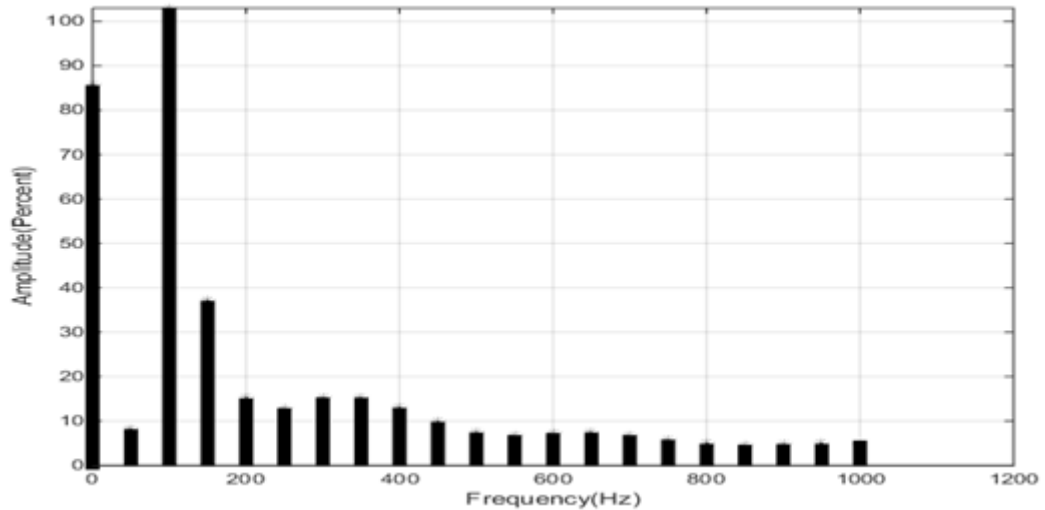


Fig. 3.58 Harmonic amplitude spectra for poly-phase 12- pulse rectifier for triggering angle $\alpha = 30^\circ$.

Table 3.4 Harmonic amplitudes for the output voltage waveform poly-phase rectifier ($p = 12$) for triggering angle $\alpha = 30^\circ$.

n	0	1	2	4	6	8	10	12	16	20
Hz	0	50	100	200	300	400	500	600	800	1000
c(n)	85.60544	8.154862	102.8878	15.09546	15.28974	12.96264	7.387619	7.265883	4.876304	4.616623

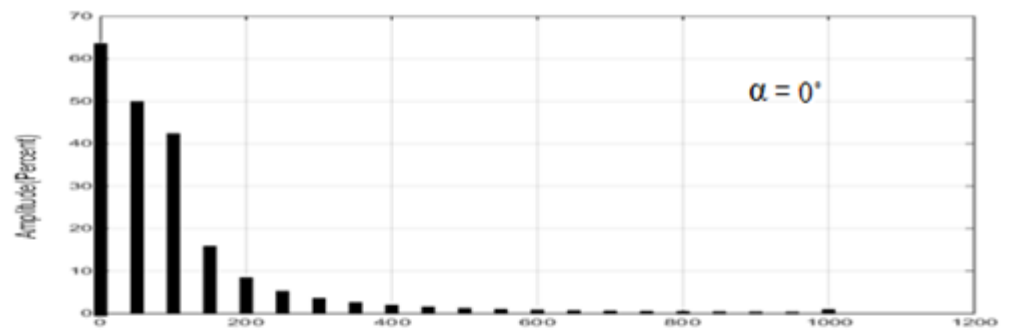
From the above results, it is evident that as the number of pulse (p) in an a.c. to d.c. converter increases, the d.c. output voltage is increases. The most significant harmonic is the second harmonic of the supply frequency which, for the same delay angle α , increases as the number of pulse p increases. On contrast, the first order harmonic is decreases as the number of pulse p increases. The variation of the d.c. component (V_{dc}) with the delay angle α is given in Table 3.5 for clarity.

Figs.3.59 to 62 show harmonic spectra for rectifiers with different number of pulses ($p = 2,3,6$, and 12) with $V_m = 100$ and $0 \leq \alpha \leq 180^\circ$,

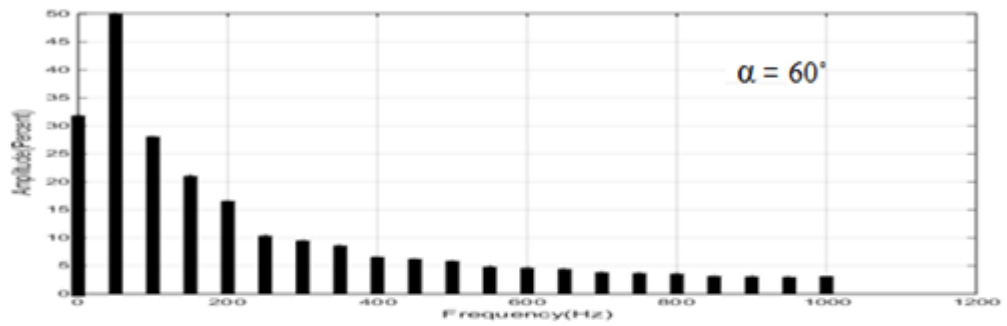
Table 3.5

Variation of V_{dc} with delay angle α for $p=2,3,6$, and 12 pulse.

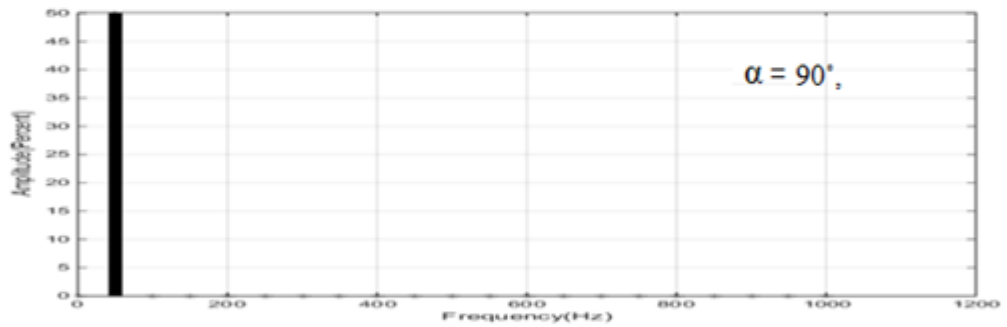
P	2						
α	0	30°	60°	90°	120°	150°	180°
$V_{dc}(V)$	63.6	55.1	31.8	0	-31.8	-55.1	-63.6
P	3						
α	0	30°	60°	90°	120°	150°	180°
$V_{dc}(V)$	82.6	71.6	41.3	0	-41.3	-71.6	-82.6
P	6						
α	0	30°	60°	90°	120°	150°	180°
$V_{dc}(V)$	95.5	82.6	47.7	0	-47.7	-82.6	-95.5
P	12						
α	0	30°	60°	90°	120°	150°	180°
$V_{dc}(V)$	98.9	85.6	49.4	0	-49.5	-85.7	-98.9



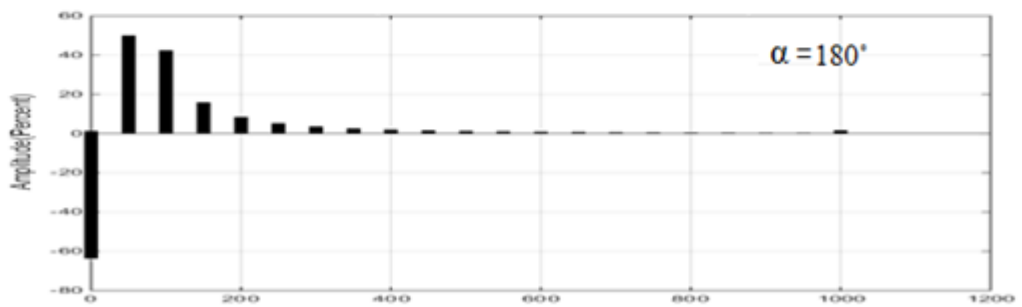
(a)



(b)



(c)



(d)

Fig.3.59 Harmonic frequency spectra for $p=2$ rectifier with different firing angles: (a) $\alpha = 0^\circ$, (b) $\alpha = 60^\circ$, (c) $\alpha = 90^\circ$, and (d) $\alpha = 180^\circ$.

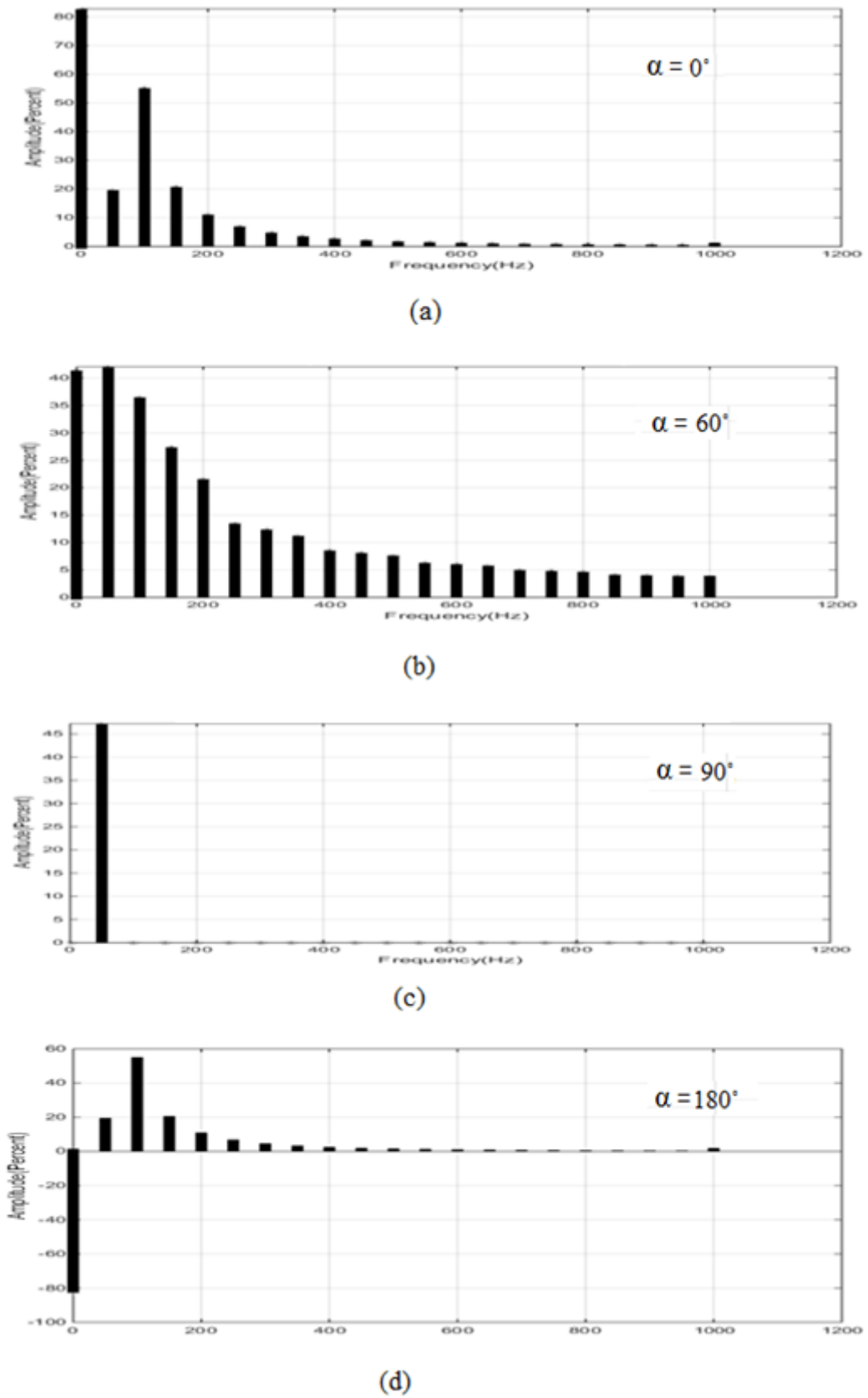
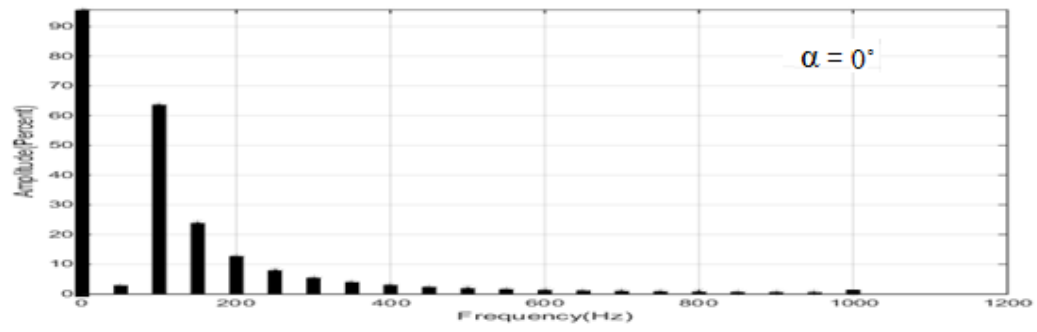
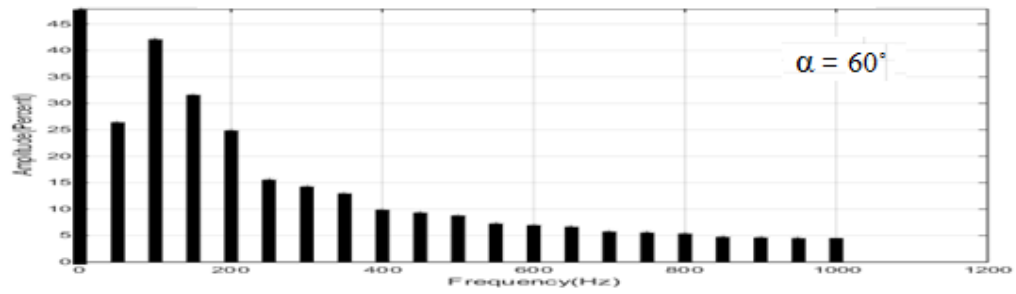


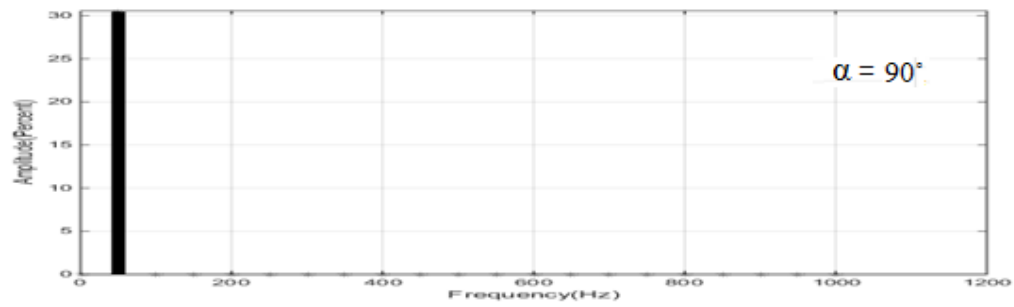
Fig.3.60 Harmonic frequency spectra for p =3 rectifier with different firing angles:
(a) $\alpha = 0^\circ$, (b) $\alpha = 60^\circ$, (c) $\alpha = 90^\circ$, and (d) $\alpha = 180^\circ$.



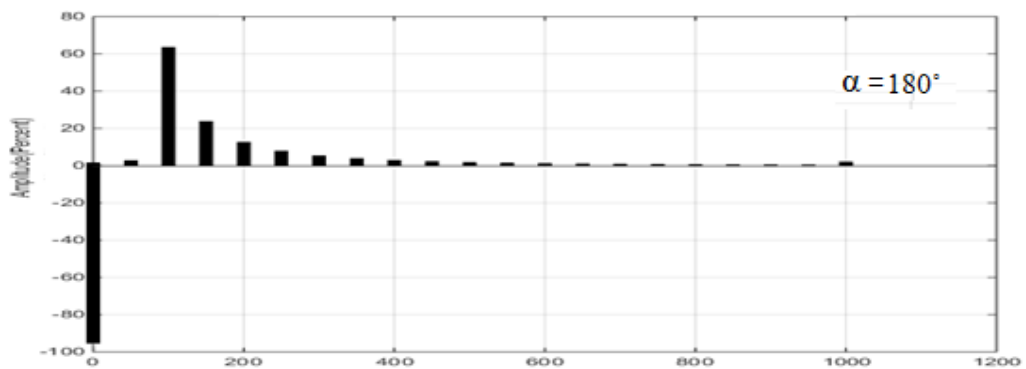
(a)



(b)

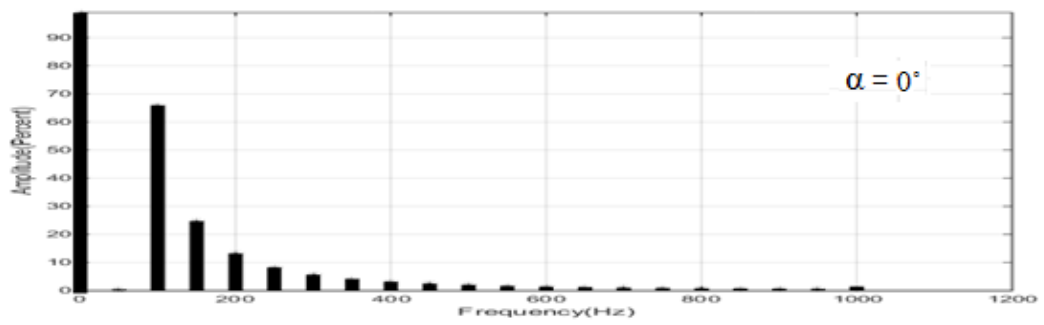


(c)

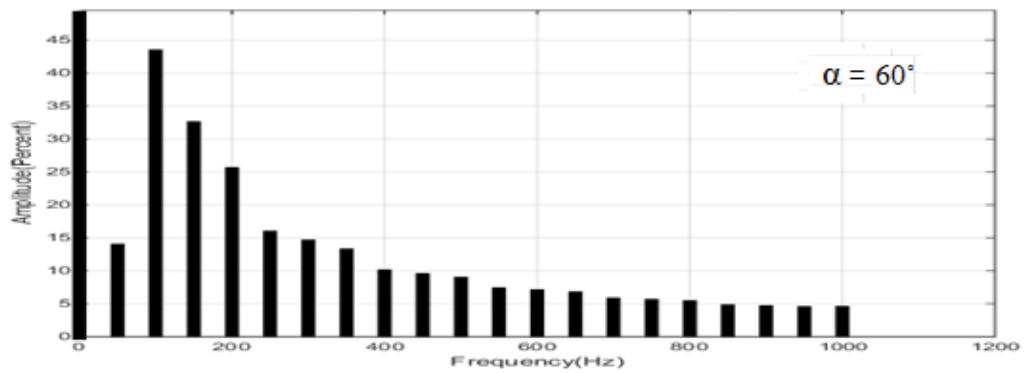


(d)

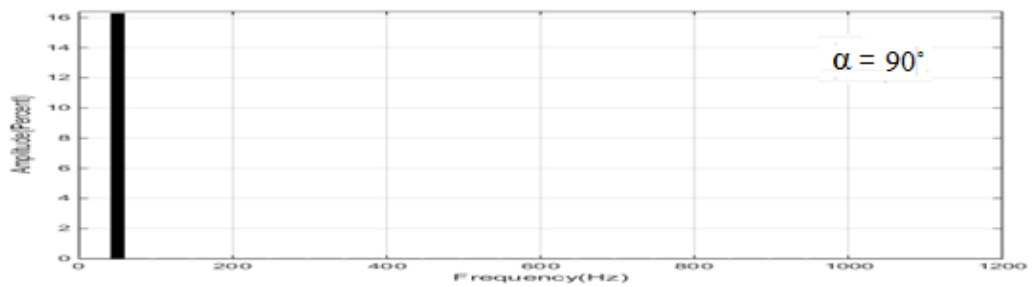
Fig.3.61 Harmonic frequency spectra for $p = 6$ rectifier with different firing angles: (a) $\alpha = 0^\circ$, (b) $\alpha = 60^\circ$, (c) $\alpha = 90^\circ$, and (d) $\alpha = 180^\circ$.



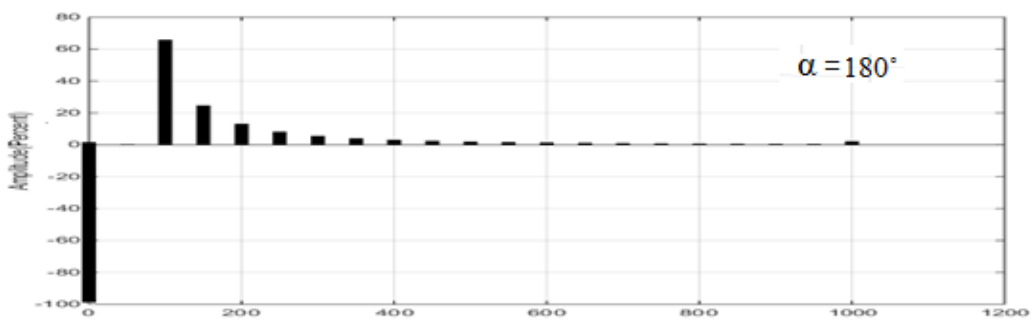
(a)



(b)



(c)



(d)

Fig.3.62 Harmonic frequency spectra for p=12 rectifier with different firing angles: (a) $\alpha = 0^\circ$, (b) $\alpha = 60^\circ$, (c) $\alpha = 90^\circ$, and (d) $\alpha = 180^\circ$.

PROBLEMS

- 3.1** A single- phase half - wave controlled rectifier shown in Fig.3.63, operates from an ideal sinusoidal source with $v_s = 325 \sin \omega t$, at 50Hz. If the load is purely resistive, and at a certain triggering α , the output d.c. voltage is $V_{dc} = 95$ V, and the average value of the load current is 2.2 A. It is required to calculate the following:

- The triggering angle α
- Load resistance
- The *rms* load voltage
- The *rms* load current
- DC power
- What are the current and voltage ratings of the thyristor used?

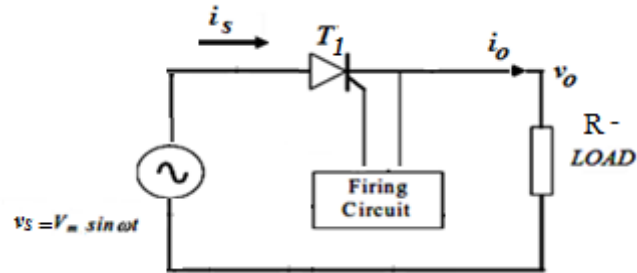


Fig.3.63.

[Ans : (a) 33.31° , (b) 43.18Ω , (c) 159.34 V , (d) 3.63 A, (e) 209 W, (f) 4 A, 400 V]

- 3.2** A single-phase half-wave controlled rectifier supplied from 230 V, 50 Hz a.c. supply is operating at $\alpha = 60^\circ$. If the load resistor is 15Ω , determine:

- The power absorbed by the load (P_{dc}), the power drawn from the supply (P_{ac}) and the efficiency of the rectifier.
- The form factor and the ripple factor.
- The volt ampere rating of the transformer and the transformer utilization factor.
- The power factor at the a.c. source.

[Ans : (a) $P_{dc} = 178.6$, $P_{ac} = 880.74$ W , $\eta = 0.202$, (b) $FF = 2.22$, $RF = 1.98$, (c) VA rating of the transformer $= 1761.78$, $TUF = 0.10$, (d) $PF = 0.1013$]

- 3.3** A bi-phase rectifier supplied from $230/110$ V, 50 Hz ideal transformer is connected to load of 10Ω resistance and infinite inductance.

- Determine the average output voltage, and current at a firing angle of 60° .
- Prove that the input power factor of the rectifier is given by

$$PF = \frac{2\sqrt{2}}{\pi} \cos \alpha$$

and find its value at $\alpha = 60^\circ$

[Ans: (a) 49.5 V, 4.95 A, (b) 0.450]

- 3.4** What are the required delay angles of a bi-phase controlled rectifier to obtain 60% and 25 % of its maximum output voltage? Assuming :
 (a) A highly inductive load.
 (b) A pure resistive load.

[Ans : (a) 53° , 75.5° , (b) 113.6° , 104.5°]

- 3.5** A single-phase mid-point (bi-phase) rectifier supplied from 240/120 V, 50 Hz ideal transformer is connected to load of $15\ \Omega$ resistance and infinite inductance. Determine the average output voltage, current and the power factor at a firing angle of 60° .

[Ans: 54 V, 3.6 A, 0.450]

- 3.6** A fully-controlled single-phase bridge rectifier is supplied from a 50 Hz 240/100 V transformer. It is supplying a highly inductive load of $5.5\ \Omega$ resistance. For a firing angle of 45° , determine the average output voltage, the average output current and the input power factor of the rectifier.

[Ans: 63.7 V, 11.57 A, 0.637]

- 3.7** If the line side of the transformer in the above example has an inductance of 5 mH, determine the output voltage, current and the angle of overlap.

[Ans: 61.4 V, 11.16 A, 9.35°]

- 3.8** Fig. 3.64 shows an SCR – Diode converter fed from 230V *rms* a.c. supply. The converter consists of one SCR and three diodes. Assume that load current is continuous and load resistance is $15\ \Omega$.

- (a) Prove that the average d.c. output voltage is given by

$$V_{dc} = \frac{V_m}{2\pi} (3 + \cos\alpha)$$

- (b) Draw waveforms of output voltage, load current and currents through thyristor and diodes for a firing angle of 60° .
 (c) Find d.c. output voltage, d.c. current and d.c. output power.

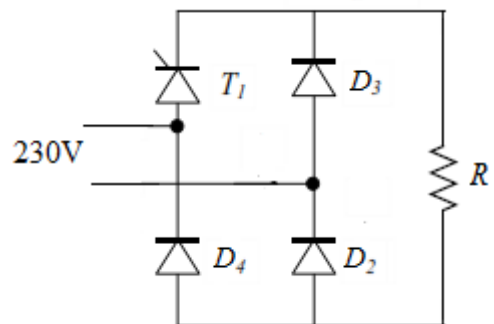
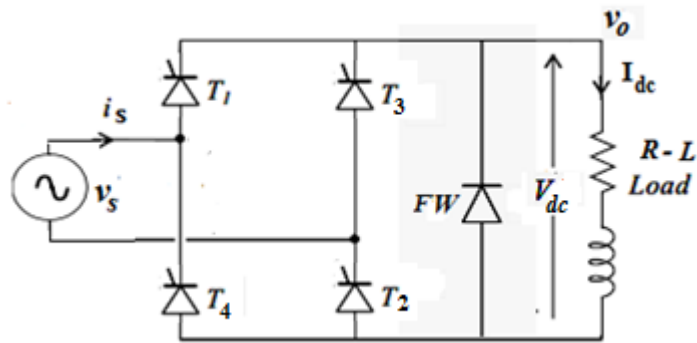


Fig.3.64.

[Ans: 181.3 V , 12 A , 2.2 kW]

- 3.9** A freewheel diode is connected across a single-phase fully-controlled, full-wave, bridge rectifier as shown in Fig. 3.65. The bridge has a source of 120 V *rms* at 50 Hz, and is feeding an *R-L* with load $R = 10\ \Omega$ and $L = 20\text{mH}$. The firing angle $\alpha = 30^\circ$ and the current extinction angle $\beta = 216^\circ$. It is required to:
- Determine whether the current is continuous or discontinuous.
 - Determine the average load voltage and current.
 - Calculate the *rms* load voltage and current.
 - Determine the a.c. and d.c. powers absorbed by the load.
 - Determine the efficiency of the converter.

Fig.3.65.



[Ans: (a) Current is continuous (b) 100.8 V, 10.08 A, (c) 118.28 V, 10.03 A, (d) 1184.14 W, 1018 W, (e) 85%]

- 3.10** A single -phase full-converter bridge circuit is feeding an *RLE* load and is fed from single-phase sinusoidal a.c. supply $v_s = 300 \sin \omega t$, 50Hz. The load current is constant at 20 A. $R = 0.25\ \Omega$, $L = 0.2\ \text{H}$. Find (a) firing angle if $E = 75\ \text{V}$, (b) firing angle if $E = -75\ \text{V}$, (c) input power factor in both cases, and (d) draw output voltage waveform.

[Ans : 65.25° , 111.49° , 0.377, 0.33]

- 3.11** A three-phase, star-connected, sinusoidal voltage supply has a peak voltage V_m per phase. This supplies power to a load resistor through a three-phase, full-wave bridge rectifier. Sketch the circuit arrangement and give the waveform of a phase current in correct time-phase to the corresponding supply phase voltage. Sketch the waveform of the load current for a supply cycle and derive an expression for its average value in terms of V_m and R . Explain what you would expect to be the lowest order of harmonic ripple frequency in the load current. What effect would be caused to the phase current waveform by (i) large load inductance, (ii) large supply inductance and large load inductance?

- 3.12** Show from first principle that the *rms* value of primary line current for a three-phase converter transformer is given by

$$I_p = \frac{\sqrt{2}}{3} I_{dc}$$

where I_{dc} is the value of well-smoothed d.c. current. The transformer has equal primary and secondary turns and star-connected primary winding.

- 3.13** Show that the line current supplying the primary winding of a 3-pulse converter with delta-connected primary, having equal primary and secondary turns and well-smoothed dc current I_d , i_p given by:

$$i_p = 3 \frac{I_{dc}}{\pi} \left[\cos \omega t - \frac{\sin 2\omega t}{2} + \frac{\sin 4\omega t}{4} + \frac{\cos 5\omega t}{5} - \frac{\cos 7\omega t}{7} + \frac{\sin 8\omega t}{8} - \frac{\sin 10\omega t}{10} - \frac{\cos 11\omega t}{11} + \frac{\cos 13\omega t}{13} \right]$$

Take the reference point $\omega t = 0$ on the waveform of i_p so as to correspond with the maximum positive voltage for the reference phase of a line carrying current i_p .

- 3.14** Show that the 'short-circuit' current flowing in the lagging phase of two involved in commutation of d.c. load current is given by

$$i_{sc} = i_{s2} = \frac{V_m}{\omega L_s}$$

- 3.15** A single-phase semiconverter bridge rectifier operating at a delay angle α and connected to highly inductive load such that the load current (I_{dc}) is continuous and ripple-free. Sketch the relevant waveforms during the development of your answer :

- (a) Prove that the *rms* value of the input current of the rectifier is given by

$$I_{in} = I_{dc} \sqrt{1 - \frac{\alpha}{\pi}}$$

- (b) Prove that the power factor of the fundamental component of the input current is

$$PF_1 = \cos \frac{\alpha}{2}$$

- (c) Prove that the power factor of the input current is

$$PF_{in} = (\sqrt{2}/\pi) \times (1 + \cos \alpha) / \sqrt{1 - \frac{\alpha}{\pi}}$$

- (d) Use the results obtained above to prove that the distortion factor of the input current can be expressed as

$$\text{Distortion factor} = \frac{0.9}{\sqrt{1 - \frac{\alpha}{\pi}}} \times \cos \frac{\alpha}{2}$$

- 3.16** A three-phase, full-wave half-controlled rectifier bridge circuit is operating at a delay angle $\alpha = 67^\circ$ when supplying full power. The per-phase inductance of the coupled transformer is 2 mH. The input voltage has an *rms* magnitude of 230 V per phase at 50 Hz. The load current in the d.c. side is 15 A at 200 V. It is required to:

- Obtain the drop in the d.c. voltage due to current overlap.
- Calculate the *rms* secondary voltage of the transformer.
- Calculate the overlap angle.
- Obtain the recovery angle while the converter is operating in the inversion mode with 15 A, 200 V d.c. input.
- Obtain the firing angle for the conditions in (d).

[Ans : (a) 4.5 V, (b) 450 V, (c) 1.06° , (d) 67.14° , (e) 111.8°]

- 3.17** In problem 3.16, if the converter (and the input transformer) is replaced by a three-phase full-wave fully-controlled one to deliver the same output power (voltage and current) when operating at the same firing angle α of that problem in rectification mode. It is required to:

- Obtain the drop in the d.c. voltage due to a.c. source internal reactance.
- Calculate the overlap angle.
- Calculate the extinction angle when the circuit is to function as inverter with 15 A and 200 V power.
- Obtain the firing angle for the case stated in (c).

[Ans: (a) 9 V, (b) 2.06° , (c) 67.14° , (e) 110.8°]

- 3.18** In a single-phase half-wave ac-dc converter, the average value of the load current is 1.78 A. If the converter is operated from a 240 V, 50 Hz supply and if the average value of the output voltage is 27% of the maximum possible value, calculate the following, assume the load to be resistive.

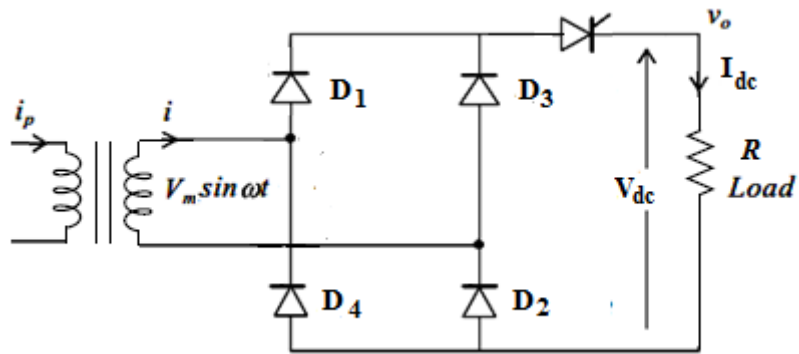
- Load resistance
- Firing angle
- Average output voltage
- The *rms* load voltage
- The *rms* load current
- DC power

- (g) AC power
- (h) Rectifier efficiency
- (i) Form factor
- (j) Ripple factor

[Ans: (a) 51.4Ω , (b) 45° , (c) 91 V , (d) 134 V , (e) 2.6 A , (f) 162 W , (g) 349 W , (h) 46.4% , (i) 1.362 , (j) 1.08]

- 3.19** For the circuit arrangement shown in Fig.3.66 and for a pure resistive load, sketch a set of waveforms for thyristor triggering $\alpha = 90^\circ$ showing the supply current, load current and thyristors voltage .
Derive an expression for the average value of the load current I_{dc} and calculate its values for $\alpha = 0^\circ$, 60° , and 120° if the the supply voltage is ideal sinusoid with $V_m = 400 \text{ V}$, 50 Hz and the load resistor $R = 10 \Omega$. How is the average load current affected if diode D_2 fails to open circuit?

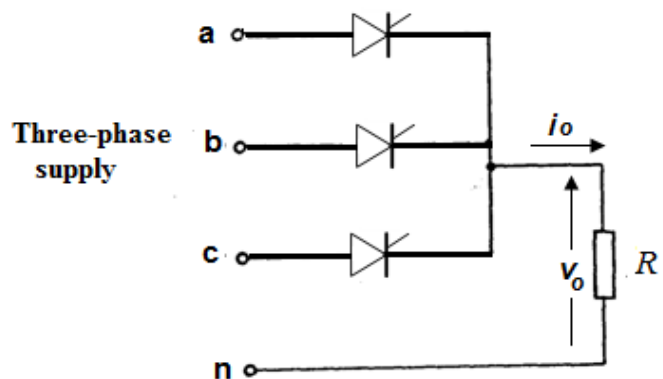
Fig.3.66.



[Ans : 12.7 A , 9.5 A , 3.17 A .When diode D_2 failed to open circuit, the circuit will act as a half-wave rectifier]

- 3.20** For the three-phase half-wave fully controlled rectifier shown in Fig.3.67, the load consists of a resistance and a very large inductance. The inductance is so large that the output current i_o can be assumed to be continuous and ripple-free. For triggering angle $\alpha = 30^\circ$.

Fig.3.67.



- (a) Draw the waveforms of v_o and i_o .
 (b) Prove that the d.c. power delivered to the load is given by:

$$P_{dc} = 0.684 \frac{V_m^2}{R} \cos^2 \alpha$$

- (c) Determine the mean value of the output voltage, current and d.c. power if phase voltage is $V_{an} = 220$ V (*rms*), $\alpha = 30^\circ$ and $R = 50$ ohms.

[Ans: (c) 121.6 V, 6.08 A , 740 W]

- 3.21** A three-phase half-wave controlled rectifier is connected to a star supply and a resistive load $R = 10 \Omega$. If the rectifier is turned-on at a firing angle of 60° , calculate average and *rms* output voltages and currents. Assume phase supply voltage to be 120 V.

[Ans: 70 V, 106.8 V, 7 A, 10.68 A]

- 3.22** A large magnet load consists of an inductance of 18.5 H and resistance of 50Ω is to be supplied from a 400 V, 50 Hz, three-phase source through a fully controlled full-wave, three-phase rectifier bridge as shown in Fig.3.68. It is required to determine:

- (a) The power delivered to the magnet at firing angle of 60° .
 (b) The maximum d.c. voltage $V_{dc,max}$ and maximum current $I_{dc,max}$ which can be supplied from this rectifier ,
 (c) The delay angle α required to give a magnet current of 4A.

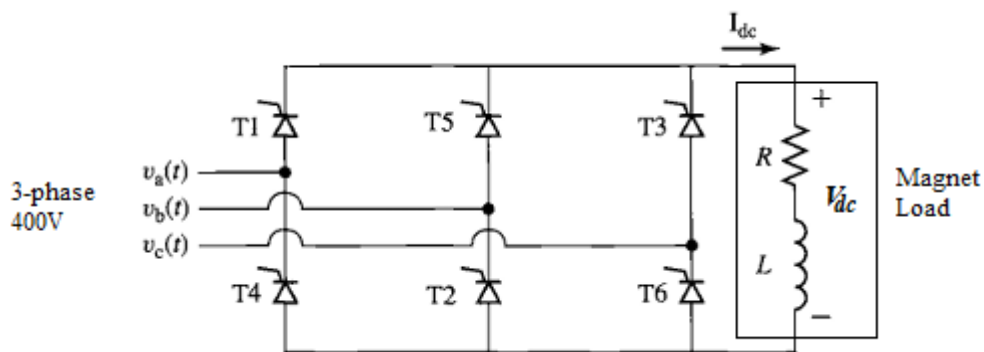


Fig.3.68.

[Ans : $P_o = 1.447$ kW, $V_{dc,max} = 538$ V , $I_{dc,max} = 10.76$ A , $\alpha = 68.18^\circ$]

- 3.23** For a p-pulse convertor, show that the average voltage V_{dc} considering the overlap period is given by:

$$V_{dc} = \frac{PV_m}{2\pi} \sin \frac{\pi}{p} (\cos \alpha + \cos(\alpha + \mu))$$

$$V_{dc} = \frac{PV_m}{\pi} \sin \frac{\pi}{p} \cos \alpha - \frac{P\omega LI}{2\pi}$$

$$\text{or } = \frac{PV_m}{\pi} \sin \frac{\pi}{p} \cos(\alpha + \mu) + \frac{P\omega LI}{2\pi}$$

- 3.24** When a controlled rectifier circuit operates with pure inductive load, the average output voltage across the load is zero but unidirectional current flows in the load. If the delay angle exceeds a certain value, dependent of the pulse number, this load current becomes discontinuous. Determine the relationship between pulse number and the minimum value of α at which load current becomes discontinuous and show that, with discontinuous current flow, the average value of the d.c. load current in p.u. of its peak value is given by :

$$\frac{I_{dc}}{I_m} = \frac{p}{2} [(\beta' - \alpha') \cos \alpha' + \sin \alpha' - \sin \beta']$$

Where α' and β' are respectively the delay angle and current extinction angle measured from the instant of zero transit into the positive half cycle for a phase about to conduct.

- 3.25** (a) Show that for a 6-pulse converter with a purely resistive load, where the triggering angle α is such that load current is discontinuous, the *rms* value of d.c. voltage is given by :

$$V_{orms} = V_m \sqrt{1 - \frac{3\alpha}{2\pi} - \frac{3}{4\pi} \sin 2\alpha}$$

where V_m is the maximum value of the transformer secondary voltage per-phase.

(b) For 400 V three-phase voltage supplying 6-pulse converter operating such that the load current is discontinuous, find the *rms* value of the output voltage and the *rms* value of the load current when the triggering angle is 75° . Assume the load is resistive with $R = 20 \Omega$.

[Ans : (b) 123.5V , 6.17A]